

Table 6.4 Approximate Lower a_w Limits for Microbial Growth

Water Activity	Environment	Bacteria	Fungi	Algae
1.00—Pure water	Blood { Vegetables, Plant wilt { meat, fruit Seawater	Most gram-negative nonhalophiles		
0.95	Bread	Most gram-positive rods	<i>Basidiomycetes</i>	Most algae
0.90	Ham	Most cocci, <i>Bacillus</i>	<i>Fusarium</i> <i>Mucor</i> , <i>Rhizopus</i> Ascomycetous yeasts <i>Saccharomyces rouxii</i> (in salt)	
0.85	Salami	<i>Staphylococcus</i>	<i>Penicillium</i>	
0.80	Preserves		<i>Aspergillus</i>	<i>Dunaliella</i>
0.75	Salt lakes Salted fish	<i>Halobacterium</i> <i>Actinospira</i>		
0.70			<i>Aspergillus</i>	
0.60	Cereals, candy, dried fruit		<i>Saccharomyces rouxii</i> (in sugars)	
	Chocolate		<i>Xeromyces bisporus</i>	
	Honey			
	Dried milk			
0.55—DNA disordered				

Adapted from A. D. Brown, "Microbial Water Stress," in *Bacteriological Reviews*, 40(4):803–846 1976. Copyright © 1976 by the American Society for Microbiology. Reprinted by permission.

Halophiles have adapted so completely to hypertonic, saline conditions that they require high levels of sodium chloride to grow, concentrations between about 2.8 M and saturation (about 6.2 M) for extreme halophilic bacteria. The archaeon *Halobacterium* can be isolated from the Dead Sea (a salt lake between Israel and Jordan and the lowest lake in the world), the Great Salt Lake in Utah, and other aquatic habitats with salt concentrations approaching saturation. *Halobacterium* and other extremely halophilic bacteria have significantly modified the structure of their proteins and membranes rather than simply increasing the intracellular concentrations of solutes, the approach used by most osmotolerant microorganisms. These extreme halophiles accumulate enormous quantities of potassium in order to remain hypertonic to their environment; the internal potassium concentration may reach 4 to 7 M. The enzymes, ribosomes, and transport proteins of these bacteria require high levels of potassium for stability and activity. In addition, the plasma membrane and cell wall of *Halobacterium* are stabilized by high concentrations of sodium ion. If the sodium concentration decreases too much, the wall and plasma membrane literally disintegrate. Extreme halophilic bacteria have successfully adapted to environmental conditions that would destroy most organisms. In the process they have become so specialized that they have lost ecological flexibility and can prosper only in a few extreme habitats. [The halobacteria \(section 20.3\)](#)

pH

pH is a measure of the hydrogen ion activity of a solution and is defined as the negative logarithm of the hydrogen ion concentration (expressed in terms of molarity).

$$\text{pH} = -\log [\text{H}^+] = \log(1/[\text{H}^+])$$

The pH scale extends from pH 0.0 (1.0 M H^+) to pH 14.0 ($1.0 \times 10^{-14} \text{ M H}^+$), and each pH unit represents a tenfold change in hydrogen ion concentration. **Figure 6.11** shows that the habitats in which microorganisms grow vary widely—from pH 1 to 2 at the acid end to alkaline lakes and soil that may have pH values between 9 and 10.

It is not surprising that pH dramatically affects microbial growth. Each species has a definite pH growth range and pH growth optimum. **Acidophiles** have their growth optimum between pH 0 and 5.5; **neutrophiles**, between pH 5.5 and 8.0; and **alkalophiles** prefer the pH range of 8.5 to 11.5. Extreme alkalophiles have growth optima at pH 10 or higher. In general, different microbial groups have characteristic pH preferences. Most bacteria and protozoa are neutrophiles. Most fungi prefer slightly acid surroundings, about pH 4 to 6; algae also seem to favor slight acidity. There are many exceptions to these generalizations. For example, the alga *Cyanidium caldarium* and the archaeon *Sulfolobus acidocaldarius* are common inhabitants of acidic hot springs; both grow well around pH 1 to 3 and at high temperatures. The Archaea *Ferroplasma acidarmanus* and *Picrophilus oshimae* can actually grow at pH 0, or very close to it.

Although microorganisms will often grow over wide ranges of pH and far from their optima, there are limits to their tolerance. Drastic variations in cytoplasmic pH can harm microorganisms by disrupting the plasma membrane or inhibiting the activity of

1. How do microorganisms adapt to hypotonic and hypertonic environments? What is plasmolysis?
2. Define water activity and briefly describe how it can be determined.
3. Why is it difficult for microorganisms to grow at low a_w values?
4. What are halophiles and why does *Halobacterium* require sodium and potassium ions?

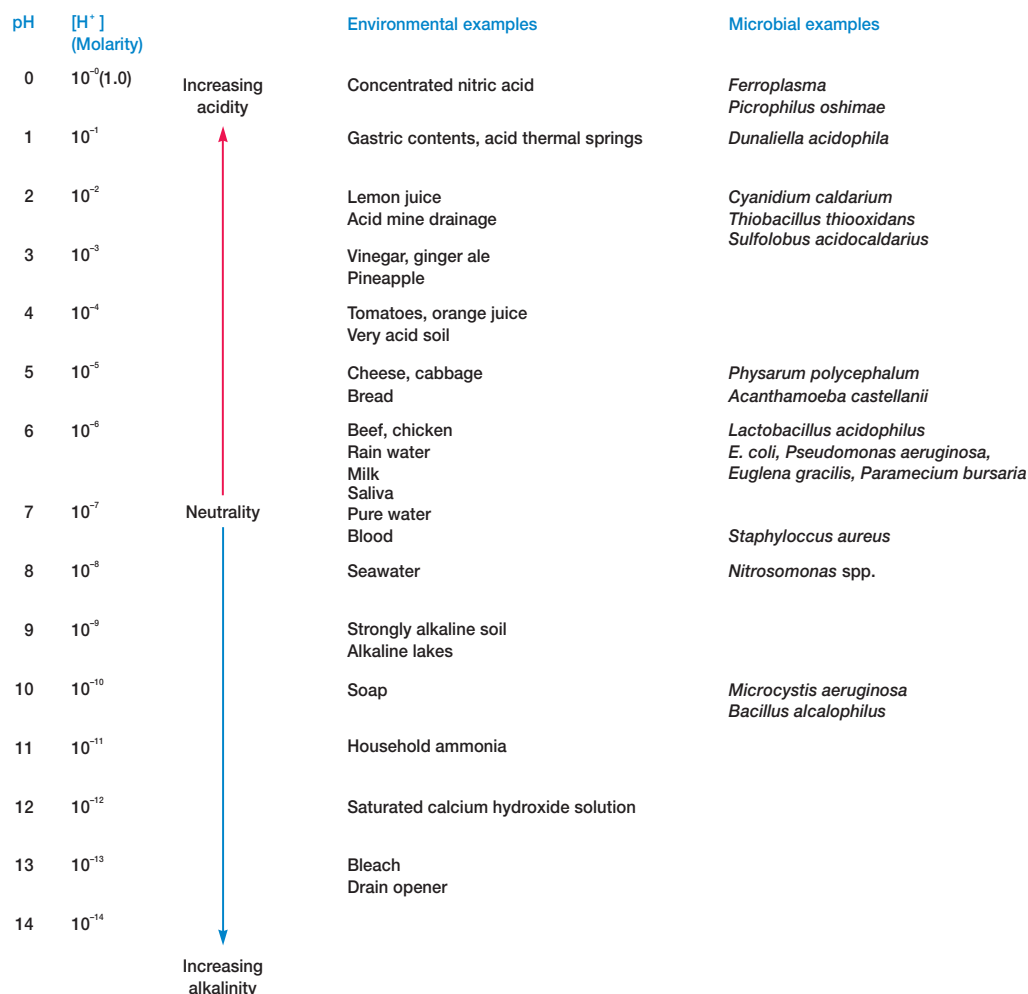


Figure 6.11 The pH Scale. The pH scale and examples of substances with different pH values. The microorganisms are placed at their growth optima.

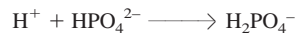
enzymes and membrane transport proteins. Prokaryotes die if the internal pH drops much below 5.0 to 5.5. Changes in the external pH also might alter the ionization of nutrient molecules and thus reduce their availability to the organism.

Several mechanisms for the maintenance of a neutral cytoplasmic pH have been proposed. The plasma membrane may be relatively impermeable to protons. Neutrophiles appear to exchange potassium for protons using an antiport transport system (*p. 102*). Extreme alkalophiles like *Bacillus alcalophilus* maintain their internal pH closer to neutrality by exchanging internal sodium ions for external protons. Internal buffering also may contribute to pH homeostasis.

Microorganisms often must adapt to environmental pH changes to survive. In bacteria, potassium/proton and sodium/proton antiport systems probably correct small variations in pH. If the pH becomes too acidic, other mechanisms come into play. When the pH drops below about 5.5 to 6.0, *Salmonella typhimurium* and *E. coli* synthesize an array of new proteins as part of what has been called their acidic tolerance response. A proton-translocating ATPase contributes to this protective response, either by making more ATP or by pumping protons out of the cell. If the external pH decreases to 4.5 or lower, chaperones such as acid shock proteins and heat shock proteins are synthesized (*see pp. 272–74*). Presumably these prevent the acid denaturation of proteins and aid in the refolding of denatured proteins.

Microorganisms frequently change the pH of their own habitat by producing acidic or basic metabolic waste products. Fermentative microorganisms form organic acids from carbohydrates, whereas chemolithotrophs like *Thiobacillus* oxidize reduced sulfur components to sulfuric acid. Other microorganisms make their environment more alkaline by generating ammonia through amino acid degradation. [Microbial fermentations \(pp. 179–81\)](#); [Sulfur-oxidizing bacteria \(pp. 496–98\)](#)

Buffers often are included in media to prevent growth inhibition by large pH changes. Phosphate is a commonly used buffer and a good example of buffering by a weak acid (H_2PO_4^-) and its conjugate base (HPO_4^{2-}).



If protons are added to the mixture, they combine with the salt form to yield a weak acid. An increase in alkalinity is resisted because the weak acid will neutralize hydroxyl ions through proton donation to give water. Peptides and amino acids in complex media also have a strong buffering effect.

Temperature

Environmental temperature profoundly affects microorganisms, like all other organisms. Indeed, microorganisms are particularly susceptible because they are usually unicellular and their temperature varies with that of the external environment. For these reasons, microbial cell temperature directly reflects that of the cell's surroundings. A most important factor influencing the effect of temperature on growth is the temperature sensitivity of enzyme-catalyzed reactions. At low temperatures a temperature rise increases the growth rate because the velocity of an enzyme-catalyzed reaction, like that of any chemical reaction, will roughly double for every 10°C rise in temperature. Because the rate of each reaction increases, metabolism as a whole is more active at higher temperatures, and the microorganism grows faster. Beyond a certain point further increases actually slow growth, and sufficiently high temperatures are lethal. High temperatures damage microorganisms by denaturing enzymes, transport carriers, and other proteins. Microbial membranes are also disrupted by temperature extremes; the lipid bilayer simply melts and disintegrates. Thus, although functional enzymes operate more rapidly at higher temperatures, the microorganism may be damaged to such an extent that growth is inhibited because the damage cannot be repaired. At very low temperatures, membranes solidify and enzymes don't work rapidly. In summary, when organisms are above the optimum temperature, both function and cell structure are affected. If temperatures are very low, function is affected but not necessarily cell chemical composition and structure. [The temperature dependence of enzyme activity \(pp. 163–64\)](#)

Because of these opposing temperature influences, microbial growth has a fairly characteristic temperature dependence with distinct **cardinal temperatures**—minimum, optimum, and max-

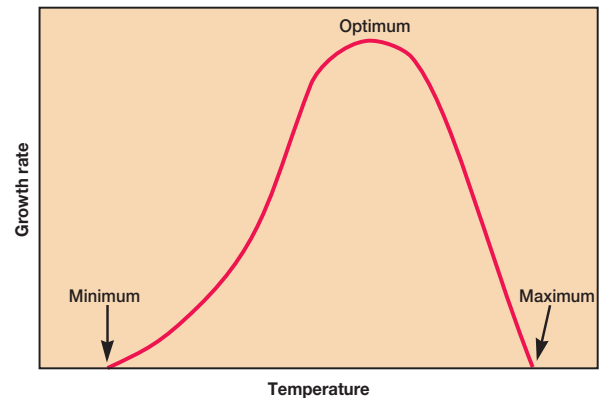


Figure 6.12 Temperature and Growth. The effect of temperature on growth rate.

imum growth temperatures (**figure 6.12**). Although the shape of the temperature dependence curve can vary, the temperature optimum is always closer to the maximum than to the minimum. The cardinal temperatures for a particular species are not rigidly fixed but often depend to some extent on other environmental factors such as pH and the available nutrients. For example, *Crithidia fasciculata*, a flagellated protozoan living in the gut of mosquitos, will grow in a simple medium at 22 to 27°C. However, it cannot be cultured at 33 to 34°C without the addition of extra metals, amino acids, vitamins, and lipids.

The cardinal temperatures vary greatly between microorganisms (table 6.5). Optima normally range from 0°C to as high as 75°C, whereas microbial growth occurs at temperatures extending from –20°C to over 100°C. The major factor determining this growth range seems to be water. Even at the most extreme temperatures, microorganisms need liquid water to grow. The growth temperature range for a particular microorganism usually spans about 30 degrees. Some species (e.g., *Neisseria gonorrhoeae*) have a small range; others, like *Enterococcus faecalis*, will grow over a wide range of temperatures. The major microbial groups differ from one another regarding their maximum growth temperature. The upper limit for protozoa is around 50°C. Some algae and fungi can grow at temperatures as high as 55 to 60°C. Prokaryotes have been found growing at or close to 100°C, the boiling point of water at sea level (*see figure 20.8*). Recently strains growing at even higher temperatures have been discovered (**Box 6.1**). Clearly, prokaryotic organisms can grow at much higher temperatures than eucaryotes. It has been suggested that eucaryotes are not able to manufacture organellar membranes that are stable and functional at temperatures above 60°C. The photosynthetic apparatus also appears to be relatively unstable because photosynthetic organisms are not found growing at very high temperatures.

Box 6.1

Life above 100°C

Until recently the highest reported temperature for procaryotic growth was 105°C. It seemed that the upper temperature limit for life was about 100°C, the boiling point of water. Now thermophilic procaryotes have been reported growing in sulfide chimneys or “black smokers,” located along rifts and ridges on the ocean floor, that spew sulfide-rich superheated vent water with temperatures above 350°C (see **Box figure**). Evidence has been presented that these microbes can grow and reproduce at 113°C. The pressure present in their habitat is sufficient to keep water liquid (at 265 atm; seawater doesn’t boil until 460°C).

The implications of this discovery are many. The proteins, membranes, and nucleic acids of these procaryotes are remarkably temperature stable and provide ideal subjects for studying the ways in which macromolecules and membranes are stabilized. In the future it may be possible to design enzymes that can operate at very high temperatures. Some thermostable enzymes from these organisms have important industrial and scientific uses. For example, the Taq polymerase from the thermophile *Thermus aquaticus* is used extensively in the polymerase chain reaction (see pp. 326–27).



Microorganisms such as those in **table 6.5** can be placed in one of five classes based on their temperature ranges for growth (**figure 6.13**).

1. **Psychrophiles** grow well at 0°C and have an optimum growth temperature of 15°C or lower; the maximum is around 20°C. They are readily isolated from Arctic and Antarctic habitats; because 90% of the ocean is 5°C or colder (see *chapter 29*), it constitutes an enormous habitat for psychrophiles. The psychrophilic alga *Chlamydomonas nivalis* can actually turn a snowfield or glacier pink with its bright red spores. Psychrophiles are widespread among bacterial taxa and found in such genera as *Pseudomonas*, *Vibrio*, *Alcaligenes*, *Bacillus*, *Arthrobacter*, *Moritella*, *Photobacterium*, and *Shewanella*. The psychrophilic archaeon *Methanogenium* has recently been isolated from Ace Lake in Antarctica. Psychrophilic microorganisms have adapted to their environment in several ways. Their enzymes, transport systems, and protein synthetic mechanisms function well at low temperatures. The cell membranes of psychrophilic microorganisms have high levels of unsaturated fatty acids and remain semifluid when cold. Indeed, many psychrophiles begin to leak cellular constituents at temperatures higher than 20°C because of cell membrane disruption.
2. Many species can grow at 0 to 7°C even though they have optima between 20 and 30°C, and maxima at about 35°C. These are called **psychrotrophs** or **facultative psychrophiles**. Psychrotrophic bacteria and fungi are major factors in the spoilage of refrigerated foods (see *chapter 41*).
3. **Mesophiles** are microorganisms with growth optima around 20 to 45°C; they often have a temperature minimum of 15 to 20°C. Their maximum is about 45°C or lower. Most microorganisms probably fall within this category. Almost all human pathogens are mesophiles, as might be expected since their environment is a fairly constant 37°C.
4. Some microorganisms are **thermophiles**; they can grow at temperatures of 55°C or higher. Their growth minimum is usually around 45°C and they often have optima between 55 and 65°C. The vast majority are procaryotes although a few algae and fungi are thermophilic (table 6.5). These organisms flourish in many habitats including composts, self-heating hay stacks, hot water lines, and hot springs. Thermophiles differ from mesophiles in having much more heat-stable enzymes and protein synthesis systems able to function at high temperatures. Their membrane lipids are also more saturated than those of mesophiles and have higher melting points; therefore thermophile membranes remain intact at higher temperatures.
5. As mentioned previously, a few thermophiles can grow at 90°C or above and some have maxima above 100°C. Procaryotes that have growth optima between 80°C and about 113°C are called **hyperthermophiles**. They usually do not grow well below 55°C. *Pyrococcus abyssi* and *Pyrodicticum occultum* are examples of marine hyperthermophiles found in hot areas of the seafloor.

Table 6.5 Temperature Ranges for Microbial Growth

Microorganism	Cardinal Temperatures (°C)		
	Minimum	Optimum	Maximum
Nonphotosynthetic Prokaryotes			
<i>Bacillus psychrophilus</i>	−10	23–24	28–30
<i>Micrococcus cryophilus</i>	−4	10	24
<i>Pseudomonas fluorescens</i>	4	25–30	40
<i>Staphylococcus aureus</i>	6.5	30–37	46
<i>Enterococcus faecalis</i>	0	37	44
<i>Escherichia coli</i>	10	37	45
<i>Neisseria gonorrhoeae</i>	30	35–36	38
<i>Thermoplasma acidophilum</i>	45	59	62
<i>Bacillus stearothermophilus</i>	30	60–65	75
<i>Thermus aquaticus</i>	40	70–72	79
<i>Sulfolobus acidocaldarius</i>	60	80	85
<i>Pyrococcus abyssi</i>	67	96	102
<i>Pyrodictum occultum</i>	82	105	110
<i>Pyrolobus fumarii</i>	90	106	113
Photosynthetic Bacteria			
<i>Rhodospirillum rubrum</i>	ND ^a	30–35	ND
<i>Anabaena variabilis</i>	ND	35	ND
<i>Oscillatoria tenuis</i>	ND	ND	45–47
<i>Synechococcus eximius</i>	70	79	84
Eucaryotic Algae			
<i>Chlamydomonas nivalis</i>	−36	0	4
<i>Fragilaria sublinearis</i>	−2	5–6	8–9
<i>Chlorella pyrenoidosa</i>	ND	25–26	29
<i>Euglena gracilis</i>	ND	23	ND
<i>Skeletonema costatum</i>	6	16–26	>28
<i>Cyanidium caldarium</i>	30–34	45–50	56
Fungi			
<i>Candida scottii</i>	0	4–15	15
<i>Saccharomyces cerevisiae</i>	1–3	28	40
<i>Mucor pusillus</i>	21–23	45–50	50–58
Protozoa			
<i>Amoeba proteus</i>	4–6	22	35
<i>Naegleria fowleri</i>	20–25	35	40
<i>Trichomonas vaginalis</i>	25	32–39	42
<i>Paramecium caudatum</i>		25	28–30
<i>Tetrahymena pyriformis</i>	6–7	20–25	33
<i>Cyclidium citrullus</i>	18	43	47

^aND, no data.

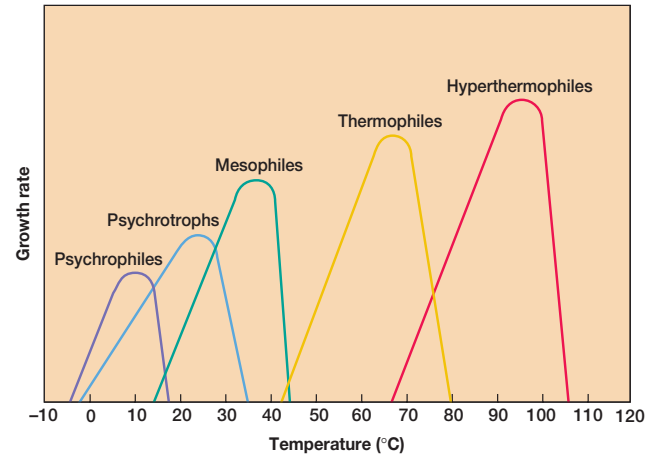


Figure 6.13 Temperature Ranges for Microbial Growth.

Microorganisms can be placed in different classes based on their temperature ranges for growth. They are ranked in order of increasing growth temperature range as psychrophiles, psychrotrophs, mesophiles, thermophiles, and hyperthermophiles. Representative ranges and optima for these five types are illustrated here.

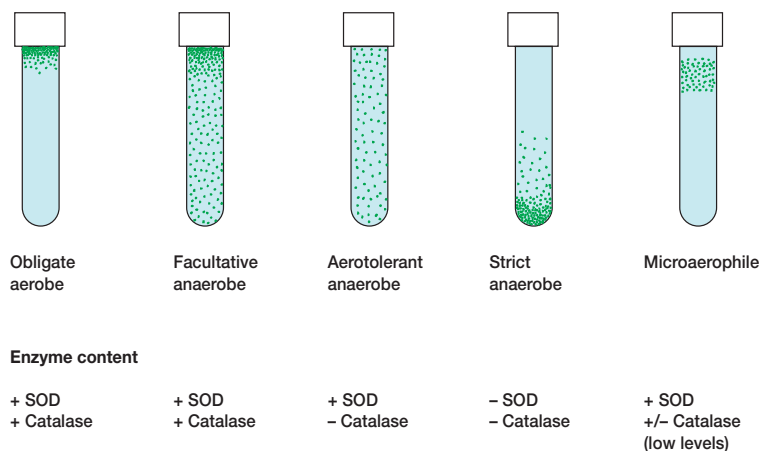
Oxygen Concentration

An organism able to grow in the presence of atmospheric O_2 is an **aerobe**, whereas one that can grow in its absence is an **anaerobe**. Almost all multicellular organisms are completely dependent on atmospheric O_2 for growth—that is, they are **obligate aerobes** (table 6.3). Oxygen serves as the terminal electron acceptor for the electron-transport chain in aerobic respiration. In addition, aerobic eucaryotes employ O_2 in the synthesis of sterols and unsaturated fatty acids. **Facultative anaerobes** do not require O_2 for growth but do grow better in its presence. In the presence of oxygen they will use aerobic respiration. **Aerotolerant anaerobes** such as *Enterococcus faecalis* simply ignore O_2 and grow equally well whether it is present or not. In contrast, **strict** or **obligate anaerobes** (e.g., *Bacteroides*, *Fusobacterium*, *Clostridium pasteurianum*, *Methanococcus*) do not tolerate O_2 at all and die in its presence. Aerotolerant and strict anaerobes cannot generate energy through respiration and must employ fermentation or anaerobic respiration pathways for this purpose. Finally, there are aerobes such as *Campylobacter*, called **microaerophiles**, that are damaged by the normal atmospheric level of O_2 (20%) and require O_2 levels below the range of 2 to 10% for growth. The nature of bacterial O_2 responses can be readily determined by growing the bacteria in culture tubes filled with a solid culture medium or a special medium like thioglycollate broth, which contains a reducing agent to lower O_2 levels (**figure 6.14**). [Electron transport and aerobic respiration \(pp. 184–89\); Fermentation \(pp. 179–81\); Anaerobic respiration \(pp. 190–91\)](#)

A microbial group may show more than one type of relationship to O_2 . All five types are found among the prokaryotes and protozoa. Fungi are normally aerobic, but a number of species—particularly among the yeasts—are facultative anaerobes. Algae

1. Define pH, acidophile, neutrophile, and alkalophile. How can microorganisms change the pH of their environment, and how does the microbiologist minimize this effect?
2. What are cardinal temperatures?
3. Why does the growth rate rise with increasing temperature and then fall again at higher temperatures?
4. Define psychrophile, psychrotroph, mesophile, thermophile, and hyperthermophile.

Figure 6.14 Oxygen and Bacterial Growth. An illustration of the growth of bacteria with varying responses to oxygen. Each dot represents an individual bacterial colony within the agar or on its surface. The surface, which is directly exposed to atmospheric oxygen, will be aerobic. The oxygen content of the medium decreases with depth until the medium becomes anaerobic toward the bottom of the tube. The presence and absence of the enzymes superoxide dismutase (SOD) and catalase for each type are shown.

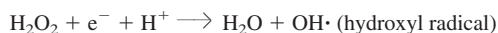
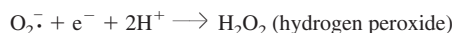
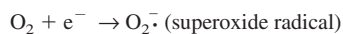


are almost always obligate aerobes. It should be noted that the ability to grow in both aerobic and anaerobic environments provides considerable flexibility and is an ecological advantage.

Although strict anaerobes are killed by O_2 , they may be recovered from habitats that appear to be aerobic. In such cases they associate with facultative anaerobes that use up the available O_2 and thus make the growth of strict anaerobes possible. For example, the strict anaerobe *Bacteroides gingivalis* lives in the mouth where it grows in the anaerobic crevices around the teeth.

These different relationships with O_2 appear due to several factors, including the inactivation of proteins and the effect of toxic O_2 derivatives. Enzymes can be inactivated when sensitive groups like sulfhydryls are oxidized. A notable example is the nitrogen-fixation enzyme nitrogenase (see section 10.4), which is very oxygen sensitive.

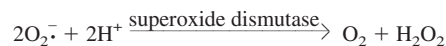
Oxygen accepts electrons and is readily reduced because its two outer orbital electrons are unpaired. Flavoproteins (see section 8.5), several other cell constituents, and radiation (pp. 130–31) promote oxygen reduction. The result is usually some combination of the reduction products **superoxide radical**, **hydrogen peroxide**, and **hydroxyl radical**.



These products of oxygen reduction are extremely toxic because they are powerful oxidizing agents and rapidly destroy cellular constituents. A microorganism must be able to protect itself against such oxygen products or it will be killed. Neutrophils and macrophages use these toxic oxygen products to destroy invading pathogens. **Oxygen-dependent killing of pathogens** (pp. 718–20)

Many microorganisms possess enzymes that afford protection against toxic O_2 products. Obligate aerobes and facultative anaerobes usually contain the enzymes **superoxide dismutase (SOD)** and **catalase**, which catalyze the destruction of superox-

ide radical and hydrogen peroxide, respectively. Peroxidase also can be used to destroy hydrogen peroxide.



Aerotolerant microorganisms may lack catalase but almost always have superoxide dismutase. The aerotolerant *Lactobacillus plantarum* uses manganous ions instead of superoxide dismutase to destroy the superoxide radical. All strict anaerobes lack both enzymes or have them in very low concentrations and therefore cannot tolerate O_2 .

Because aerobes need O_2 and anaerobes are killed by it, radically different approaches must be used when growing the two types of microorganisms. When large volumes of aerobic microorganisms are cultured, either the culture vessel is shaken to aerate the medium or sterile air must be pumped through the culture vessel. Precisely the opposite problem arises with anaerobes; all O_2 must be excluded. This can be accomplished in several ways. (1) Special anaerobic media containing reducing agents such as thioglycollate or cysteine may be used. The medium is boiled during preparation to dissolve its components; boiling also drives off oxygen very effectively. The reducing agents will eliminate any dissolved O_2 remaining within the medium so that anaerobes can grow beneath its surface. (2) Oxygen also may be eliminated from an anaerobic system by removing air with a vacuum pump and flushing out residual O_2 with nitrogen gas (figure 6.15). Often CO_2 as well as nitrogen is added to the chamber since many anaerobes require a small amount of CO_2 for best growth. (3) One of the most popular ways of culturing small numbers of anaerobes is by use of a Gas-Pak jar (figure 6.16). In this procedure the environment is made anaerobic by using hydrogen and a palladium catalyst to remove

O₂ through the formation of water. The reducing agents in anaerobic agar also remove oxygen, as mentioned previously. (4) Plastic bags or pouches make convenient containers when only a few samples are to be incubated anaerobically. These have a catalyst and calcium carbonate to produce an anaerobic, carbon-dioxide-rich atmosphere. A special solution is added to the pouch's reagent compartment; petri dishes or other containers are placed

in the pouch; it then is clamped shut and placed in an incubator. A laboratory may make use of all these techniques since each is best suited for different purposes.

1. Describe the five types of O₂ relationships seen in microorganisms.
2. For what do aerobes use O₂? Why is O₂ toxic to many microorganisms and how do they protect themselves?
3. Describe four ways in which anaerobes may be cultured.



Figure 6.15 An Anaerobic Work Chamber and Incubator. This anaerobic system contains an oxygen-free work area and an incubator. The interchange compartment on the right of the work area allows materials to be transferred inside without exposing the interior to oxygen. The anaerobic atmosphere is maintained largely with a vacuum pump and nitrogen purges. The remaining oxygen is removed by a palladium catalyst and hydrogen. The oxygen reacts with hydrogen to form water, which is absorbed by desiccant.

Pressure

Most organisms spend their lives on land or on the surface of water, always subjected to a pressure of 1 atmosphere (atm), and are never affected significantly by pressure. Yet the deep sea (ocean of 1,000 m or more in depth) is 75% of the total ocean volume. The hydrostatic pressure can reach 600 to 1,100 atm in the deep sea, while the temperature is about 2 to 3°C. Despite these extremes, bacteria survive and adapt. Many are **barotolerant**: increased pressure does adversely affect them but not as much as it does nontolerant bacteria. Some bacteria in the gut of deep-sea invertebrates such as amphipods and holothurians are truly **barophilic**—they grow more rapidly at high pressures. These gut bacteria may play an important role in nutrient recycling in the deep sea. One barophile has been recovered from the Mariana trench near the Philippines (depth about 10,500 m) that is actually unable to grow at pressures below about 400 to 500 atm when incubated at 2°C. Thus far, barophiles have been found among several bacterial genera (e.g., *Photobacterium*, *Shewanella*, *Colwellia*). Some members of the Archaea are thermobarophiles (e.g., *Pyrococcus* spp., *Methanococcus jannaschii*). [The marine environment \(pp. 644–48\)](#)

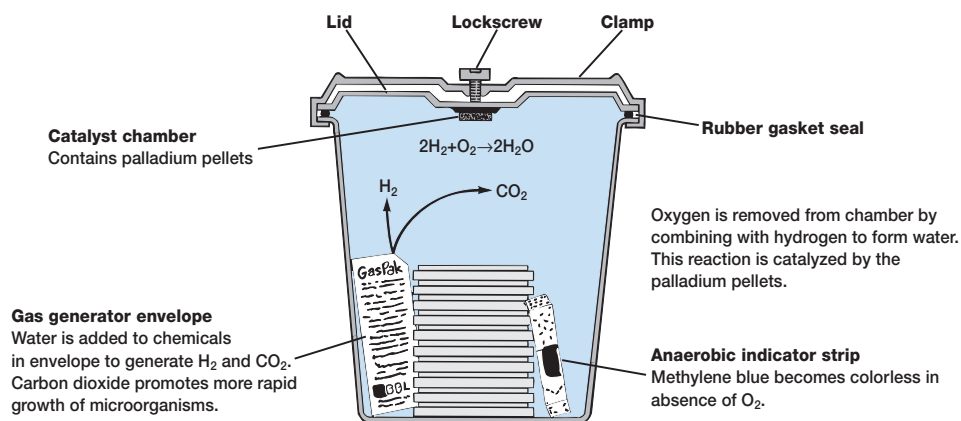


Figure 6.16 The GasPak Anaerobic System. Hydrogen and carbon dioxide are generated by a GasPak envelope. The palladium catalyst in the chamber lid catalyzes the formation of water from hydrogen and oxygen, thereby removing oxygen from the sealed chamber.

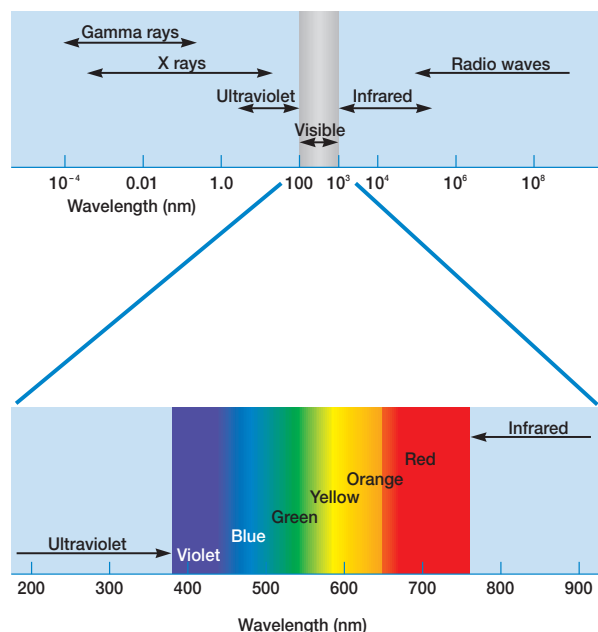


Figure 6.17 The Electromagnetic Spectrum. The visible portion of the spectrum is expanded at the bottom of the figure.

Radiation

Our world is bombarded with electromagnetic radiation of various types (**figure 6.17**). This radiation often behaves as if it were composed of waves moving through space like waves traveling on the surface of water. The distance between two wave crests or troughs is the wavelength. As the wavelength of electromagnetic radiation decreases, the energy of the radiation increases—gamma rays and X rays are much more energetic than visible light or infrared waves. Electromagnetic radiation also acts like a stream of energy packets called photons, each photon having a quantum of energy whose value will depend on the wavelength of the radiation.

Sunlight is the major source of radiation on the Earth. It includes visible light, ultraviolet (UV) radiation, infrared rays, and radio waves. Visible light is a most conspicuous and important aspect of our environment: all life is dependent on the ability of photosynthetic organisms to trap the light energy of the sun. Almost 60% of the sun's radiation is in the infrared region rather than the visible portion of the spectrum. Infrared is the major source of the Earth's heat. At sea level, one finds very little ultraviolet radiation below about 290 to 300 nm. UV radiation of wavelengths shorter than 287 nm is absorbed by O_2 in the Earth's atmosphere; this process forms a layer of ozone between 25 and 30 miles above the Earth's surface. The ozone layer then absorbs somewhat longer UV rays and reforms O_2 . This elimination of UV radiation is crucial because it is quite damaging to living systems (*see chapter 11*). The fairly even distribution of sunlight throughout the visible spectrum accounts for the fact that sunlight is generally "white." [Microbial photosynthesis \(pp. 195–201\)](#)

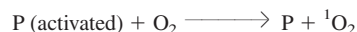
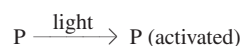
Many forms of electromagnetic radiation are very harmful to microorganisms. This is particularly true of **ionizing radiation**, radiation of very short wavelength or high energy, which can cause atoms to lose electrons or ionize. Two major forms of ionizing radiation are (1) X rays, which are artificially produced, and (2) gamma rays, which are emitted during radioisotope decay. Low levels of ionizing radiation will produce mutations and may indirectly result in death, whereas higher levels are directly lethal. Although microorganisms are more resistant to ionizing radiation than larger organisms, they will still be destroyed by a sufficiently large dose. Ionizing radiation can be used to sterilize items. Some procaryotes (e.g., *Deinococcus radiodurans*) and bacterial endospores can survive large doses of ionizing radiation. [Use of radiation in destroying microorganisms \(p. 144\); *Deinococcus* \(p. 468\)](#)

A variety of changes in cells are due to ionizing radiation; it breaks hydrogen bonds, oxidizes double bonds, destroys ring structures, and polymerizes some molecules. Oxygen enhances these destructive effects, probably through the generation of hydroxyl radicals ($OH\cdot$). Although many types of constituents can be affected, it is reasonable to suppose that destruction of DNA is the most important cause of death.

Ultraviolet (UV) radiation, mentioned earlier, kills all kinds of microorganisms due to its short wavelength (approximately from 10 to 400 nm) and high energy. The most lethal UV radiation has a wavelength of 260 nm, the wavelength most effectively absorbed by DNA. The primary mechanism of UV damage is the formation of thymine dimers in DNA (*see pp. 248–49*). Two adjacent thymines in a DNA strand are covalently joined to inhibit DNA replication and function. This damage is repaired in several ways. In **photoreactivation**, blue light is used by a photoreactivating enzyme to split thymine dimers. A short sequence containing the thymine dimer can also be excised and replaced. This process occurs in the absence of light and is called **dark reactivation**. Damage also can be repaired by the recA protein in recombination repair and SOS repair. When UV exposure is too heavy, the damage is so extensive that repair is impossible. [DNA repair mechanisms \(pp. 254–56\)](#)

Although very little UV radiation below 290 to 300 nm reaches the earth's surface, near-UV radiation between 325 and 400 nm can harm microorganisms. Exposure to near-UV radiation induces tryptophan breakdown to toxic photoproducts. It appears that these toxic tryptophan photoproducts plus the near-UV radiation itself produce breaks in DNA strands. The precise mechanism is not known, although it is different from that seen with 260 nm UV.

Visible light is immensely beneficial because it is the source of energy for photosynthesis. Yet even visible light, when present in sufficient intensity, can damage or kill microbial cells. Usually pigments called photosensitizers and O_2 are required. All microorganisms possess pigments like chlorophyll, bacteriochlorophyll, cytochromes, and flavins, which can absorb light energy, become excited or activated, and act as photosensitizers. The excited photosensitizer (P) transfers its energy to O_2 generating **singlet oxygen** (1O_2).



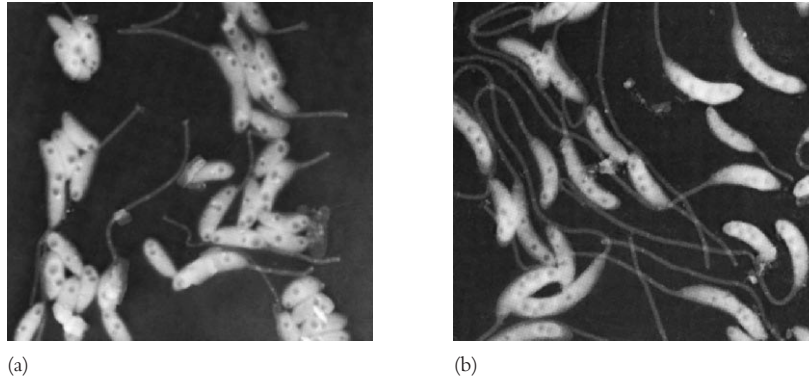


Figure 6.18 Morphology and Nutrient Absorption. Microorganisms can change their morphology in response to starvation and different limiting factors to improve their ability to survive. (a) *Caulobacter* has relatively short stalks when nitrogen is limiting. (b) The stalks are extremely long under phosphorus-limited conditions.

Singlet oxygen is a very reactive, powerful oxidizing agent that will quickly destroy a cell. It is probably the major agent employed by phagocytes to destroy engulfed bacteria (see section 31.8).

Many microorganisms that are airborne or live on exposed surfaces use carotenoid pigments for protection against photooxidation. Carotenoids effectively quench singlet oxygen—that is, they absorb energy from singlet oxygen and convert it back into the unexcited ground state. Both photosynthetic and nonphotosynthetic microorganisms employ pigments in this way.

1. What are barotolerant and barophilic bacteria? Where would you expect to find them?
2. List the types of electromagnetic radiation in the order of decreasing energy or increasing wavelength.
3. Why is it so important that the Earth receives an adequate supply of sunlight? What is the importance of ozone formation?
4. How do ionizing radiation, ultraviolet radiation, and visible light harm microorganisms? How do microorganisms protect themselves against damage from UV and visible light?

6.5 Microbial Growth in Natural Environments

The previous section surveyed the effects on microbial growth of individual environmental factors such as water availability, pH, and temperature. Although microbial ecology will be introduced in more detail at a later point, we will now briefly consider the effect of the environment as a whole on microbial growth. [Microbial ecology \(chapters 28–30\)](#)

Growth Limitation by Environmental Factors

The microbial environment is complex and constantly changing. Characteristically microorganisms in a particular location are exposed to many overlapping gradients of nutrients and various other environmental factors. This is particularly true of microorganisms growing in biofilms. Microorganisms will grow in “microenvironments” until an environmental or nutritional factor limits growth. **Liebig’s law of the minimum** states that the

total biomass of an organism will be determined by the nutrient present in the lowest concentration relative to the organism’s requirements. This law applies in both the laboratory (figure 6.2) and in terrestrial and aquatic environments. An increase in a limiting essential nutrient such as phosphate will result in an increase in the microbial population until some other nutrient becomes limiting. If a specific nutrient is limiting, changes in other nutrients will have no effect. The situation may be even more complex than this. Multiple limiting factors can influence a population over time. Furthermore, as we have seen, factors such as temperature, pH, light, and salinity influence microbial populations and limit growth. **Shelford’s law of tolerance** states that there are limits to environmental factors below and above which a microorganism cannot survive and grow, regardless of the nutrient supply. This can readily be seen for temperature in figure 6.13. Each microorganism has a specific temperature range in which it can grow. The same rule applies to other factors such as pH, oxygen level, and hydrostatic pressure in the marine environment. The growth of a microorganism depends on both the nutrient supply and its tolerance of the environmental conditions. [Biofilms \(pp. 620–22\)](#)

Most microorganisms are confronted with deficiencies that limit their activities except when excess nutrients allow unlimited growth. Such rapid growth will quickly deplete nutrients and possibly result in the release of toxic waste products, which will limit further growth.

In response to low nutrient levels (**oligotrophic environments**) and intense competition, many microorganisms become more competitive in nutrient capture and exploitation of available resources. Often the organism’s morphology will change in order to increase its surface area and ability to absorb nutrients. This can involve conversion of rod-shaped procaryotes to “mini” and “ultramicro” cells or changes in the morphology of prosthecate (see pp. 490–92) procaryotes (figure 6.18), in response to starvation. Nutrient deprivation induces many other changes as discussed previously. For example, microorganisms can undergo a step-by-step shutdown of metabolism except for housekeeping maintenance genes.

Many factors can alter nutrient levels in oligotrophic environments. Microorganisms may sequester critical limiting nutrients, such as iron, making them less available to competitors. The atmosphere can contribute essential nutrients and support

microbial growth. This is seen in the laboratory as well as natural environments. Airborne organic substances have been found to stimulate microbial growth in dilute media, and enrichment of growth media by airborne organic matter can allow significant populations of microorganisms to develop. Even distilled water, which already contains traces of organic matter, can absorb one-carbon compounds from the atmosphere and grow microorganisms. The presence of such airborne nutrients and microbial growth, if not detected, can affect experiments in biochemistry and molecular biology, as well as studies of microorganisms growing in oligotrophic environments.

Natural substances also can directly inhibit microbial growth and reproduction in low-nutrient environments. These agents include phenolics, tannins, ammonia, ethylene, and volatile sulfur compounds. This may be a means by which microorganisms avoid expending limited energy reserves until an adequate supply of nutrients becomes available. Such chemicals are also important in plant pathology and may aid in controlling soil-borne microbial diseases.

Counting Viable But Nonculturable Vegetative Procaryotes

In order to study the growth of natural procaryotic populations outside the laboratory, it is essential to determine the number of viable microorganisms present. For most of microbiology's history, a viable microorganism has been defined as one that is able to grow actively, resulting in the formation of a colony or visible turbidity in a liquid medium. John R. Postgate of the University of Sussex in England was one of the first to note that microorganisms stressed by survival in natural habitats—or in many selective laboratory media—were particularly sensitive to secondary stresses. Such stresses can produce viable microorganisms without the ability to grow on media normally used for their cultivation. To determine the growth potential of such microorganisms, Postgate developed what is now called the Postgate Microviability Assay, in which microorganisms are cultured in a thin agar film under a coverslip. The ability of a cell to change its morphology, even if it does not grow beyond the single-cell stage, indicates that the microorganism does show “life signs.”

Since that time many workers have developed additional sensitive microscopic and isotopic procedures to evaluate the presence and significance of these viable but nonculturable bacteria in both lab and field. For example, levels of fluorescent antibody and acridine orange-stained cells often are compared with population counts obtained by the most probable number (MPN) method (see pp. 654–55) and plate counts using selective and nonselective media. The release of radioactive-labeled cell materials also is used to monitor stress effects on microorganisms. Despite these advances the estimation of substrate-responsive viable cells by Postgate's method is still important. These studies show that even when bacteria such as *Escherichia coli*, *Vibrio cholerae*, *Klebsiella pneumoniae*, *Enterobacter aerogenes*, and *Enterococcus faecalis* have lost their ability to grow on conventional laboratory media using standard cultural techniques, they still might be able to play a role in infectious disease.

The situation in natural environments with mixed populations is much more complex. Here, where often only 1 to 10% of observable cells are able to form colonies, the microbiologist is attempting to grow microorganisms that perhaps never have been cultured or characterized. In the future it is possible that media or proper environmental conditions for their growth will be developed. At present, molecular techniques involving PCR amplification and small subunit ribosomal RNA analysis are increasingly used to analyze the diversity of uncultured microbial populations (see pp. 626–29).

Quorum Sensing and Microbial Populations

For decades microbiologists tended to think of bacterial populations as collections of individuals growing and behaving independently. More recently it has become clear that many bacteria can communicate with one another and behave cooperatively. A major way in which this cooperation is accomplished is by a process known as **quorum sensing** or autoinduction. This is a phenomenon in which bacteria monitor their own population density through sensing the levels of signal molecules, sometimes called autoinducers because they can stimulate the cell that releases them. The concentration of these signal molecules increases along with the bacterial population until it rises to a specific threshold and signals the bacteria that the population density has reached a critical level or quorum. The bacteria then begin expressing sets of quorum-dependent genes. Quorum sensing has been found among both gram-negative and gram-positive bacteria.

Quorum sensing makes good practical sense. Take the production and release of extracellular enzymes as an example. If such enzymes were released by only a few bacteria, they would diffuse away and be rendered ineffective because of dilution. With control by quorum sensing, the bacteria reach a high population density before they release enzymes, and as a consequence enzyme levels are concentrated enough to have significant effects. This is an advantage within a host's body as well as in the soil or an aquatic habitat. If a pathogen can reach high levels at a particular site before producing virulence factors and escaping into surrounding host tissues, it has a much better chance of counteracting host defenses and successfully spreading throughout the host's body. This explains another pattern in quorum sensing. It seems to be very important in many bacteria that establish symbiotic or parasitic relationships with hosts.

Quorum sensing was first discovered in gram-negative bacteria and is best understood in these microorganisms. The most common signals in gram-negative bacteria are acyl homoserine lactones (HSLs). These are small molecules composed of a 4- to 14-carbon acyl chain attached by an amide bond to homoserine lactone (figure 6.19a). The acyl chain may have a keto group or hydroxyl group on its third carbon. Acyl HSLs diffuse into the target cell (figure 6.19b). Once they reach a sufficiently high level, acyl HSLs bind to special receptor proteins and trigger a conformational change. Usually the activated complexes act as inducers—that is, they bind to target sites on the DNA and stimulate transcription of quorum-sensitive genes. The gene needed to synthesize acyl HSL is also produced frequently, thus amplifying

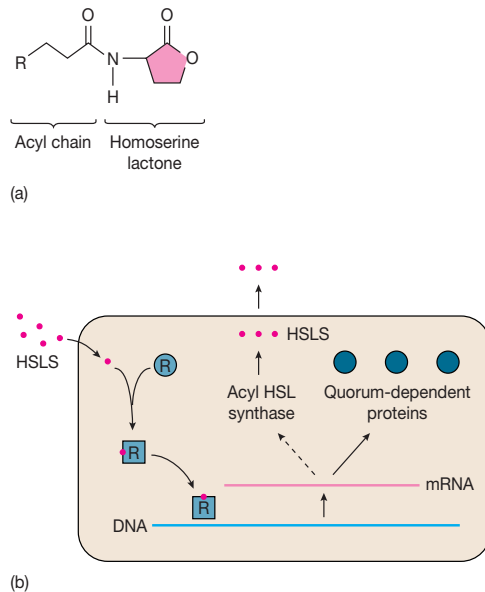


Figure 6.19 Quorum Sensing in Gram-Negative Bacteria.

(a) A generalized structure for acyl homoserine lactone, the best-known quorum sensing signal or autoinducer. (b) A schematic diagram giving an overview of the way in which quorum sensing functions in many gram-negative bacteria. The receptor protein that acts as an inducer is labeled R. The dashed lines indicate that acyl HSL synthase is not always made in response to the autoinducer. See text for more details.

the effect by the production and release of more autoinducer molecules. [Induction and repression of genetic activity \(pp. 275–78\)](#)

Many different processes are sensitive to acyl HSL signals and quorum sensing in gram-negative bacteria. Some well-studied examples are (1) bioluminescence production by *Vibrio fischeri*, (2) *Pseudomonas aeruginosa* synthesis and release of virulence factors, (3) conjugal transfer of genetic material by

Agrobacterium tumefaciens, and (4) antibiotic production by *Erwinia carotovora* and *Pseudomonas aureofaciens*.

Gram-positive bacteria also regulate activities by quorum sensing, often using an oligopeptide signal. Good examples are mating in *Enterococcus faecalis*, competence induction in *Streptococcus pneumoniae*, stimulation of sporulation by *Bacillus subtilis*, and production of many toxins and other virulence factors by *Staphylococcus aureus*. Quorum sensing even stimulates the development of aerial mycelia and the production of streptomycin by *Streptomyces griseus*. In this case, the signal seems to be γ -butyrolactone rather than an oligopeptide.

An interesting and important function of quorum sensing is to promote the formation of mature biofilms by the pathogen *Pseudomonas aeruginosa*, and it may play a role in cystic fibrosis. Biofilm formation makes sense for the pathogen because biofilms protect against antibiotics and detergents. Quorum sensing should be very effective within biofilms because there will be less dilution and acyl HSL levels will increase rapidly. Under such circumstances, two different bacteria might stimulate each other by releasing similar signals; this appears to be the case in biofilms containing the pathogens *P. aeruginosa* and *Burkholderia cepacia*.

Quorum sensing is an example of what might be called multicellular behavior in that many individual cells communicate and coordinate their activities to act as a unit. Other examples of such complex behavior is pattern formation in colonies (*see pp. 108–10*) and fruiting body formation in the myxobacteria (*see pp. 512–13*).

1. How are Liebig's law of the minimum and Shelford's law of tolerance related?
2. Describe how microorganisms respond to oligotrophic environments.
3. Briefly discuss the Postgate microviability assay and other ways in which viable but nonculturable microorganisms can be counted or studied.
4. What is quorum sensing? Describe how it occurs and briefly discuss its importance to microorganisms.

Summary

1. Growth is an increase in cellular constituents and results in an increase in cell size, cell number, or both.
2. When microorganisms are grown in a closed system or batch culture, the resulting growth curve usually has four phases: the lag, exponential or log, stationary, and death phases (**figure 6.1**).
3. In the exponential phase, the population number doubles at a constant interval called the doubling or generation time (**figure 6.3**). The mean growth rate constant (k) is the reciprocal of the generation time.
4. Exponential growth is balanced growth, cell components are synthesized at constant rates relative to one another. Changes in culture conditions (e.g., in shift-up and shift-down experiments) lead to unbalanced growth. A portion of the available nutrients is used to supply maintenance energy.
5. Microbial populations can be counted directly with counting chambers, electronic counters, or fluorescence microscopy. Viable counting techniques such as the spread plate, the pour plate, or the membrane filter can be employed.
6. Population changes also can be followed by determining variations in microbial mass through the measurement of dry weight, turbidity, or the amount of a cell component.
7. Microorganisms can be grown in an open system in which nutrients are constantly provided and wastes removed.
8. A continuous culture system is an open system that can maintain a microbial population in the log phase. There are two types of these systems: chemostats and turbidostats.
9. Most bacteria, algae, and fungi have rigid cell walls and are hypertonic to the habitat because of solutes such as amino acids, polyols, and potassium ions. The amount of water actually available to microorganisms is expressed in terms of the water activity (a_w).
10. Although most microorganisms will not grow well at water activities below 0.98 due to plasmolysis and associated effects, osmotolerant organisms survive and even flourish at low a_w values. Halophiles actually require high sodium chloride concentrations for growth (**table 6.3**).

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11. Each species of microorganism has an optimum pH for growth and can be classified as an acidophile, neutrophile, or alkalophile.
12. Microorganisms can alter the pH of their surroundings, and most culture media must be buffered to stabilize the pH.
13. Microorganisms have distinct temperature ranges for growth with minima, maxima, and optima—the cardinal temperatures. These ranges are determined by the effects of temperature on the rates of catalysis, protein denaturation, and membrane disruption.
14. There are five major classes of microorganisms with respect to temperature preferences: (1) psychrophiles, (2) facultative psychrophiles or psychrotrophs, (3) mesophiles, (4) thermophiles and (5) hyperthermophiles (figure 6.13 and table 6.3).
15. Microorganisms can be placed into at least five different categories based on their response to the presence of O₂: obligate aerobes, facultative anaerobes, aerotolerant anaerobes, strict or obligate anaerobes, and microaerophiles (figure 6.14 and table 6.3).
16. Oxygen can become toxic because of the production of hydrogen peroxide, superoxide radical, and hydroxyl radical. These are destroyed by the enzymes superoxide dismutase, catalase, and peroxidase.
17. Most deep-sea microorganisms are barotolerant, but some are barophilic and require high pressure for optimal growth.
18. High-energy or short-wavelength radiation harms organisms in several ways. Ionizing radiation—X rays and gamma rays—ionizes molecules and destroys DNA and other cell components.
19. Ultraviolet (UV) radiation induces the formation of thymine dimers and strand breaks in DNA. Such damage can be repaired by photoreactivation or dark reactivation mechanisms.
20. Visible light can provide energy for the formation of reactive singlet oxygen, which will destroy cells.
21. Microbial growth in natural environments is profoundly affected by nutrient limitations and other adverse factors. Some microorganisms can be viable but unculturable and must be studied with special techniques.
22. Often, bacteria will communicate with one another in a density-dependent way and carry out a particular activity only when a certain population density is reached. This phenomenon is called quorum sensing.

Key Terms

acidophile 123	extremophiles 121	obligate aerobe 127
aerobe 127	facultative anaerobe 127	obligate anaerobe 127
aerotolerant anaerobe 127	facultative psychrophiles 126	oligotrophic environment 131
alkalophile 123	generation time 115	osmotolerant 122
anaerobe 127	growth 113	photoreactivation 130
balanced growth 114	halophile 123	psychrophile 126
barophilic 129	hydrogen peroxide 128	psychrotroph 126
barotolerant 129	hydroxyl radical 128	quorum sensing 132
batch culture 113	hyperthermophile 126	Shelford's law of tolerance 131
cardinal temperatures 125	ionizing radiation 130	singlet oxygen 130
catalase 128	lag phase 113	starvation proteins 115
chemostat 120	Liebig's law of the minimum 131	stationary phase 114
coenocytic 113	log phase 114	strict anaerobe 127
colony forming units (CFU) 118	maintenance energy 121	superoxide dismutase (SOD) 128
compatible solutes 121	mean generation time 116	superoxide radical 128
continuous culture system 120	mean growth rate constant (<i>k</i>) 116	thermophile 126
dark reactivation 130	membrane filter 118	turbidostat 121
death phase 115	mesophile 126	ultraviolet (UV) radiation 130
doubling time 115	microaerophile 127	unbalanced growth 114
exponential phase 114	neutrophile 123	water activity (<i>a_w</i>) 122

Questions for Thought and Review

1. Discuss the reasons why a culture might have a long lag phase after inoculation.
2. Why can't one always tell when a culture enters the death phase by the use of total cell counts?
3. Calculate the mean growth rate and generation time of a culture that increases in the exponential phase from 5×10^2 to 1×10^8 cells in 12 hours.
4. If the generation time is 90 minutes and the initial population contains 10^3 cells, how many bacteria will there be after 8 hours of exponential growth?
5. Why are continuous culture systems so useful to microbiologists?
6. How do bacterial populations respond in shift-up and shift-down experiments? Account for their behavior in molecular terms.
7. Does the internal pH remain constant despite changes in the external pH? How might this be achieved? Explain how extreme pH values might harm microorganisms.
8. What metabolic and structural adaptations for extreme temperatures have psychrophiles and thermophiles made?
9. Why are generation times in nature usually much longer than in culture?

Critical Thinking Questions

1. As an alternative to diffusable signals, suggest another mechanism by which bacteria can quorum sense.
2. Design an "enrichment" culture medium and a protocol for the isolation and purification of a soil bacterium (e.g., *Bacillus subtilis*) from a sample of soil. Note possible contaminants and competitors. How will you adjust conditions of growth and what conditions will be adjusted to differentially enhance the growth of the *Bacillus*?

Additional Reading

General

- Atlas, R. M., and Bartha, R. 1997. *Microbial ecology: Fundamentals and applications*, 4th ed. Menlo Park, Calif.: Benjamin/Cummings.
- Caldwell, D. R. 2000. *Microbial physiology and metabolism*. Belmont, Calif.: Star Publishing.
- Cavicchioli, R., and Thomas, T. 2000. Extremophiles. In *Encyclopedia of microbiology*, 2d ed., vol. 2, J. Lederberg, editor-in-chief, 317–37. San Diego: Academic Press.
- Gerhardt, P.; Murray, R. G. E.; Wood, W. A.; and Krieg, N. R., editors. 1994. *Methods for general and molecular bacteriology*, chaps. 6–12. Washington, D.C.: American Society for Microbiology.
- Koch, A. L. 1995. *Bacterial growth and form*. New York: Chapman & Hall.
- Kushner, D. J., editor. 1978. *Microbial life in extreme environments*. New York: Academic Press.
- Madigan, M. T., and Marrs, B. L. 1997. Extremophiles. *Sci. Am.* 276(4):82–87.
- Moat, A. G., and Foster, J. W. 1995. *Microbial physiology*, 3d ed. New York: John Wiley and Sons.
- Neidhardt, F. C.; Ingraham, J. L.; and Schaechter, M. 1990. *Physiology of the bacterial cell: A molecular approach*. Sunderland, Mass.: Sinauer Associates.
- Postgate, J. 1994. *The outer reaches of life*. New York: Cambridge University Press.
- Schlegel, H. G., and Jannasch, H. W. 1992. Prokaryotes and their habitats. In *The prokaryotes*, 2d ed. A. Balows et al., editors, 75–125. New York: Springer-Verlag.

6.1 The Growth Curve

- Kolter, R.; Siegle, D. A.; and Tormo, A. 1993. The stationary phase of the bacterial life cycle. *Annu. Rev. Microbiol.* 47:855–74.
- Lazazzera, B. A. 2000. Quorum sensing and starvation: Signals for entry into stationary phase. *Curr. Opin. Microbiol.* 3:177–82.
- Marr, A. G. 2000. Growth kinetics, bacterial. In *Encyclopedia of microbiology*, 2d ed., vol. 2, J. Lederberg, editor-in-chief, 584–89. San Diego: Academic Press.
- Martin, A. 2000. Starvation, bacterial. In *Encyclopedia of microbiology*, 2d ed., vol. 4, J. Lederberg, editor-in-chief, 394–403. San Diego: Academic Press.
- Prosser, J. I., and Tough, A. J. 1991. Growth mechanisms and growth kinetics of filamentous microorganisms. *Critical reviews in biotechnology* 10(4):253–74.
- Russell, J. B., and Cook, G. M. 1995. Energetics of bacterial growth: Balance of anabolic and catabolic reactions. *Microbiol. Rev.* 59(1):48–62.
- Stouthamer, A. H. 1977. Energetic aspects of the growth of micro-organisms. In *Microbial energetics*, B. A. Haddock and W. A.

Hamilton, editors, 285–315. New York: Cambridge University Press.

- Tempest, D. W. 1978. Dynamics of microbial growth. In *Essays in microbiology*, J. R. Norris and M. H. Richmond, editors, 7/1–7/32. New York: John Wiley and Sons.
- Zambrano, M. M., and Kolter, R. 1996. GASping for life in stationary phase. *Cell* 86:181–84.

6.4 The Influence of Environmental Factors on Growth

- Abe, F.; Kato, C.; and Horikoshi, K. 1999. Pressure-regulated metabolism in microorganisms. *Trends Microbiol.* 7(11):447–53.
- Adams, M. W. W. 1993. Enzymes and proteins from organisms that grow near and above 100°C. *Annu. Rev. Microbiol.* 47:627–58.
- Blomberg, A., and Adler, L. 1992. Physiology of osmotolerance in fungi. In *Advances in microbial physiology*, vol. 33, A. H. Rose., editor, 145–212. San Diego: Academic Press.
- Blount, P., and Moe, P. C. 1999. Bacterial mechanosensitive channels: integrating physiology, structure and function. *Trends Microbiol.* 7(10):420–24.
- Booth, I. R. 1985. Regulation of cytoplasmic pH in bacteria. *Microbiol. Rev.* 49(4):359–78.
- Brown, A. D. 1976. Microbial water stress. *Bacteriol. Rev.* 40(4):803–46.
- Csonka, L. N. 1989. Physiological and genetic responses of bacteria to osmotic stress. *Microbiol. Rev.* 53(1):121–47.
- Fridovich, I. 1977. Oxygen is toxic! *BioScience* 27(7):462–66.
- Friedman, S. M. 1992. Thermophilic microorganisms. In *Encyclopedia of microbiology*, 1st ed., vol. 4, J. Lederberg, editor-in-chief, 217–29. San Diego: Academic Press.
- Gaill, F. 1993. Aspects of life development at deep sea hydrothermal vents. *FASEB J.* 7:558–65.
- Gillis, A. M. 1994. A pressure-filled life. *BioScience* 44(9):584–86.
- Gottschal, J. C., and Prins, R. A. 1991. Thermophiles: A life at elevated temperatures. *Trends Ecol. & Evol.* 6(5):157–62.
- Inlag, J. A., and Linn, S. 1988. DNA damage and oxygen radical toxicity. *Science* 240:1302–9.
- Jannasch, H. W., and Taylor, C. D. 1984. Deep-sea microbiology. *Annu. Rev. Microbiol.* 38:487–514.
- Jannasch, H. W., and Wirsén, C. O. 1977. Microbial life in the deep sea. *Sci. Am.* 236(6):42–52.
- Kelly, R. M., and Adams, M. W. W. 1994. Metabolism in hyperthermophilic microorganisms. *Antonie van Leeuwenhoek* 66:247–70.
- Krieg, N. R., and Hoffman, P. S. 1986. Microaerophily and oxygen toxicity. *Annu. Rev. Microbiol.* 40:107–30.

Krulwich, T. A., and Guffanti, A. A. 1989.

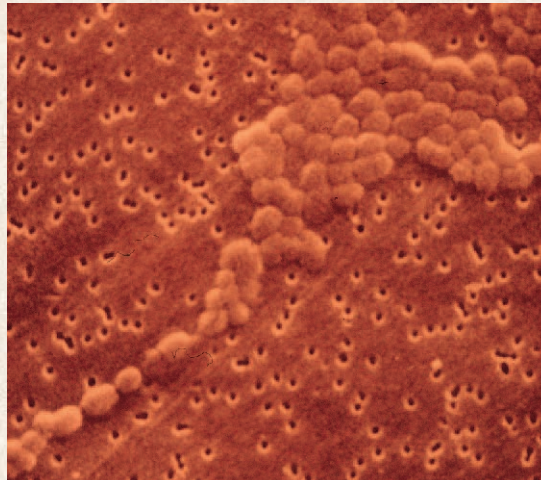
- Alkalophilic bacteria. *Annu. Rev. Microbiol.* 43:435–63.
- Le Rudulier, D.; Strom, A. R.; Dandekar, A. M.; Smith, L. T.; and Valentine, R. C. 1984. Molecular biology of osmoregulation. *Science* 224:1064–68.
- Morita, R. Y. 2000. Low-temperature environments. In *Encyclopedia of microbiology*, 2d ed., vol. 3, J. Lederberg, editor-in-chief, 93–98. San Diego: Academic Press.
- Potts, M. 1994. Desiccation tolerance of prokaryotes. *Microbiol. Rev.* 58(4):755–805.
- Stetter, K. O. 1995. Microbial life in hyperthermal environments. *ASM News* 61(6):285–90.
- Yancey, P. H.; Clark, M. E.; Hand, S. C.; Bowlus, R. D.; and Somero, G. N. 1982. Living with water stress: Evolution of osmolyte systems. *Science* 217:1214–22.

6.5 Microbial Growth in Natural Environments

- Dunny, G. M., and Leonard, B. A. B. 1997. Cell-cell communication in gram-positive bacteria. *Annu. Rev. Microbiol.* 51:527–64.
- Dunny, G. M., and Winans, S. C., editors. 1999. *Cell-cell signaling in bacteria*. Washington, D.C.: ASM Press.
- Fuqua, C. 2000. Quorum sensing in gram-negative bacteria. In *Encyclopedia of microbiology*, 2d ed., vol. 4, J. Lederberg, editor-in-chief, 1–13. San Diego: Academic Press.
- Gray, K. M. 1997. Intercellular communication and group behavior in bacteria. *Trends Microbiol.* 5(5):184–88.
- Greenberg, E. P. 1999. Quorum sensing in gram-negative bacteria: An important signaling mechanism in symbiosis and disease. In *Microbial ecology and infectious disease*, E. Rosenberg, editor, 112–22. Washington, D.C.: ASM Press.
- Lazazzera, B. A., and Grossman, A. D. 1998. The ins and outs of peptide signaling. *Trends Microbiol.* 6(7):288–94.
- Losick, R., and Kaiser, D. 1997. Why and how bacteria communicate. *Sci. Am.* 276(2):68–73.
- Morita, R. Y. 1997. *Bacteria in oligotrophic environments: Starvation-survival life style*. New York: Chapman & Hall.
- Shapiro, J. A. 1988. Bacteria as multicellular organisms. *Sci. Am.* 258(6):82–89.
- Shapiro, J. A. 1998. Thinking about bacterial populations as multicellular organisms. *Annu. Rev. Microbiol.* 52:81–104.
- Watnick, P., and Kolter, R. 2000. Biofilm, city of microbes. *J. Bacteriol.* 182(10):2675–79.
- Wirth, R.; Muscholl, A.; and Wanner, G. 1996. The role of pheromones in bacterial interactions. *Trends Microbiol.* 4(3):96–103.

CHAPTER 7

Control of Microorganisms by Physical and Chemical Agents



Bacteria are trapped on the surface of a membrane filter used to remove microorganisms from fluids.

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Concepts

- 1. Microbial population death is exponential, and the effectiveness of an agent is not fixed but influenced by many environmental factors.
- 2. Solid objects can be sterilized by physical agents such as heat and radiation; liquids and gases are sterilized by heat, radiation, and filtration through the proper filter.
- 3. Most chemical agents do not readily destroy bacterial endospores and therefore cannot sterilize objects; they are used as disinfectants, sanitizers, and antiseptics. Objects can be sterilized by gases like ethylene oxide that destroy endospores.
- 4. A knowledge of methods used for microbial control is essential for personal and public safety.

We all labour against our own cure, for death is the cure of all diseases.

—Sir Thomas Browne

The chapters in Part II are concerned with the nutrition, growth, and control of microorganisms. This chapter addresses the subject of the nonspecific control and destruction of microorganisms, a topic of immense practical importance. Although many microorganisms are beneficial and necessary for human well-being, microbial activities may have undesirable consequences, such as food spoilage and disease. Therefore it is essential to be able to kill a wide variety of microorganisms or inhibit their growth to minimize their destructive effects. The goal is twofold: (1) to destroy pathogens and prevent their transmission, and (2) to reduce or eliminate microorganisms responsible for the contamination of water, food, and other substances.

This chapter focuses on the control of microorganisms by non-specific physical and chemical agents. Chapter 35 introduces the use of antimicrobial chemotherapy to control microbial disease.

From the beginning of recorded history, people have practiced disinfection and sterilization, even though the existence of microorganisms was long unsuspected. The Egyptians used fire to sterilize infectious material and disinfectants to embalm bodies, and the Greeks burned sulfur to fumigate buildings. Mosaic law commanded the Hebrews to burn any clothing suspected of being contaminated with the leprosy bacterium. Today the ability to de-

stroy microorganisms is no less important: it makes possible the aseptic techniques used in microbiological research, the preservation of food, and the prevention of disease. The techniques described in this chapter are also essential to personal safety in both the laboratory and hospital (**Box 7.1**).

There are several ways to control microbial growth that have not been included in this chapter, but they should be considered for a more complete picture of how microorganisms are controlled. Chapter 6 describes the effects of osmotic activity, pH, temperature, O₂, and radiation on microbial growth and survival (*see pp. 121–31*). Chapter 41 discusses the use of physical and chemical agents in food preservation (*see pp. 970–73*).

7.1 Definition of Frequently Used Terms

Terminology is especially important when the control of microorganisms is discussed because words like disinfectant and antiseptic often are used loosely. The situation is even more confusing because a particular treatment can either inhibit growth or kill depending on the conditions.

The ability to control microbial populations on inanimate objects, like eating utensils and surgical instruments, is of considerable practical importance. Sometimes it is necessary to eliminate all microorganisms from an object, whereas only partial destruction of the microbial population may be required in other situations. **Sterilization** [Latin *sterilis*, unable to produce offspring or barren] is the process by which all living cells, viable spores, viruses, and viroids (*see chapter 18*) are either destroyed or removed from an object or habitat. A sterile object is totally free of viable microorganisms, spores, and other infectious agents. When sterilization is achieved by a chemical agent, the chemical is called a sterilant. In

Box 7.1

Safety in the Microbiology Laboratory

Personnel safety should be of major concern in all microbiology laboratories. It has been estimated that thousands of infections have been acquired in the laboratory, and many persons have died because of such infections. The two most common laboratory-acquired bacterial diseases are typhoid fever and brucellosis. Most deaths have come from typhoid fever (20 deaths) and Rocky Mountain spotted fever (13 deaths). Infections by fungi (histoplasmosis) and viruses (Venezuelan equine encephalitis and hepatitis B virus from monkeys) are also not uncommon. Hepatitis is the most frequently reported laboratory-acquired viral infection, especially in people working in clinical laboratories and with blood. In a survey of 426 U.S. hospital workers, 40% of those in clinical chemistry and 21% in microbiology had antibodies to hepatitis B virus, indicating their previous exposure (though only about 19% of these had disease symptoms).

Efforts have been made to determine the causes of these infections in order to enhance the development of better preventive measures. Although often it is not possible to determine the direct cause of infection,

some major potential hazards are clear. One of the most frequent causes of disease is the inhalation of an infectious aerosol. An aerosol is a gaseous suspension of liquid or solid particles that may be generated by accidents and laboratory operations such as spills, centrifuge accidents, removal of closures from shaken culture tubes, and plunging of contaminated loops into a flame. Accidents with hypodermic syringes and needles, such as self-inoculation and spraying solutions from the needle, also are common. Hypodermics should be employed only when necessary and then with care. Pipette accidents involving the mouth are another major source of infection; pipettes should be filled with the use of pipette aids and operated in such a way as to avoid creating aerosols.

People must exercise care and common sense when working with microorganisms. Operations that might generate infectious aerosols should be carried out in a biological safety cabinet. Bench tops and incubators should be disinfected regularly. Autoclaves must be maintained and operated properly to ensure adequate sterilization. Laboratory personnel should wash their hands thoroughly before and after finishing work.

contrast, **disinfection** is the killing, inhibition, or removal of microorganisms that may cause disease. The primary goal is to destroy potential pathogens, but disinfection also substantially reduces the total microbial population. **Disinfectants** are agents, usually chemical, used to carry out disinfection and are normally used only on inanimate objects. A disinfectant does not necessarily sterilize an object because viable spores and a few microorganisms may remain. **Sanitization** is closely related to disinfection. In sanitization, the microbial population is reduced to levels that are considered safe by public health standards. The inanimate object is usually cleaned as well as partially disinfected. For example, sanitizers are used to clean eating utensils in restaurants.

It is frequently necessary to control microorganisms on living tissue with chemical agents. **Antisepsis** [Greek *anti*, against, and *sepsis*, putrefaction] is the prevention of infection or sepsis and is accomplished with **antiseptics**. These are chemical agents applied to tissue to prevent infection by killing or inhibiting pathogen growth; they also reduce the total microbial population. Because they must not destroy too much host tissue, antiseptics are generally not as toxic as disinfectants.

A suffix can be employed to denote the type of antimicrobial agent. Substances that kill organisms often have the suffix *-cide* [Latin *cida*, to kill]: a **germicide** kills pathogens (and many nonpathogens) but not necessarily endospores. A disinfectant or antiseptic can be particularly effective against a specific group, in which case it may be called a **bactericide**, **fungicide**, **algicide**, or **viricide**. Other chemicals do not kill, but they do prevent growth. If these agents are removed, growth will resume. Their names end in *-static* [Greek *statikos*, causing to stand or stopping]—for example, **bacteriostatic** and **fungistatic**.

Although these agents have been described in terms of their effects on pathogens, it should be noted that they also kill or inhibit the growth of nonpathogens as well. Their ability to reduce the total microbial population, not just to affect pathogen levels, is quite important in many situations.

1. Define the following terms: sterilization, sterilant, disinfection, disinfectant, sanitization, antisepsis, antiseptic, germicide, bactericide, bacteriostatic.

7.2 The Pattern of Microbial Death

A microbial population is not killed instantly when exposed to a lethal agent. Population death, like population growth, is generally exponential or logarithmic—that is, the population will be reduced by the same fraction at constant intervals (**table 7.1**). If the logarithm of the population number remaining is plotted against the time of exposure of the microorganism to the agent, a straight line plot will result (compare **figure 7.1** with figure 6.2). When the population has been greatly reduced, the rate of killing may slow due to the survival of a more resistant strain of the microorganism.

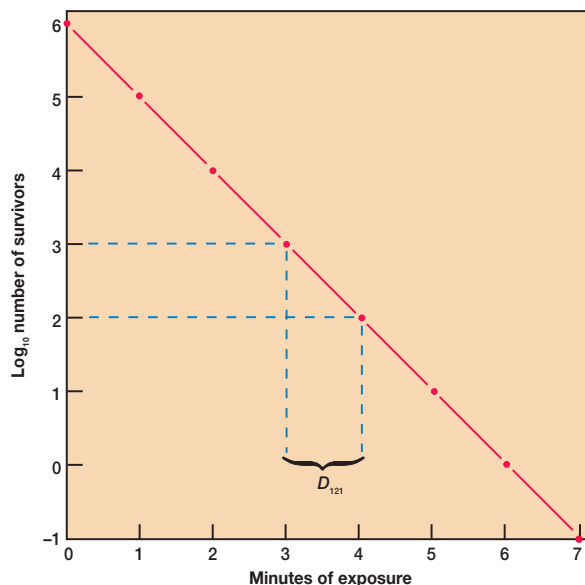


Figure 7.1 The Pattern of Microbial Death. An exponential plot of the survivors versus the minutes of exposure to heating at 121°C. In this example the D_{121} value is 1 minute. The data are from table 7.1.

Table 7.1 A Theoretical Microbial Heat-Killing Experiment

Minute	Microbial Number at Start of Minute ^a	Microorganisms Killed in 1 Minute (90% of total) ^a	Microorganisms at End of 1 Minute	Log ₁₀ of Survivors
1	10 ⁶	9 × 10 ⁵	10 ⁵	5
2	10 ⁵	9 × 10 ⁴	10 ⁴	4
3	10 ⁴	9 × 10 ³	10 ³	3
4	10 ³	9 × 10 ²	10 ²	2
5	10 ²	9 × 10 ¹	10	1
6	10 ¹	9	1	0
7	1	0.9	0.1	-1

^aAssume that the initial sample contains 10⁶ vegetative microorganisms per ml and that 90% of the organisms are killed during each minute of exposure. The temperature is 121°C.

To study the effectiveness of a lethal agent, one must be able to decide when microorganisms are dead, a task by no means as easy as with macroorganisms. It is hardly possible to take a bacterium's pulse. A bacterium is defined as dead if it does not grow and reproduce when inoculated into culture medium that would normally support its growth. In like manner an inactive virus cannot infect a suitable host.

-
1. Describe the pattern of microbial death and how one decides whether microorganisms are actually dead.
-

7.3 Conditions Influencing the Effectiveness of Antimicrobial Agent Activity

Destruction of microorganisms and inhibition of microbial growth are not simple matters because the efficiency of an **antimicrobial agent** (an agent that kills microorganisms or inhibits their growth) is affected by at least six factors.

1. Population size. Because an equal fraction of a microbial population is killed during each interval, a larger population requires a longer time to die than a smaller one. This can be seen in the theoretical heat-killing experiment shown in table 7.1 and figure 7.1. The same principle applies to chemical antimicrobial agents.
2. Population composition. The effectiveness of an agent varies greatly with the nature of the organisms being treated because microorganisms differ markedly in susceptibility. Bacterial endospores are much more resistant to most antimicrobial agents than are vegetative forms, and younger cells are usually more readily destroyed than mature organisms. Some species are able to withstand adverse conditions better than others. *Mycobacterium tuberculosis*, which causes tuberculosis, is much more resistant to antimicrobial agents than most other bacteria.
3. Concentration or intensity of an antimicrobial agent. Often, but not always, the more concentrated a chemical agent or intense a physical agent, the more rapidly microorganisms are destroyed. However, agent effectiveness usually is not directly related to concentration or intensity. Over a short range a small increase in concentration leads to an exponential rise in effectiveness; beyond a certain point, increases may not raise the killing rate much at all. Sometimes an agent is more effective at lower concentrations. For example, 70% ethanol is more effective than 95% ethanol because its activity is enhanced by the presence of water.
4. Duration of exposure. The longer a population is exposed to a microbicidal agent, the more organisms are killed (figure 7.1). To achieve sterilization, an exposure duration sufficient to reduce the probability of survival to 10^{-6} or less should be used.

5. Temperature. An increase in the temperature at which a chemical acts often enhances its activity. Frequently a lower concentration of disinfectant or sterilizing agent can be used at a higher temperature.
6. Local environment. The population to be controlled is not isolated but surrounded by environmental factors that may either offer protection or aid in its destruction. For example, because heat kills more readily at an acid pH, acid foods and beverages such as fruits and tomatoes are easier to pasteurize than foods with higher pHs like milk. A second important environmental factor is organic matter that can protect microorganisms against heating and chemical disinfectants. Biofilms are a good example. The organic matter in a surface biofilm will protect the biofilm's microorganisms; furthermore, the biofilm and its microbes often will be hard to remove. It may be necessary to clean an object before it is disinfected or sterilized. Syringes and medical or dental equipment should be cleaned before sterilization because the presence of too much organic matter could protect pathogens and increase the risk of infection. The same care must be taken when pathogens are destroyed during the preparation of drinking water. When a city's water supply has a high content of organic material, more chlorine must be added to disinfect it.

-
1. Briefly explain how the effectiveness of antimicrobial agents varies with population size, population composition, concentration or intensity of the agent, treatment duration, temperature, and local environmental conditions.
-

7.4 The Use of Physical Methods in Control

Heat and other physical agents are normally used to control microbial growth and sterilize objects, as can be seen from the continual operation of the autoclave in every microbiology laboratory. The four most frequently employed physical agents are heat, low temperatures, filtration, and radiation.

Heat

Fire and boiling water have been used for sterilization and disinfection since the time of the Greeks, and heating is still one of the most popular ways to destroy microorganisms. Either moist or dry heat may be applied.

Moist heat readily kills viruses, bacteria, and fungi (**table 7.2**). Exposure to boiling water for 10 minutes is sufficient to destroy vegetative cells and eucaryotic spores. Unfortunately the temperature of boiling water (100°C or 212°F) is not high enough to destroy bacterial endospores that may survive hours of boiling. Therefore boiling can be used for disinfection of drinking water and objects not harmed by water, but boiling does not sterilize.

Because heat is so useful in controlling microorganisms, it is essential to have a precise measure of the heat-killing efficiency.

Table 7.2 Approximate Conditions for Moist Heat Killing

Organism	Vegetative Cells	Spores
Yeasts	5 minutes at 50–60°C	5 minutes at 70–80°C
Molds	30 minutes at 62°C	30 minutes at 80°C
Bacteria ^a	10 minutes at 60–70°C	2 to over 800 minutes at 100°C
		0.5–12 minutes at 121°C
Viruses	30 minutes at 60°C	

^aConditions for mesophilic bacteria.

Initially effectiveness was expressed in terms of thermal death point (TDP), the lowest temperature at which a microbial suspension is killed in 10 minutes. Because TDP implies that a certain temperature is immediately lethal despite the conditions, **thermal death time (TDT)** is now more commonly used. This is the shortest time needed to kill all organisms in a microbial suspension at a specific temperature and under defined conditions. However, such destruction is logarithmic, and it is theoretically not possible to “completely destroy” microorganisms in a sample, even with extended heating. Therefore an even more precise figure, the **decimal reduction time (D)** or **D value** has gained wide acceptance. The decimal reduction time is the time required to kill 90% of the microorganisms or spores in a sample at a specified temperature. In a semilogarithmic plot of the population remaining versus the time of heating (figure 7.1), the *D* value is the time required for the line to drop by one log cycle or tenfold. The *D* value is usually written with a subscript, indicating the temperature for which it applies. *D* values are used to estimate the relative resistance of a microorganism to different temperatures through calculation of the ***z* value**. The *z* value is the increase in temperature required to reduce *D* to 1/10 its value or to reduce it by one log cycle when log *D* is plotted against temperature (figure 7.2). Another way to describe heating effectiveness is with the *F* value. The ***F* value** is the time in minutes at a specific temperature (usually 250°F or 121.1°C) needed to kill a population of cells or spores.

The food processing industry makes extensive use of *D* and *z* values. After a food has been canned, it must be heated to eliminate the risk of botulism arising from *Clostridium botulinum* spores. Heat treatment is carried out long enough to reduce a population of 10^{12} *C. botulinum* spores to 10^0 (one spore); thus there is a very small chance of any can having a viable spore. The *D* value for these spores at 121°C is 0.204 minutes. Therefore it would take 12*D* or 2.5 minutes to reduce 10^{12} spores to one spore by heating at 121°C. The *z* value for *C. botulinum* spores is 10°C—that is, it takes a 10°C change in temperature to alter the *D* value tenfold. If the cans were to be processed at 111°C rather than at 121°C, the *D* value would increase by tenfold to 2.04 minutes and the 12*D* value to 24.5 minutes. *D* values and *z* values for some common food-borne pathogens are given in table 7.3. Three *D* values are included for *Staphylococcus aureus* to illustrate the

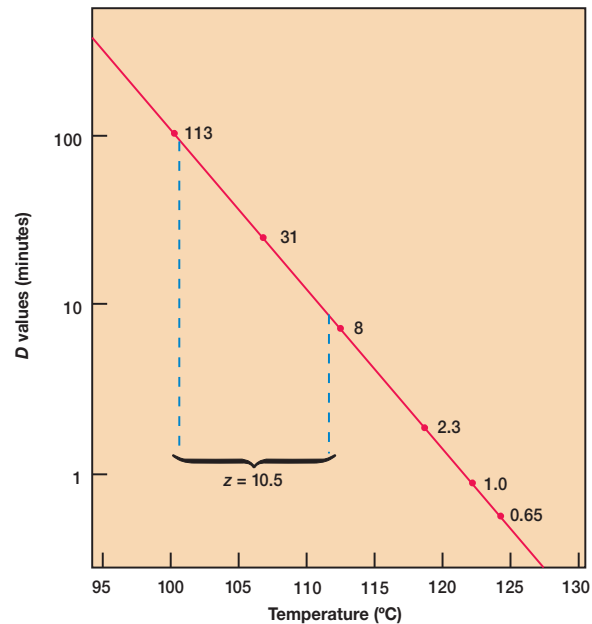


Figure 7.2 *z* Value Calculation. The *z* value used in calculation of time-temperature relationships for survival of a test microorganism, based on *D* value responses at various temperatures. The *z* value is the increase in temperature needed to reduce the decimal reduction time (*D*) to 10% of the original value. For this homogeneous sample of a test microorganism the *z* value is 10.5°. The *D* values are plotted on a logarithmic scale.

variation of killing rate with environment and the protective effect of organic material. [Food processing \(pp. 970–73\); Botulism \(p. 929\)](#)

Moist heat sterilization must be carried out at temperatures above 100°C in order to destroy bacterial endospores, and this requires the use of saturated steam under pressure. Steam sterilization is carried out with an **autoclave (figure 7.3)**, a device somewhat like a fancy pressure cooker. The development of the autoclave by Chamberland in 1884 tremendously stimulated the growth of microbiology. Water is boiled to produce steam, which is released through the jacket and into the autoclave’s chamber. The air initially present in the chamber is forced out until the chamber is filled with saturated steam and the outlets are closed. Hot, saturated steam continues to enter until the chamber reaches the desired temperature and pressure, usually 121°C and 15 pounds of pressure. At this temperature saturated steam destroys all vegetative cells and endospores in a small volume of liquid within 10 to 12 minutes. Treatment is continued for about 15 minutes to provide a margin of safety. Of course, larger containers of liquid such as flasks and carboys will require much longer treatment times.

Moist heat is thought to kill so effectively by degrading nucleic acids and by denaturing enzymes and other essential proteins. It also may disrupt cell membranes.

Autoclaving must be carried out properly or the processed materials will not be sterile. If all air has not been flushed out of the chamber, it will not reach 121°C even though it may reach a

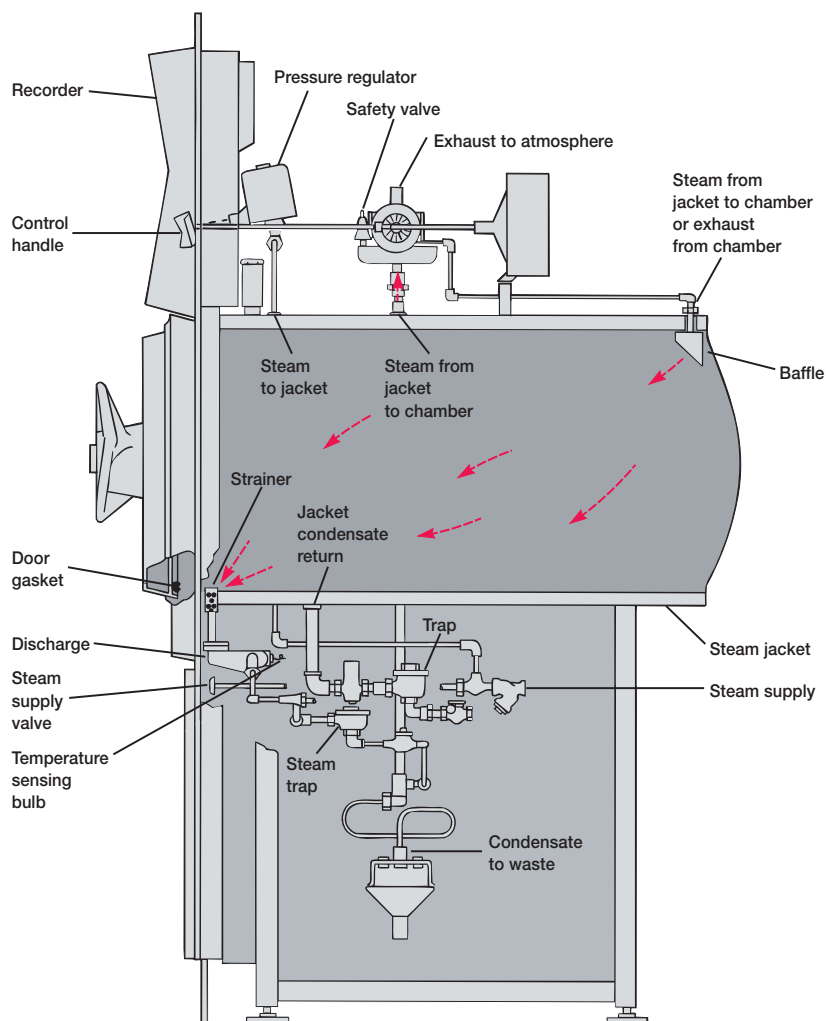
Table 7.3 *D* Values and *z* Values for Some Food-Borne Pathogens

Organism	Substrate	<i>D</i> Value (°C) in Minutes	<i>z</i> Value (°C)
<i>Clostridium botulinum</i>	Phosphate buffer	$D_{121} = 0.204$	10
<i>Clostridium perfringens</i> (heat-resistant strain)	Culture media	$D_{90} = 3-5$	6-8
<i>Salmonella</i>	Chicken à la king	$D_{60} = 0.39-0.40$	4.9-5.1
<i>Staphylococcus aureus</i>	Chicken à la king	$D_{60} = 5.17-5.37$	5.2-5.8
	Turkey stuffing	$D_{60} = 15.4$	6.8
	0.5% NaCl	$D_{60} = 2.0-2.5$	5.6

Values taken from F. L. Bryan, 1979, "Processes That Affect Survival and Growth of Microorganisms," *Time-Temperature Control of Foodborne Pathogens*, 1979. Atlanta: Centers for Disease Control and Prevention, Atlanta, GA.



(a)



(b)

Figure 7.3 The Autoclave or Steam Sterilizer. (a) A modern, automatically controlled autoclave or sterilizer. (b) Longitudinal cross section of a typical autoclave showing some of its parts and the pathway of steam. (b) From John J. Perkins, *Principles and Methods of Sterilization in Health Science*, 2nd edition, 1969. Courtesy of Charles C. Thomas, Publisher, Springfield, Illinois.

pressure of 15 pounds. The chamber should not be packed too tightly because the steam needs to circulate freely and contact everything in the autoclave. Bacterial endospores will be killed only if they are kept at 121°C for 10 to 12 minutes. When a large volume of liquid must be sterilized, an extended sterilization time will be needed because it will take longer for the center of the liquid to reach 121°C; 5 liters of liquid may require about 70 minutes. In view of these potential difficulties, a biological indicator is often autoclaved along with other material. This indicator commonly consists of a culture tube containing a sterile ampule of medium and a paper strip covered with spores of *Bacillus stearothermophilus* or *Clostridium* PA3679. After autoclaving, the ampule is aseptically broken and the culture incubated for several days. If the test bacterium does not grow in the medium, the sterilization run has been successful. Sometimes either special tape that spells out the word *sterile* or a paper indicator strip that changes color upon sufficient heating is autoclaved with a load of material. If the word appears on the tape or if the color changes after autoclaving, the material is supposed to be sterile. These approaches are convenient and save time but are not as reliable as the use of bacterial endospores.

Many substances, such as milk, are treated with controlled heating at temperatures well below boiling, a process known as **pasteurization** in honor of its developer Louis Pasteur. In the 1860s the French wine industry was plagued by the problem of wine spoilage, which made wine storage and shipping difficult. Pasteur examined spoiled wine under the microscope and detected microorganisms that looked like the bacteria responsible for lactic acid and acetic acid fermentations. He then discovered that a brief heating at 55 to 60°C would destroy these microorganisms and preserve wine for long periods. In 1886 the German chemists V. H. and F. Soxhlet adapted the technique for preserving milk and reducing milk-transmissible diseases. Milk pasteurization was introduced into the United States in 1889. Milk, beer, and many other beverages are now pasteurized. Pasteurization does not sterilize a beverage, but it does kill any pathogens present and drastically slows spoilage by reducing the level of nonpathogenic spoilage microorganisms.

Milk can be pasteurized in two ways. In the older method the milk is held at 63°C for 30 minutes. Large quantities of milk are now usually subjected to **flash pasteurization** or high-temperature short-term (HTST) pasteurization, which consists of quick heating to about 72°C for 15 seconds, then rapid cooling. The dairy industry also sometimes uses **ultrahigh-temperature (UHT) sterilization**. Milk and milk products are heated at 140 to 150°C for 1 to 3 seconds. UHT-processed milk does not require refrigeration and can be stored at room temperature for about 2 months without flavor changes. The small coffee creamer portions provided by restaurants often are prepared using UHT sterilization. [Pasteurization and the dairy industry \(pp. 970–71\)](#)

Many objects are best sterilized in the absence of water by **dry heat sterilization**. The items to be sterilized are placed in an oven at 160 to 170°C for 2 to 3 hours. Microbial death apparently results from the oxidation of cell constituents and denaturation of proteins. Although dry air heat is less effective than moist heat—*Clostridium botulinum* spores are killed in 5 minutes at 121°C by moist heat but only after 2 hours at 160°C with dry heat—it has

some definite advantages. Dry heat does not corrode glassware and metal instruments as moist heat does, and it can be used to sterilize powders, oils, and similar items. Most laboratories sterilize glass petri dishes and pipettes with dry heat. Despite these advantages, dry heat sterilization is slow and not suitable for heat-sensitive materials like many plastic and rubber items.

Low Temperatures

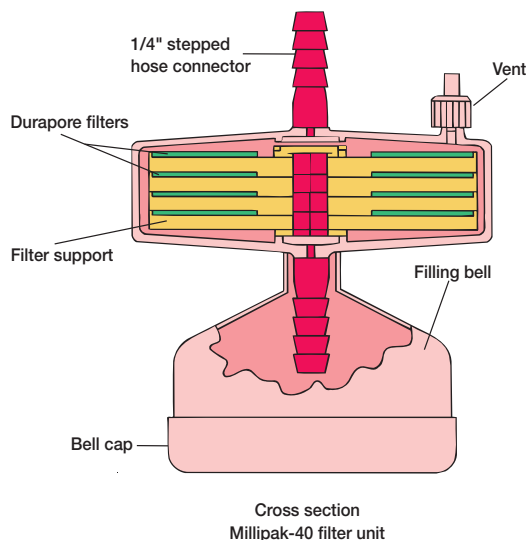
Although our emphasis is on the destruction of microorganisms, often the most convenient control technique is to inhibit their growth and reproduction by the use of either freezing or refrigeration. This approach is particularly important in food microbiology (*see p. 970*). Freezing items at –20°C or lower stops microbial growth because of the low temperature and the absence of liquid water. Some microorganisms will be killed by ice crystal disruption of cell membranes, but freezing does not destroy contaminating microbes. In fact, freezing is a very good method for long-term storage of microbial samples when carried out properly, and many laboratories have a low-temperature freezer for culture storage at –30 or –70°C. Because frozen food can contain many microorganisms, it should be prepared and consumed promptly after thawing in order to avoid spoilage and pathogen growth. [Effect of temperature on microbial growth \(pp. 125–27\)](#)

Refrigeration greatly slows microbial growth and reproduction, but does not halt it completely. Fortunately most pathogens are mesophilic and do not grow well at temperatures around 4°C. Refrigerated items may be ruined by growth of psychrophilic and psychrotrophic microorganisms, particularly if water is present. Thus refrigeration is a good technique only for shorter-term storage of food and other items.

Filtration

Filtration is an excellent way to reduce the microbial population in solutions of heat-sensitive material, and sometimes it can be used to sterilize solutions. Rather than directly destroying contaminating microorganisms, the filter simply removes them. There are two types of filters. **Depth filters** consist of fibrous or granular materials that have been bonded into a thick layer filled with twisting channels of small diameter. The solution containing microorganisms is sucked through this layer under vacuum, and microbial cells are removed by physical screening or entrapment and also by adsorption to the surface of the filter material. Depth filters are made of diatomaceous earth (Berkefeld filters), unglazed porcelain (Chamberlain filters), asbestos, or other similar materials.

Membrane filters have replaced depth filters for many purposes. These circular filters are porous membranes, a little over 0.1 mm thick, made of cellulose acetate, cellulose nitrate, polycarbonate, polyvinylidene fluoride, or other synthetic materials. Although a wide variety of pore sizes are available, membranes with pores about 0.2 µm in diameter are used to remove most vegetative cells, but not viruses, from solutions ranging in volume from 1 ml to many liters. The membranes are held in special holders (**figure 7.4**) and often preceded by depth filters made of glass fibers to remove larger particles that might clog the membrane filter. The solution is pulled or forced through the filter with a vacuum



(a)

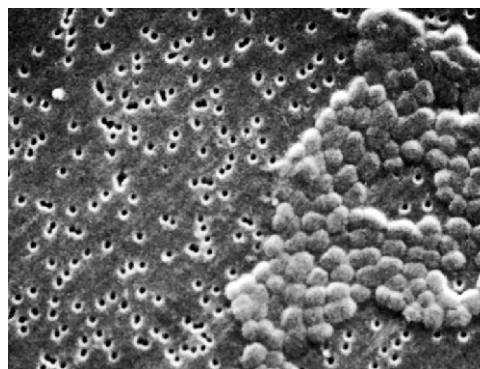


(b)

Figure 7.4 Membrane Filter Sterilization. A membrane filter outfit for sterilizing medium volumes of solution. (a) Cross section of the membrane filtering unit. Several membranes are used to increase capacity. (b) A complete filtering setup. The solution to be sterilized is kept in the Erlenmeyer flask, 1, and forced through the filter by a peristaltic pump, 2. The solution is sterilized by flowing through a membrane filter unit, 3, and into a sterile container. A wide variety of other kinds of filtering outfits are also available.



(a)



(b)

Figure 7.5 Membrane Filter Types. (a) *Bacillus megaterium* on an Ultipor nylon membrane with a bacterial removal rating of $0.2\ \mu\text{m}$ ($\times 2,000$). (b) *Enterococcus faecalis* resting on a polycarbonate membrane filter with $0.4\ \mu\text{m}$ pores ($\times 5,900$).

or with pressure from a syringe, peristaltic pump, or nitrogen gas bottle, and collected in previously sterilized containers. Membrane filters remove microorganisms by screening them out much as a sieve separates large sand particles from small ones (**figure 7.5**). These filters are used to sterilize pharmaceuticals, ophthalmic solutions, culture media, oils, antibiotics, and other heat-sensitive solutions. [The use of membrane filters in microbial counting \(p. 118\)](#)

Air also can be sterilized by filtration. Two common examples are surgical masks and cotton plugs on culture vessels that let air in

but keep microorganisms out. **Laminar flow biological safety cabinets** employing **high-efficiency particulate air (HEPA) filters**, which remove 99.97% of $0.3\ \mu\text{m}$ particles, are one of the most important air filtration systems. Laminar flow biological safety cabinets force air through HEPA filters, then project a vertical curtain of sterile air across the cabinet opening. This protects a worker from microorganisms being handled within the cabinet and prevents contamination of the room (**figure 7.6**). A person uses these cabinets when working with dangerous agents such as *Mycobacterium tuberculosis*,

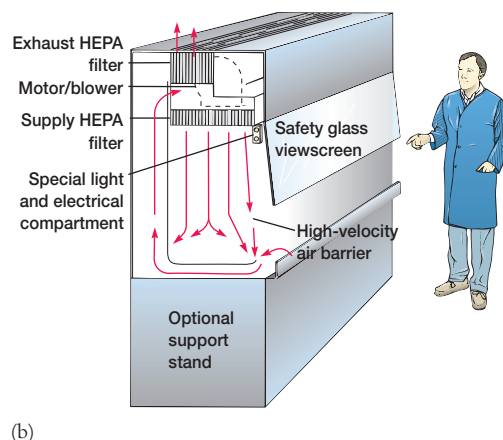


Figure 7.6 A Laminar Flow Biological Safety Cabinet. (a) A technician pipetting potentially hazardous material in a safety cabinet. (b) A schematic diagram showing the airflow pattern.

tumor viruses, and recombinant DNA. They are also employed in research labs and industries, such as the pharmaceutical industry, when a sterile working surface is needed for conducting assays, preparing media, examining tissue cultures, and the like.

Radiation

The types of radiation and the ways in which radiation damages or destroys microorganisms have already been discussed. The practical uses of ultraviolet and ionizing radiation in sterilizing objects are briefly described next. [Radiation and its effects on microorganisms \(pp. 130–31\)](#)

Ultraviolet (UV) radiation around 260 nm (see figure 6.17) is quite lethal but does not penetrate glass, dirt films, water, and other substances very effectively. Because of this disadvantage, UV radiation is used as a sterilizing agent only in a few specific situations. UV lamps are sometimes placed on the ceilings of rooms or in biological safety cabinets to sterilize the air and any exposed surfaces. Because UV radiation burns the skin and damages eyes, people working in such areas must be certain the UV lamps are off when the areas are in use. Commercial UV units are available for water treatment. Pathogens and other microorganisms are destroyed when a thin layer of water is passed under the lamps.

Ionizing radiation is an excellent sterilizing agent and penetrates deep into objects. It will destroy bacterial endospores and vegetative cells, both procaryotic and eucaryotic; however, ionizing radiation is not always as effective against viruses. Gamma radiation from a cobalt 60 source is used in the cold sterilization of antibiotics, hormones, sutures, and plastic disposable supplies such

as syringes. Gamma radiation has also been used to sterilize and “pasteurize” meat and other food. Irradiation can eliminate the threat of such pathogens as *Escherichia coli* O157:H7, *Staphylococcus aureus*, and *Campylobacter jejuni*. Both the Food and Drug Administration and the World Health Organization have approved food irradiation and declared it safe. A commercial irradiation plant operates near Tampa, Florida. However, this process has not yet been widely employed in the United States because of the cost and concerns about the effects of gamma radiation on food. The U.S. government currently approves the use of radiation to treat poultry, beef, pork, veal, lamb, fruits, vegetables, and spices. It will probably be more extensively employed in the future.

1. Define thermal death point (TDP), thermal death time (TDT), decimal reduction time (D) or D value, z value, and the F value.
2. Describe how an autoclave works. What conditions are required for sterilization by moist heat, and what three things must one do when operating an autoclave to help ensure success?
3. How are pasteurization, flash pasteurization, ultrahigh temperature sterilization, and dry heat sterilization carried out? Give some practical applications for each of these procedures.
4. How can low temperature be used to control microorganisms?
5. What are depth filters and membrane filters, and how are they used to sterilize liquids? Describe the operation of a biological safety cabinet.
6. Give the advantages and disadvantages of ultraviolet light and ionizing radiation as sterilizing agents. Provide a few examples of how each is used for this purpose.

Box 7.2

Universal Precautions for Microbiology Laboratories

Blood and other body fluids from all patients should be considered infective.

1. All specimens of blood and body fluids should be put in a well-constructed container with a secure lid to prevent leaking during transport. Care should be taken when collecting each specimen to avoid contaminating the outside of the container and of the laboratory form accompanying the specimen.
2. All persons processing blood and body-fluid specimens should wear gloves. Masks and protective eyewear should be worn if mucous membrane contact with blood or body fluids is anticipated. Gloves should be changed and hands washed after completion of specimen processing.
3. For routine procedures, such as histologic and pathological studies or microbiologic culturing, a biological safety cabinet is not necessary. However, biological safety cabinets should be used whenever procedures are conducted that have a high potential for generating droplets. These include activities such as blending, sonicating, and vigorous mixing.
4. Mechanical pipetting devices should be used for manipulating all liquids in the laboratory. Mouth pipetting must not be done.
5. Use of needles and syringes should be limited to situations in which there is no alternative, and the recommendations for preventing injuries with needles outlined under universal precautions (*see Box 36.1, p. 829*) should be followed.
6. Laboratory work surfaces should be decontaminated with an appropriate chemical germicide after a spill of blood or other body fluids and when work activities are completed.
7. Contaminated materials used in laboratory tests should be decontaminated before reprocessing or be placed in bags and disposed of in accordance with institutional policies for disposal of infective waste.
8. Scientific equipment that has been contaminated with blood or other body fluids should be decontaminated and cleaned before being repaired in the laboratory or transported to the manufacturer.
9. All persons should wash their hands after completing laboratory activities and should remove protective clothing before leaving the laboratory.
10. There should be no eating, drinking, or smoking in the work area.

Source: Adapted from *Morbidity and Mortality Weekly Report*, 36 (Suppl. 2S) 5S–10S, 1987, the Centers for Disease Control and Prevention Guidelines.

7.5 The Use of Chemical Agents in Control

Although objects are sometimes disinfected with physical agents, chemicals are more often employed in disinfection and antiseptics. Many factors influence the effectiveness of chemical disinfectants and antiseptics as previously discussed. Factors such as the kinds of microorganisms potentially present, the concentration and nature of the disinfectant to be used, and the length of treatment should be considered. Dirty surfaces must be cleaned before a disinfectant or antiseptic is applied. The proper use of chemical agents is essential to laboratory and hospital safety (**Box 7.2**; *see also Box 36.1*). It should be noted that chemicals also are employed to prevent microbial growth in food. This is discussed in the chapter on food microbiology (*see pp. 971–72*).

Many different chemicals are available for use as disinfectants, and each has its own advantages and disadvantages. In selecting an agent, it is important to keep in mind the characteristics of a desirable disinfectant. Ideally the disinfectant must be effective against a wide variety of infectious agents (gram-positive and gram-negative bacteria, acid-fast bacteria, bacterial endospores, fungi, and viruses) at high dilutions and in the presence of organic matter. Although the chemical must be toxic for infectious agents, it should not be toxic to people or corrosive for common materials. In practice, this balance between effectiveness and low toxicity for animals is hard to achieve. Some chemicals are used despite their low effectiveness because they are

relatively nontoxic. The disinfectant should be stable upon storage, odorless or with a pleasant odor, soluble in water and lipids for penetration into microorganisms, and have a low surface tension so that it can enter cracks in surfaces. If possible the disinfectant should be relatively inexpensive.

One potentially serious problem is the overuse of triclosan and other germicides. This antibacterial agent is now found in products such as deodorants, mouthwashes, soaps, cutting boards, and baby toys. Triclosan seems to be everywhere. Unfortunately we are already seeing the emergence of triclosan-resistant bacteria. *Pseudomonas aeruginosa* actively pumps the antiseptic out the cell. Bacteria seem to be responding to antiseptic overuse in the same way as they reacted to antibiotic overuse (*see pp. 818–20*). There is now some evidence that extensive use of triclosan also increases the frequency of antibiotic resistance in bacteria. Thus overuse of antiseptics can have unintended harmful consequences.

The properties and uses of several groups of common disinfectants and antiseptics are surveyed next. Many of their characteristics are summarized in **tables 7.4** and **7.5**. Structures of some common agents are given in **figure 7.7**.

Phenolics

Phenol was the first widely used antiseptic and disinfectant. In 1867 Joseph Lister employed it to reduce the risk of infection during operations. Today phenol and phenolics (phenol

Table 7.4 Activity Levels of Selected Germicides

Class	Use Concentration of Active Ingredient	Activity Level ^a
Gas		
Ethylene oxide	450–500 mg/liter ^b	High
Liquid		
Glutaraldehyde, aqueous	2%	High to intermediate
Formaldehyde + alcohol	8 + 70%	High
Stabilized hydrogen peroxide	6–30%	High to intermediate
Formaldehyde, aqueous	6–8%	High to intermediate
Iodophors	750–5,000 mg/liter ^c	High to intermediate
Iodophors	75–150 mg/liter ^c	Intermediate to low
Iodine + alcohol	0.5 + 70%	Intermediate
Chlorine compounds	0.1–0.5% ^d	Intermediate
Phenolic compounds, aqueous	0.5–3%	Intermediate to low
Iodine, aqueous	1%	Intermediate
Alcohols (ethyl, isopropyl)	70%	Intermediate
Quaternary ammonium compounds	0.1–0.2% aqueous	Low
Chlorhexidine	0.75–4%	Low
Hexachlorophene	1–3%	Low
Mercurial compounds	0.1–0.2%	Low

Source: From Seymour S. Block, *Disinfection, Sterilization and Preservation*. Copyright © 1983 Lea & Febiger, Malvern, Pa. 1983. Reprinted by permission.

^aHigh-level disinfectants destroy vegetative bacterial cells including *M. tuberculosis*, bacterial endospores, fungi, and viruses. Intermediate-level disinfectants destroy all of the above except endospores. Low-level agents kill bacterial vegetative cells except for *M. tuberculosis*, fungi, and medium-sized lipid-containing viruses (but not bacterial endospores or small, nonlipid viruses).

^bIn autoclave-type equipment at 55 to 60°C.

^cAvailable iodine.

^dFree chlorine.

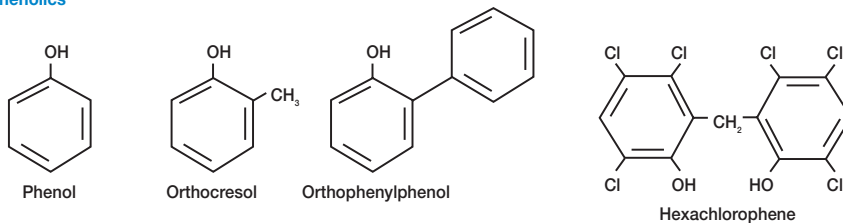
Table 7.5 Relative Efficacy of Commonly Used Disinfectants and Antiseptics

Class	Disinfectant	Antiseptic	Comment
Gas			
Ethylene oxide	3–4 ^a	0 ^a	Sporicidal; toxic; good penetration; requires relative humidity of 30% or more; microbicidal activity varies with apparatus used; absorbed by porous material; dry spores highly resistant; moisture must be present, and presoaking is most desirable
Liquid			
Glutaraldehyde, aqueous	3	0	Sporicidal; active solution unstable; toxic
Stabilized hydrogen peroxide	3	0	Sporicidal; use solution stable up to 6 weeks; toxic orally and to eyes; mildly skin toxic; little inactivated by organic matter
Formaldehyde + alcohol	3	0	Sporicidal; noxious fumes; toxic; volatile
Formaldehyde, aqueous	1–2	0	Sporicidal; noxious fumes; toxic
Phenolic compounds	3	0	Stable; corrosive; little inactivation by organic matter; irritates skin
Chlorine compounds	1–2	0	Fast action; inactivation by organic matter; corrosive; irritates skin
Alcohol	1	3	Rapidly microbicidal except for bacterial spores and some viruses; volatile; flammable; dries and irritates skin
Iodine + alcohol	0	4	Corrosive; very rapidly microbicidal; causes staining; irritates skin; flammable
Iodophors	1–2	3	Somewhat unstable; relatively bland; staining temporary; corrosive
Iodine, aqueous	0	2	Rapidly microbicidal; corrosive; stains fabrics; stains and irritates skin
Quaternary ammonium compounds	1	0	Bland; inactivated by soap and anionics; compounds absorbed by fabrics; old or dilute solution can support growth of gram-negative bacteria
Hexachlorophene	0	2	Bland; insoluble in water, soluble in alcohol; not inactivated by soap; weakly bactericidal
Chlorhexidine	0	3	Bland; soluble in water and alcohol; weakly bactericidal
Mercurial compounds	0	±	Bland; much inactivated by organic matter; weakly bactericidal

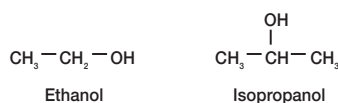
Source: From Seymour S. Block, *Disinfection, Sterilization and Preservation*. Copyright © 1983 Lea & Febiger, Malvern, Pa. 1983. Reprinted by permission.

^aSubjective ratings of practical usefulness in a hospital environment—4 is maximal usefulness; 0 is little or no usefulness; ± signifies that the substance is sometimes useful but not always.

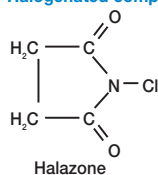
Phenolics



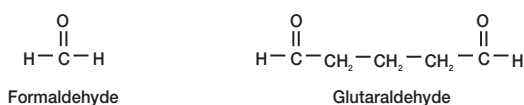
Alcohols



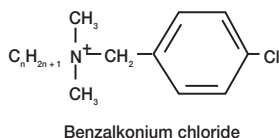
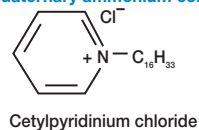
Halogenated compound



Aldehydes



Quaternary ammonium compounds



Gases



Figure 7.7 Disinfectants and Antiseptics. The structures of some frequently used disinfectants and antiseptics.

derivatives) such as cresols, xylenols, and orthophenylphenol are used as disinfectants in laboratories and hospitals. The commercial disinfectant Lysol is made of a mixture of phenolics. Phenolics act by denaturing proteins and disrupting cell membranes. They have some real advantages as disinfectants: phenolics are tuberculocidal, effective in the presence of organic material, and remain active on surfaces long after application. However, they do have a disagreeable odor and can cause skin irritation.

Hexachlorophene (figure 7.7) has been one of the most popular antiseptics because it persists on the skin once applied and reduces skin bacteria for long periods. However, it can cause brain damage and is now used in hospital nurseries only in response to a staphylococcal outbreak.

Alcohols

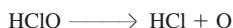
Alcohols are among the most widely used disinfectants and antiseptics. They are bactericidal and fungicidal but not sporicidal; some lipid-containing viruses are also destroyed. The two most popular alcohol germicides are ethanol and isopropanol, usually used in about 70 to 80% concentration. They act by denaturing proteins and possibly by dissolving membrane lipids. A 10 to 15 minute soaking is sufficient to disinfect thermometers and small instruments.

Halogens

A halogen is any of the five elements (fluorine, chlorine, bromine, iodine, and astatine) in group VIIA of the periodic table. They exist as diatomic molecules in the free state and form saltlike compounds

with sodium and most other metals. The halogens iodine and chlorine are important antimicrobial agents. Iodine is used as a skin antiseptic and kills by oxidizing cell constituents and iodinating cell proteins. At higher concentrations, it may even kill some spores. Iodine often has been applied as tincture of iodine, 2% or more iodine in a water-ethanol solution of potassium iodide. Although it is an effective antiseptic, the skin may be damaged, a stain is left, and iodine allergies can result. More recently iodine has been complexed with an organic carrier to form an **iodophor**. Iodophors are water soluble, stable, and nonstaining, and release iodine slowly to minimize skin burns and irritation. They are used in hospitals for preoperative skin degerming and in hospitals and laboratories for disinfecting. Some popular brands are Wescodyne for skin and laboratory disinfection and Betadine for wounds.

Chlorine is the usual disinfectant for municipal water supplies and swimming pools and is also employed in the dairy and food industries. It may be applied as chlorine gas, sodium hypochlorite, or calcium hypochlorite, all of which yield hypochlorous acid (HClO) and then atomic oxygen. The result is oxidation of cellular materials and destruction of vegetative bacteria and fungi, although not spores.



Death of almost all microorganisms usually occurs within 30 minutes. Since organic material interferes with chlorine action by reacting with chlorine and its products, an excess of chlorine is added to ensure microbial destruction. One potential problem is that chlorine reacts with organic compounds to form carcinogenic trihalomethanes, which must be monitored in drinking water. Ozone sometimes has been used successfully as an alternative to chlorination in Europe and Canada. [Municipal water purification \(pp. 651–53\)](#)

Chlorine is also an excellent disinfectant for individual use because it is effective, inexpensive, and easy to employ. Small quantities of drinking water can be disinfected with halazone tablets. Halazone (parasulfone dichloramidobenzoic acid) slowly releases chloride when added to water and disinfects it in about a half hour. It is frequently used by campers lacking access to uncontaminated drinking water.

Chlorine solutions make very effective laboratory and household disinfectants. An excellent disinfectant-detergent combination can be prepared if a 1/100 dilution of household bleach (e.g., 1.3 fl oz of Clorox or Purex bleach in 1 gal or 10 ml/liter) is combined with sufficient nonionic detergent (about 1 oz/gal or 7.8 ml/liter) to give a 0.8% detergent concentration. This mixture will remove both dirt and bacteria.

Heavy Metals

For many years the ions of heavy metals such as mercury, silver, arsenic, zinc, and copper were used as germicides. More recently these have been superseded by other less toxic and more effective

germicides (many heavy metals are more bacteriostatic than bactericidal). There are a few exceptions. A 1% solution of silver nitrate is often added to the eyes of infants to prevent ophthalmic gonorrhea (in many hospitals, erythromycin is used instead of silver nitrate because it is effective against *Chlamydia* as well as *Neisseria*). Silver sulfadiazine is used on burns. Copper sulfate is an effective algicide in lakes and swimming pools.

Heavy metals combine with proteins, often with their sulfhydryl groups, and inactivate them. They may also precipitate cell proteins.

Quaternary Ammonium Compounds

Detergents [Latin *detergere*, to wipe off or away] are organic molecules that serve as wetting agents and emulsifiers because they have both polar hydrophilic and nonpolar hydrophobic ends. Due to their amphipathic nature (*see section 3.2*), detergents solubilize otherwise insoluble residues and are very effective cleansing agents. They are different than soaps, which are derived from fats.

Although anionic detergents have some antimicrobial properties, only cationic detergents are effective disinfectants. The most popular of these disinfectants are quaternary ammonium compounds characterized by a positively charged quaternary nitrogen and a long hydrophobic aliphatic chain (figure 7.7). They disrupt microbial membranes and may also denature proteins.

Cationic detergents like benzalkonium chloride and cetylpyridinium chloride kill most bacteria but not *M. tuberculosis* or endospores. They do have the advantages of being stable, nontoxic, and bland but they are inactivated by hard water and soap. Cationic detergents are often used as disinfectants for food utensils and small instruments and as skin antiseptics. Several brands are on the market. Zephiran contains benzalkonium chloride and Ceepryn, cetylpyridinium chloride.

Aldehydes

Both of the commonly used aldehydes, formaldehyde and glutaraldehyde, are highly reactive molecules that combine with nucleic acids and proteins and inactivate them, probably by cross-linking and alkylating molecules (figure 7.7). They are sporicidal and can be used as chemical sterilants. Formaldehyde is usually dissolved in water or alcohol before use. A 2% buffered solution of glutaraldehyde is an effective disinfectant. It is less irritating than formaldehyde and is used to disinfect hospital and laboratory equipment. Glutaraldehyde usually disinfects objects within about 10 minutes but may require as long as 12 hours to destroy all spores.

Sterilizing Gases

Many heat-sensitive items such as disposable plastic petri dishes and syringes, heart-lung machine components, sutures, and catheters are now sterilized with ethylene oxide gas (figure 7.7). Ethylene oxide (EtO) is both microbicidal and sporicidal and kills by combining with cell proteins. It is a particularly effective sterilizing agent because it rapidly penetrates packing materials, even plastic wraps.

Sterilization is carried out in a special ethylene oxide sterilizer, very much resembling an autoclave in appearance, that controls the EtO concentration, temperature, and humidity. Because pure EtO is explosive, it is usually supplied in a 10 to 20% concentration mixed with either CO₂ or dichlorodifluoromethane. The ethylene oxide concentration, humidity, and temperature influence the rate of sterilization. A clean object can be sterilized if treated for 5 to 8 hours at 38°C or 3 to 4 hours at 54°C when the relative humidity is maintained at 40 to 50% and the EtO concentration at 700 mg/liter. Extensive aeration of the sterilized materials is necessary to remove residual EtO because it is so toxic.

Betapropiolactone (BPL) is occasionally employed as a sterilizing gas. In the liquid form it has been used to sterilize vaccines and sera. BPL decomposes to an inactive form after several hours and is therefore not as difficult to eliminate as EtO. It also destroys microorganisms more readily than ethylene oxide but does not penetrate materials well and may be carcinogenic. For these reasons, BPL has not been used as extensively as EtO.

Recently vapor-phase hydrogen peroxide has been used to decontaminate biological safety cabinets.

1. Why are most antimicrobial chemical agents disinfectants rather than sterilants? What general characteristics should one look for in a disinfectant?
2. Describe each of the following agents in terms of its chemical nature, mechanism of action, mode of application, common uses and effectiveness, and advantages and disadvantages: phenolics, alcohols, halogens (iodine and chlorine), heavy metals, quaternary ammonium compounds, aldehydes, and ethylene oxide.

7.6 Evaluation of Antimicrobial Agent Effectiveness

Testing of antimicrobial agents is a complex process regulated by two different federal agencies. The U.S. Environmental Protection Agency regulates disinfectants, whereas agents used on humans and animals are under the control of the Food and Drug Administration. Testing of antimicrobial agents often begins with an initial screening test to see if they are effective and at what concentrations. This may be followed by more realistic in-use testing.

The best-known disinfectant screening test is the **phenol coefficient test** in which the potency of a disinfectant is compared with that of phenol. A series of dilutions of phenol and the experimental disinfectant are inoculated with the test bacteria *Salmonella typhi* and *Staphylococcus aureus*, then placed in a 20 or 37°C water bath. These inoculated disinfectant tubes are next subcultured to regular fresh medium at 5 minute intervals, and the subcultures are incubated for two or more days. The highest dilutions that kill the bacteria after a 10 minute exposure, but not after 5 minutes, are used to calculate the phenol coefficient. The reciprocal of the appropriate test disinfectant dilution is divided by

Table 7.6 Phenol Coefficients for Some Disinfectants

Disinfectant	Phenol Coefficients ^a	
	<i>Salmonella typhi</i>	<i>Staphylococcus aureus</i>
Phenol	1	1
Cetylpyridinium chloride	228	337
<i>O</i> -phenylphenol	5.6 (20°C)	4.0
<i>p</i> -cresol	2.0–2.3	2.3
Hexachlorophene	5–15	15–40
Merthiolate	600	62.5
Mercurochrome	2.7	5.3
Lysol	1.9	3.5
Isopropyl alcohol	0.6	0.5
Ethanol	0.04	0.04
2% I ₂ solution in EtOH	4.1–5.2 (20°C)	4.1–5.2 (20°C)

^aAll values were determined at 37°C except where indicated.

that for phenol to obtain the coefficient. Suppose that the phenol dilution was 1/90 and maximum effective dilution for disinfectant X was 1/450. The phenol coefficient of X would be 5. The higher the phenol coefficient value, the more effective the disinfectant under these test conditions. A value greater than 1 means that the disinfectant is more effective than phenol. A few representative phenol coefficient values are given in **table 7.6**.

The phenol coefficient test is a useful initial screening procedure, but the phenol coefficient can be misleading if taken as a direct indication of disinfectant potency during normal use. This is because the phenol coefficient is determined under carefully controlled conditions with pure bacterial strains, whereas disinfectants are normally used on complex populations in the presence of organic matter and with significant variations in environmental factors like pH, temperature, and presence of salts.

To more realistically estimate disinfectant effectiveness, other tests are often used. The rates at which selected bacteria are destroyed with various chemical agents may be experimentally determined and compared. A **use dilution test** can also be carried out. Stainless steel cylinders are contaminated with specific bacterial species under carefully controlled conditions. The cylinders are dried briefly, immersed in the test disinfectants for 10 minutes, transferred to culture media, and incubated for two days. The disinfectant concentration that kills the organisms in the sample with a 95% level of confidence under these conditions is determined. Disinfectants also can be tested under conditions designed to simulate normal in-use situations. In-use testing techniques allow a more accurate determination of the proper disinfectant concentration for a particular situation.

1. Briefly describe the phenol coefficient test.
2. Why might it be necessary to employ procedures like the use dilution and in-use tests?

Summary

1. Sterilization is the process by which all living cells, viable spores, viruses, and viroids are either destroyed or removed from an object or habitat. Disinfection is the killing, inhibition, or removal of microorganisms (but not necessarily endospores) that can cause disease.
2. Many terms are used to define how microorganisms are controlled: sterilant, disinfectant, sanitization, antisepsis, and antiseptic.
3. Antimicrobial agents that kill organisms often have the suffix *-cide*, whereas agents that prevent growth and reproduction have the suffix *-static*.
4. Microbial death is usually exponential or logarithmic (**figure 7.1**).
5. The effectiveness of a disinfectant or sterilizing agent is influenced by population size, population composition, concentration or intensity of the agent, exposure duration, temperature, and nature of the local environment.
6. The efficiency of heat killing is often indicated by the thermal death time or the decimal reduction time.
7. Although treatment with boiling water for 10 minutes kills vegetative forms, an autoclave must be used to destroy endospores by heating at 121°C and 15 pounds of pressure (**figure 7.3**).
8. Moist heat kills by degrading nucleic acids, denaturing enzymes and other proteins, and disrupting cell membranes.
9. Heat-sensitive liquids can be pasteurized by heating at 63°C for 30 minutes or at 72°C for 15 seconds (flash pasteurization). Heating at 140 to 150°C for 1 to 3 seconds (ultrahigh-temperature sterilization) may be used.
10. Glassware and other heat-stable items may be sterilized by dry heat at 160 to 170°C for 2 to 3 hours.
11. Refrigeration and freezing can be used to control microbial growth and reproduction.
12. Microorganisms can be efficiently removed by filtration with either depth filters or membrane filters.
13. Biological safety cabinets with high-efficiency particulate filters sterilize air by filtration (**figure 7.6**).
14. Radiation of short wavelength or high-energy ultraviolet and ionizing radiation can be used to sterilize objects.
15. Chemical agents usually act as disinfectants because they cannot readily destroy bacterial endospores. Disinfectant effectiveness depends on concentration, treatment duration, temperature, and presence of organic material (**tables 7.4 and 7.5**).
16. Phenolics and alcohols are popular disinfectants that act by denaturing proteins and disrupting cell membranes (**figure 7.7**).
17. Halogens (iodine and chlorine) kill by oxidizing cellular constituents; cell proteins may also be iodinated. Iodine is applied as a tincture or iodophor. Chlorine may be added to water as a gas, hypochlorite, or an organic chlorine derivative.
18. Heavy metals tend to be bacteriostatic agents. They are employed in specialized situations such as the use of silver nitrate in the eyes of newborn infants and copper sulfate in lakes and pools.
19. Cationic detergents are often used as disinfectants and antiseptics; they disrupt membranes and denature proteins.
20. Aldehydes such as formaldehyde and glutaraldehyde can sterilize as well as disinfect because they kill spores.
21. Ethylene oxide gas penetrates plastic wrapping material and destroys all life forms by reacting with proteins. It is used to sterilize packaged, heat-sensitive materials.
22. A variety of procedures can be used to determine the effectiveness of disinfectants, among them the following: phenol coefficient test, measurement of killing rates with germicides, use dilution testing, and in-use testing.

Key Terms

algicide 138
antimicrobial agent 139
antisepsis 138
antiseptics 138
autoclave 140
bactericide 138
bacteriostatic 138
D value 140
decimal reduction time (*D*) 140
depth filters 142
detergent 148
disinfectant 138

disinfection 138
dry heat sterilization 142
F value 140
flash pasteurization 142
fungicide 138
fungistatic 138
germicide 138
high-efficiency particulate air (HEPA) filters 143
iodophor 148
ionizing radiation 144
laminar flow biological safety cabinets 143
membrane filters 142

pasteurization 142
phenol coefficient test 149
sanitization 138
sterilization 137
thermal death time (TDT) 140
ultrahigh-temperature (UHT) sterilization 142
ultraviolet (UV) radiation 144
use dilution test 149
viricide 138
z value 140

Questions for Thought and Review

1. How can the *D* value be used to estimate the time required for sterilization? Suppose that you wanted to eliminate the risk of *Salmonella* poisoning by heating your food ($D_{60} = 0.40$ minutes, z value = 5.0). Calculate the 12*D* value at 60°C. How long would it take to achieve the same results by heating at 50, 55, and 65°C?
2. How can one alter treatment conditions to increase the effectiveness of a disinfectant?
3. How would the following be best sterilized: glass pipettes and petri plates, tryptic soy broth tubes, nutrient agar, antibiotic solution, interior of a biological safety cabinet, wrapped package of plastic petri plates?
4. Which disinfectants or antiseptics would be used to treat the following: oral thermometer, laboratory bench top, drinking water, patch of skin before surgery, small medical instruments (probes, forceps, etc.)? List all chemicals suitable for each task.
5. Until relatively recently, spoiled milk was responsible for a significant proportion of infant death.
 - a. Why is untreated milk easily spoiled?
 - b. Why is boiling milk over prolonged periods not desirable?
6. In table 7.3 why is the *D* value so different for the three conditions in which *S. aureus* might be found?

Critical Thinking Questions

- Throughout history, spices have been used as preservatives and to cover up the smell/taste of food that is slightly spoiled. The success of some spices led to a magical, ritualized use of many of them and possession of spices was often limited to priests or other powerful members of the community.
 - Choose a spice and trace its use geographically and historically. What is its common-day use today?
 - Spices grow and tend to be used predominantly in warmer climates. Explain.
- Design an experiment to determine whether an antimicrobial agent is acting as a cidal or static agent. How would you determine whether an agent is suitable for use as an antiseptic rather than as a disinfectant?
- Suppose that you are testing the effectiveness of disinfectants with the phenol coefficient test and obtained the following results.

Dilution	Bacterial Growth after Treatment		
	Disinfectant A	Disinfectant B	Disinfectant C
1/20	—	—	—
1/40	+	—	—
1/80	+	—	+
1/160	+	+	+
1/320	+	—	+

What disinfectant can you safely say is the most effective? Can you determine its phenol coefficient from these results?

Additional Reading

General

- Barkley, W. E., and Richardson, J. H. 1994. Laboratory safety. In *Methods for general and molecular bacteriology*, P. Gerhardt, et al., editors, 715–34. Washington, D.C.: American Society for Microbiology.
- Block, S. S. 1992. Sterilization. In *Encyclopedia of microbiology*, 1st ed., vol. 4, J. Lederberg, editor-in-chief, 87–103. San Diego: Academic Press.
- Block, S. S., editor. 1991. *Disinfection, sterilization and preservation*, 4th ed. Philadelphia: Lea and Febiger.
- Centers for Disease Control. 1987. Recommendations for prevention of HIV transmission in health-care settings. *Morbidity and Mortality Weekly Rep.* 36(Suppl. 2):3S–18S.
- Centers for Disease Control. 1988. Update: Universal precautions for prevention of transmission of human immunodeficiency virus, hepatitis B virus, and other bloodborne pathogens in health-care settings. *Morbidity and Mortality Weekly Rep.* 37(24):377–88.
- Centers for Disease Control. 1989. Guidelines for prevention of transmission of human immunodeficiency virus and hepatitis B virus to health-care and public-safety workers. *Morbidity and Mortality Weekly Rep.* 38(Suppl. 6):1–37.
- Centers for Disease Control and National Institutes of Health. 1992. *Biosafety in microbiological and biomedical laboratories*, 3d ed. Washington, D.C.: U.S. Government Printing Office.
- Collins, C. H., and Lyne, P. M. 1976. *Microbiological methods*, 4th ed. Boston: Butterworths.

- Henderson, D. K. 1995. HIV-1 in the health-care setting. In *Principles and practice of infectious diseases*, 4th ed., G. L. Mandell, J. E. Bennett, and R. Dolin editors, 2632–56. New York: Churchill Livingstone.
- Martin, M. A., and Wenzel, R. P. 1995. Sterilization, disinfection, and disposal of infectious waste. In *Principles and practice of infectious diseases*, 4th ed., G. L. Mandell, J. E. Bennett, and R. Dolin editors, 2579–87. New York: Churchill Livingstone.
- Perkins, J. J. 1969. *Principles and methods of sterilization in health sciences*, 2d ed. Springfield, Ill.: Charles C. Thomas.
- Pike, R. M. 1979. Laboratory-associated infections: Incidence, fatalities, causes, and prevention. *Annu. Rev. Microbiol.* 33:41–66.
- Russell, A. D.; Hugo, W. B.; and Ayliffe, G. A. J., editors. 1992. *Principles and practice of disinfection, preservation and sterilization*, 2d ed. Oxford: Blackwell Scientific Publications.
- Sewell, D. L. 1995. Laboratory-associated infections and biosafety. *Clin. Microbiol. Rev.* 8(3):389–405.
- Strain, B. A., and Gröschel, D. H. M. 1995. Laboratory safety and infectious waste management. In *Manual of clinical microbiology*, 6th ed., P. R. Murray, editor, 75–85. Washington, D.C.: American Society for Microbiology.
- Warren, E. 1981. Laboratory safety. In *Laboratory procedures in clinical microbiology*, J. A. Washington, editor, 729–45. New York: Springer-Verlag.

- Widmer, A. F., and Frei, R. 1999. Decontamination, disinfection, and sterilization. In *Manual of clinical microbiology*, 7th ed., P. R. Murray, et al., editors, 138–64. Washington, D.C.: ASM Press.

7.4 The Use of Physical Methods in Control

- Brock, T. D. 1983. *Membrane filtration: A user's guide and reference manual*. Madison, Wis.: Science Tech Publishers.
- Sørhaug, T. 1992. Temperature control. In *Encyclopedia of microbiology*, 1st ed., vol. 4, J. Lederberg, editor-in-chief, 201–11. San Diego: Academic Press.

7.5 The Use of Chemical Agents in Control

- Belkin, S.; Dukan, S.; Levi, Y.; and Touati, D. 1999. Death by disinfection: Molecular approaches to understanding bacterial sensitivity and resistance to free chlorine. In *Microbial ecology and infectious disease*, E. Rosenberg, editor, 133–42. Washington, D.C.: ASM Press.
- Borick, P. M. 1973. *Chemical sterilization*. Stroudsburg, Pa.: Dowden, Hutchinson and Ross.
- McDonnell, G., and Russell, A. D. 1999. Antiseptics and disinfectants: Activity, action, and resistance. *Clin. Microbiol. Rev.* 12(1):147–79.
- Russell, A. D. 1990. Bacterial spores and chemical sporicidal agents. *Clin. Microbiol. Rev.* 3(2):99–119.
- Rutala, W. A., and Weber, D. J. 1997. Uses of inorganic hypochlorite (bleach) in health-care facilities. *Clin. Microbiol. Rev.* 10(4):597–610.

PART III

Microbial Metabolism

Chapter 8

Metabolism: Energy, Enzymes, and
Regulation

Chapter 9

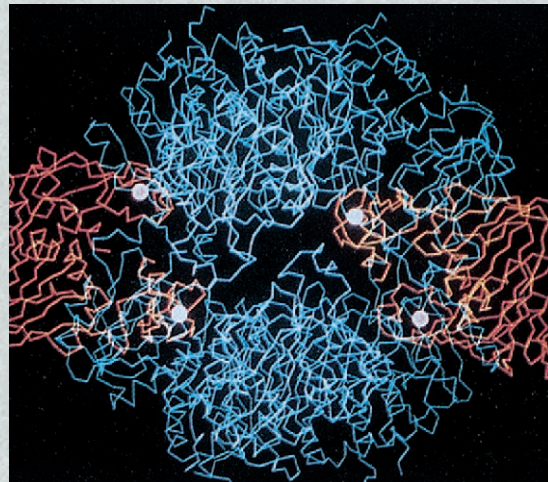
Metabolism: Energy Release and
Conservation

Chapter 10

Metabolism: The Use of Energy in
Biosynthesis

CHAPTER 8

Metabolism: Energy, Enzymes, and Regulation



This diagram shows *E. coli* aspartate carbamoyltransferase in the less active T state. The catalytic polypeptide chains are in blue and the regulatory chains are colored red.

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Concepts

1. Energy is the capacity to do work. Living organisms can perform three major types of work: chemical work, transport work, and mechanical work.
2. Most energy used by living organisms originally comes from sunlight trapped during photosynthesis by photoautotrophs. Chemoheterotrophs then consume autotrophic organic materials and use them as sources of energy and as building blocks.
3. An energy currency is needed to connect energy-yielding exergonic reactions with energy-requiring endergonic reactions. The most commonly used currency is ATP.
4. All living systems obey the laws of thermodynamics.
5. When electrons are transferred from a reductant with a more negative reduction potential to an oxidant with a more positive potential, energy is made available. A reversal of the direction of electron transfer—for example, during photosynthesis—requires energy input.
6. Enzymes are protein catalysts that make life possible by increasing the rate of reactions at ambient temperatures. Enzymes do not change chemical equilibria or violate the laws of thermodynamics but accelerate reactions by lowering their activation energy.

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7. Metabolism is regulated in such a way that (a) cell components are maintained at the proper concentrations, even in the face of a changing environment, and (b) energy and material are conserved.
8. The localization of enzymes and metabolites in separate compartments of a cell regulates and coordinates metabolic activity.
9. The activity of regulatory enzymes may be changed through reversible binding of effectors to a regulatory site separate from the catalytic site or through covalent modification of the enzyme. Regulation of enzyme activity operates rapidly and serves as a fine-tuning mechanism to adjust metabolism from moment to moment.
10. A pathway's activity is often controlled by its end products through feedback inhibition of regulatory enzymes located at the start of the sequence and at branch points.

Living cells are self-regulating chemical engines, tuned to operate on the principle of maximum economy.

—A. L. Lehninger

Chapters 3 and 4 contain many examples of an important principle: that a cell's structure is intimately related to its function. In each instance one can readily relate an organelle's construction to its function (and vice versa). A second unifying principle in biology is that life is sustained by the trapping and use of energy, a process made possible by the action of enzymes. Because this is so crucial to our understanding of microbial function, considerable attention is given to energy and enzymes in this chapter.

The organization of microbial metabolism will be briefly described in chapters 8 to 10. Metabolic pathways are treated as a sequence of enzymes functioning as a unit, with each enzyme using as its substrate a product of the preceding enzyme-catalyzed reaction. This picture of metabolic pathways is incomplete because we will usually ignore the regulation of pathway operation for the sake of space and simplicity. However, one should keep in mind that both regulation of the activity of individual pathways and coordination of the action of separate sequences are essential to the existence of life. Cells become disorganized and die without adequate control of metabolism, and regulation is just as important to life as is the efficient use of energy. Thus the last part of this chapter will be devoted to the regulation of metabolism as a foundation for the subsequent discussion of pathways.

This chapter begins with a brief survey of the nature of energy and the laws of thermodynamics. The participation of energy in metabolism and the role of ATP as an energy currency is considered next. An introduction to the nature and function of enzymes follows. The chapter ends with an overview of metabolic regulation, including an introduction to metabolic channeling and the regulation of the activity of critical enzymes.

8.1 Energy and Work

Energy may be most simply defined as the capacity to do work or to cause particular changes. Thus all physical and chemical processes are the result of the application or movement of en-

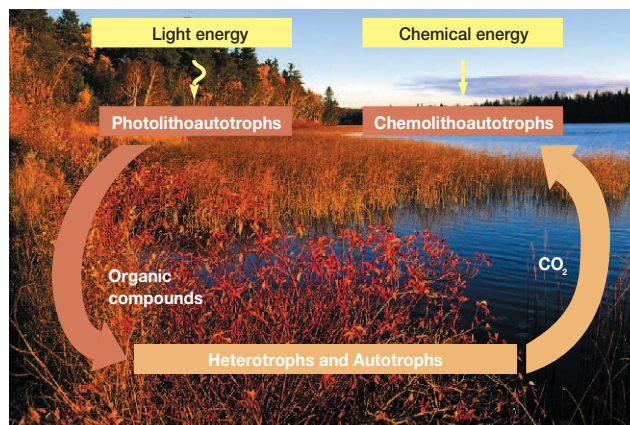


Figure 8.1 The Flow of Carbon and Energy in an Ecosystem. This diagram depicts the flow of energy and carbon in general terms. See text for discussion.

ergy. Living cells carry out three major types of work, and all are essential to life processes. **Chemical work** involves the synthesis of complex biological molecules required by cells from much simpler precursors; energy is needed to increase the molecular complexity of a cell. Molecules and ions often must be transported across cell membranes against an electro-chemical gradient. For example, a molecule sometimes moves into a cell even though its concentration is higher internally. Similarly a solute may be expelled from the cell against a concentration gradient. This process is **transport work** and requires energy input in order to take up nutrients, eliminate wastes, and maintain ion balances. The third type of work is **mechanical work**, perhaps the most familiar of the three. Energy is required to change the physical location of organisms, cells, and structures within cells.

The ultimate source of most biological energy is the visible sunlight impinging on the earth's surface. Light energy is trapped by phototrophs during **photosynthesis**, in which it is absorbed by chlorophyll and other pigments and converted to chemical energy. As noted in chapter 5, chemolithoautotrophs derive energy by oxidizing inorganic compounds rather than obtaining it from light absorption. Chemical energy from photosynthesis and chemolithotrophy can then be used by photolithoautotrophs and chemolithoautotrophs to transform CO₂ into biological molecules such as glucose (**figure 8.1**). [Nutritional types \(pp. 97–98\)](#)

The complex molecules manufactured by autotrophic organisms (both plant and microbial producers) serve as a carbon source for chemoheterotrophs and other consumers that use complex organic molecules as a source of material and energy for building their own cellular structures (it should be remembered that autotrophs also use complex organic molecules). Chemoheterotrophs often employ O₂ as an electron acceptor when oxidizing glucose and other organic molecules to CO₂. This process, in which O₂ acts as the final electron acceptor and is reduced to water, is called **aerobic respiration**. Much energy is released

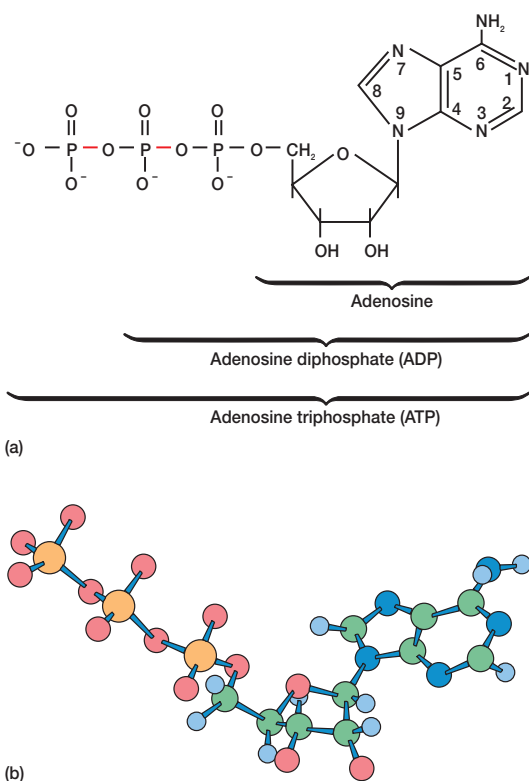


Figure 8.2 Adenosine Triphosphate and Adenosine Diphosphate. (a) Structure of ATP and ADP. The two red bonds are more easily broken or have a high phosphate group transfer potential (see text). The pyrimidine ring atoms have been numbered. (b) A model of ATP. Carbon is in green; hydrogen in light blue; nitrogen in dark blue; oxygen in red; and phosphorus in orange.

during this process. Thus, in the ecosystem, energy is trapped by photoautotrophs and chemolithoautotrophs; some of this energy subsequently flows to chemoheterotrophs when they use nutrients derived from autotrophs (figure 8.1; *see also* figure 28.32). The CO_2 produced during aerobic respiration can be incorporated again into complex organic molecules during photosynthesis and chemolithoautotrophy. Clearly the flow of carbon and energy in the ecosystem is intimately related.

Cells must efficiently transfer energy from their energy-generating or trapping apparatus to the systems actually carrying out work. That is, cells must have a practical form of energy currency. In living organisms the major currency is **adenosine 5'-triphosphate (ATP; figure 8.2)**. When ATP breaks down to **adenosine diphosphate (ADP)** and orthophosphate (P_i), energy is made available for useful work. Later, energy from photosynthesis, aerobic respiration, anaerobic respiration, and fermentation is used to resynthesize ATP from ADP and P_i . An energy cycle is created in the cell (**figure 8.3**). [Fermentation \(pp. 179–81\);](#) [Anaerobic respiration \(pp. 190–91\)](#)

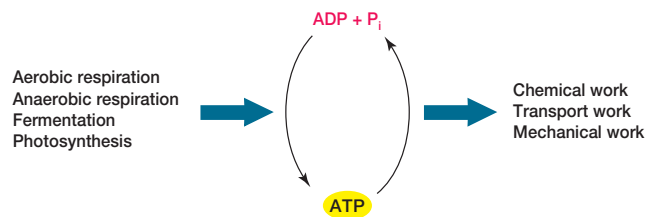


Figure 8.3 The Cell's Energy Cycle. ATP is formed from energy made available during aerobic respiration, anaerobic respiration, fermentation, and photosynthesis. Its breakdown to ADP and phosphate (P_i) makes chemical, transport, and mechanical work possible.

8.2 The Laws of Thermodynamics

To understand how energy is trapped or generated and how ATP functions as an energy currency, some knowledge of the basic principles of thermodynamics is required. The science of **thermodynamics** analyzes energy changes in a collection of matter (e.g., a cell or a plant) called a system. All other matter in the universe is called the surroundings. Thermodynamics focuses on the energy differences between the initial state and the final state of a system. It is not concerned with the rate of the process. For instance, if a pan of water is heated to boiling, only the condition of the water at the start and at boiling is important in thermodynamics, not how fast it is heated or on what kind of stove. Two important laws of thermodynamics must be understood. The **first law of thermodynamics** says that energy can be neither created nor destroyed. The total energy in the universe remains constant although it can be redistributed. For example, many energy exchanges do occur during chemical reactions (e.g., heat is given off by exothermic reactions and absorbed during endothermic reactions), but these heat exchanges do not violate the first law.

It is necessary to specify quantitatively the amount of energy used in or evolving from a particular process, and two types of energy units are employed. A **calorie (cal)** is the amount of heat energy needed to raise one gram of water from 14.5 to 15.5°C. The amount of energy also may be expressed in terms of **joules (J)**, the units of work capable of being done. One cal of heat is equivalent to 4.1840 J of work. One thousand calories or a kilocalorie (kcal) is enough energy to boil 1.9 ml of water. A kilojoule is enough energy to boil about 0.44 ml of water, or enable a person weighing 70 kg to climb 35 steps. The joule is normally used by chemists and physicists. Because biologists most often speak of energy in terms of calories, this text will employ calories when discussing energy changes.

Although it is true that energy is conserved in the universe, the first law of thermodynamics does not account for many physical and chemical processes. A simple example may help make this clear. Suppose a full gas cylinder is connected to an empty one by a tube with a valve (**figure 8.4**). If the valve is opened, gas flows from the full to the empty cylinder until the gas pressure is equal on

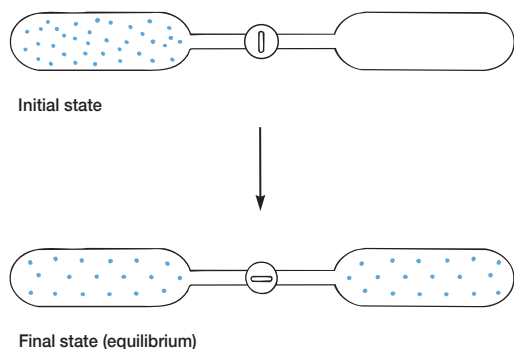


Figure 8.4 A Second Law Process. The expansion of gas into an empty cylinder simply redistributes the gas molecules until equilibrium is reached. The total number of molecules remains unchanged.

both sides. Energy has not only been redistributed but also conserved. The expansion of gas is explained by the **second law of thermodynamics** and a condition of matter called entropy. **Entropy** may be considered a measure of the randomness or disorder of a system. The greater the disorder of a system, the greater is its entropy. The second law states that physical and chemical processes proceed in such a way that the randomness or disorder of the universe (the system and its surroundings) increases to the maximum possible. Gas will always expand into an empty cylinder.

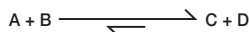
8.3 Free Energy and Reactions

The first and second laws can be combined in a useful equation, relating the changes in energy that can occur in chemical reactions and other processes.

$$\Delta G = \Delta H - T\Delta S$$

ΔG is the change in free energy, ΔH is the change in enthalpy, T is the temperature in Kelvin ($^{\circ}\text{C} + 273$), and ΔS is the change in entropy occurring during the reaction. The change in **enthalpy** is the change in heat content. Cellular reactions occur under conditions of constant pressure and volume. Thus the change in enthalpy is about the same as the change in total energy during the reaction. The **free energy change** is the amount of energy in a system available to do useful work at constant temperature and pressure. Therefore the change in entropy is a measure of the proportion of the total energy change that the system cannot use in performing work. Free energy and entropy changes do not depend on how the system gets from start to finish. A reaction will occur spontaneously at constant temperature and pressure if the free energy of the system decreases during the reaction or, in other words, if ΔG is negative. It follows from the equation that a reaction with a large positive change in entropy will normally tend to have a negative ΔG value and therefore occur spontaneously. A decrease in entropy will tend to make ΔG more positive and the reaction less favorable.

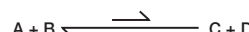
Exergonic reactions



$$K_{\text{eq}} = \frac{[C][D]}{[A][B]} > 1.0$$

$\Delta G^{\circ'}$ is negative.

Endergonic reactions

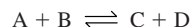


$$K_{\text{eq}} = \frac{[C][D]}{[A][B]} < 1.0$$

$\Delta G^{\circ'}$ is positive.

Figure 8.5 $\Delta G^{\circ'}$ and Equilibrium. The relationship of $\Delta G^{\circ'}$ to the equilibrium of reactions. Note the differences between exergonic and endergonic reactions.

The change in free energy has a definite, concrete relationship to the direction of chemical reactions. Consider the following simple reaction:



If the molecules A and B are mixed, they will combine to form the products C and D. Eventually C and D will become concentrated enough to combine and produce A and B at the same rate as they are formed from A and B. The reaction is now at **equilibrium**: the rates in both directions are equal and no further net change occurs in the concentrations of reactants and products. This situation is described by the **equilibrium constant** (K_{eq}), relating the equilibrium concentrations of products and substrates to one another.

$$K_{\text{eq}} = \frac{[C][D]}{[A][B]}$$

If the equilibrium constant is greater than one, the products are in greater concentration than the reactants at equilibrium—that is, the reaction tends to go to completion as written.

The equilibrium constant of a reaction is directly related to its change in free energy. When the free energy change for a process is determined at carefully defined standard conditions of concentration, pressure, pH, and temperature, it is called the **standard free energy change** (ΔG°). If the pH is set at 7.0 (which is close to the pH of living cells), the standard free energy change is indicated by the symbol $\Delta G^{\circ'}$. The change in standard free energy may be thought of as the maximum amount of energy available from the system for useful work under standard conditions. Using $\Delta G^{\circ'}$ values allows one to compare reactions without worrying about variations in the ΔG due to differences in environmental conditions. The relationship between $\Delta G^{\circ'}$ and K_{eq} is given by the following equation:

$$\Delta G^{\circ'} = -2.303RT \cdot \log K_{\text{eq}}$$

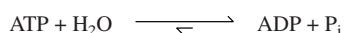
R is the gas constant (1.9872 cal/mole-degree or 8.3145 J/mole-degree), and T is the absolute temperature. Inspection of this equation shows that when $\Delta G^{\circ'}$ is negative, the equilibrium constant is greater than one and the reaction goes to completion as written. It is said to be an **exergonic reaction** (figure 8.5). In an **endergonic reaction** $\Delta G^{\circ'}$ is positive and the equilibrium constant is less than one. That is, the reaction is not favorable, and little product will be formed at equilibrium under standard conditions. Keep in mind

that the $\Delta G^{\circ'}$ value shows only where the reaction lies at equilibrium, not how fast the reaction reaches equilibrium.

1. What is energy and what kinds of work are carried out in a cell? Describe the energy cycle and ATP's role in it.
2. What is thermodynamics? Summarize the first and second laws. Define free energy, entropy, and enthalpy.
3. How is the change in standard free energy related to the equilibrium constant for a reaction? What are exergonic and endergonic reactions?

8.4 The Role of ATP in Metabolism

Many reactions in the cell are endergonic and will not proceed far toward completion without outside assistance. One of ATP's major roles is to drive such endergonic reactions more to completion. ATP is a **high-energy molecule**. That is, it breaks down or hydrolyzes almost completely to the products ADP and P_i with a $\Delta G^{\circ'}$ of -7.3 kcal/mole.



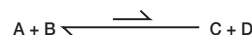
In reference to ATP the term high-energy molecule does not mean that there is a great deal of energy stored in a particular bond of ATP. It simply indicates that the removal of the terminal phosphate goes to completion with a large negative standard free energy change, or the reaction is strongly exergonic. In other words, ATP has a high **phosphate group transfer potential**; it readily transfers its phosphate to water. The phosphate group transfer potential is defined as the negative of $\Delta G^{\circ'}$ for the hydrolytic removal of phosphate. A molecule with a higher group transfer potential will donate phosphate to one with a lower potential.

Thus ATP is ideally suited for its role as an energy currency. It is formed in energy-trapping and -generating processes such as photosynthesis, fermentation, and aerobic respiration. In the cell's economy, exergonic ATP breakdown is coupled with various endergonic reactions to promote their completion (**figure 8.6**). In other words ATP links energy-generating reactions, which liberate free energy, with energy-using reactions, which require free energy input to proceed toward completion. Facilitation of chemical work is the focus of the preceding example, but the same principles apply when ATP is coupled with endergonic processes involving transport work and mechanical work (**figure 8.3**).

8.5 Oxidation-Reduction Reactions and Electron Carriers

Free energy changes are not only related to the equilibria of "regular" chemical reactions but also to the equilibria of oxidation-reduction reactions. The release of energy normally involves oxidation-reduction reactions. **Oxidation-reduction (redox) reactions** are those in which electrons move from a donor, the **reducing agent** or **reductant**, to an electron acceptor, the

Endergonic reaction alone

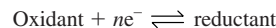


Endergonic reaction coupled to ATP breakdown

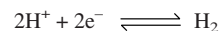


Figure 8.6 ATP as a Coupling Agent. The use of ATP to make endergonic reactions more favorable. It is formed by exergonic reactions and then used to drive endergonic reactions.

oxidizing agent or **oxidant**. By convention such a reaction is written with the reductant to the right of the oxidant and the number (n) of electrons (e^-) transferred.



The oxidant and reductant pair is referred to as a redox couple (**table 8.1**). When an oxidant accepts electrons, it becomes the reductant of the couple. The equilibrium constant for the reaction is called the **standard reduction potential** (E'_0) and is a measure of the tendency of the reducing agent to lose electrons. The reference standard for reduction potentials is the hydrogen system with an E'_0 (the reduction potential at pH 7.0) of -0.42 volts or -420 millivolts.

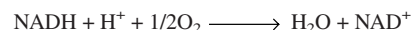


In this reaction each hydrogen atom provides one proton (H^+) and one electron (e^-).

The reduction potential has a concrete meaning. Redox couples with more negative reduction potentials will donate electrons to couples with more positive potentials and greater affinity for electrons. Thus electrons will tend to move from reductants at the top of the list in **table 8.1** to oxidants at the bottom because they have more positive potentials. This may be expressed visually in the form of an electron tower in which the most negative reduction potentials are at the top (**figure 8.7**). Electrons move from donors to acceptors down the potential gradient or fall down the tower to more positive potentials. Consider the case of the electron carrier **nicotinamide adenine dinucleotide** (NAD^+). The NAD^+/NADH couple has a very negative E'_0 and can therefore give electrons to many acceptors, including O_2 .



Because NAD^+/NADH is more negative than $1/2 \text{O}_2/\text{H}_2\text{O}$, electrons will flow from NADH (the reductant) to O_2 (the oxidant) as shown in **figure 8.7**.



When electrons move from a reductant to an acceptor with a more positive redox potential, free energy is released. The $\Delta G^{\circ'}$ of the reaction is directly related to the magnitude of the difference

Table 8.1 Selected Biologically Important Redox Couples

Redox Couple	E'_0 (Volts) ^a
$2\text{H}^+ + 2\text{e}^- \longrightarrow \text{H}_2$	-0.42
Ferredoxin(Fe^{3+}) + $\text{e}^- \longrightarrow$ ferredoxin(Fe^{2+})	-0.42
$\text{NAD(P)}^+ + \text{H}^+ + 2\text{e}^- \longrightarrow \text{NAD(P)H}$	-0.32
$\text{S} + 2\text{H}^+ + 2\text{e}^- \longrightarrow \text{H}_2\text{S}$	-0.274
Acetaldehyde + $2\text{H}^+ + 2\text{e}^- \longrightarrow$ ethanol	-0.197
Pyruvate ⁻ + $2\text{H}^+ + 2\text{e}^- \longrightarrow$ lactate ²⁻	-0.185
$\text{FAD} + 2\text{H}^+ + 2\text{e}^- \longrightarrow \text{FADH}_2$	-0.18 ^b
Oxaloacetate ²⁻ + $2\text{H}^+ + 2\text{e}^- \longrightarrow$ malate ²⁻	-0.166
Fumarate ²⁻ + $2\text{H}^+ + 2\text{e}^- \longrightarrow$ succinate ²⁻	0.031
Cytochrome <i>b</i> (Fe^{3+}) + $\text{e}^- \longrightarrow$ cytochrome <i>b</i> (Fe^{2+})	0.075
Ubiquinone + $2\text{H}^+ + 2\text{e}^- \longrightarrow$ ubiquinol	0.10
Cytochrome <i>c</i> (Fe^{3+}) + $\text{e}^- \longrightarrow$ cytochrome <i>c</i> (Fe^{2+})	0.254
$\text{NO}_3^- + 2\text{H}^+ + 2\text{e}^- \longrightarrow \text{NO}_2^- + \text{H}_2\text{O}$	0.421
$\text{NO}_2^- + 8\text{H}^+ + 6\text{e}^- \longrightarrow \text{NH}_4^+ + 2\text{H}_2\text{O}$	0.44
$\text{Fe}^{3+} + \text{e}^- \longrightarrow \text{Fe}^{2+}$	0.771
$\text{O}_2 + 4\text{H}^+ + 4\text{e}^- \longrightarrow 2\text{H}_2\text{O}$	0.815

^a E'_0 is the standard reduction potential at pH 7.0.

^bThe value for FAD/FADH₂ applies to the free cofactor because it can vary considerably when bound to an apoenzyme.

between the reduction potentials of the two couples ($\Delta E'_0$). The larger the $\Delta E'_0$, the greater the amount of free energy made available, as is evident from the equation

$$\Delta G^{\circ'} = -nF \cdot \Delta E'_0$$

in which n is the number of electrons transferred and F is the Faraday constant (23,062 cal/mole-volt or 96,494 J/mole-volt). For every 0.1 volt change in $\Delta E'_0$, there is a corresponding 4.6 kcal change in $\Delta G^{\circ'}$ when a two-electron transfer takes place. This is similar to the relationship of $\Delta G^{\circ'}$ and K_{eq} in other chemical reactions—the larger the equilibrium constant, the greater the $\Delta G^{\circ'}$. The difference in reduction potentials between NAD^+/NADH and $1/2\text{O}_2/\text{H}_2\text{O}$ is 1.14 volts, a large $\Delta E'_0$ value. When electrons move from NADH to O_2 during aerobic respiration, a large amount of free energy is made available to synthesize ATP (figure 8.8). If energy is released when electrons flow from negative to positive reduction potentials, then an input of energy is required to move electrons in the opposite direction, from more positive to more negative potentials. This is precisely what occurs during photosynthesis (figure 8.8). Light energy is trapped and used to move electrons from water to the electron carrier **nicotinamide adenine dinucleotide phosphate (NADP⁺)**.

The cycle of energy flow discussed earlier and illustrated in figure 8.1 can be understood from a different perspective, if the preceding concept is kept in mind. Photosynthetic organisms capture light energy and use it to move electrons from water (and other electron donors such as H_2S) to electron acceptors, such as

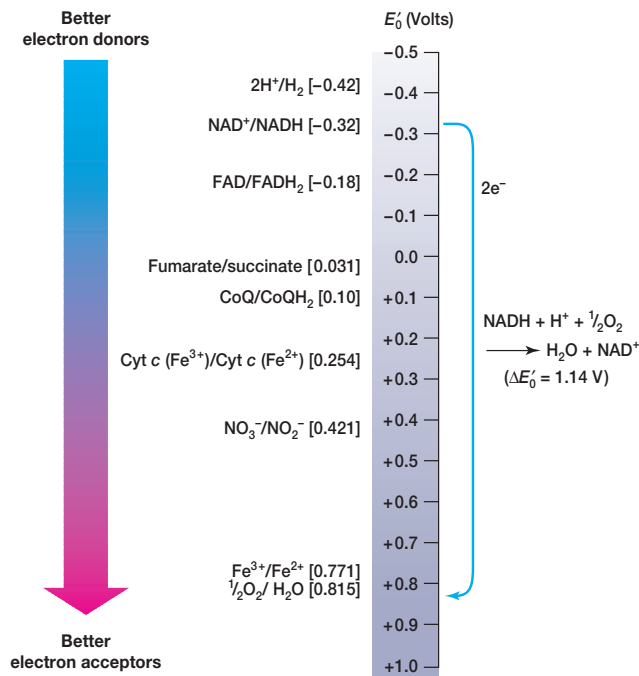


Figure 8.7 Electron Movement and Reduction Potentials. The vertical electron tower in this illustration has the most negative reduction potentials at the top. Electrons will spontaneously move from donors higher on the tower (more negative potentials) to acceptors lower on the tower (more positive potentials). That is, the reductant is always higher on the tower than the oxidant. For example, NADH will donate electrons to oxygen and form water in the process. Some typical donors and acceptors are shown on the left, and their redox potentials are given in brackets.

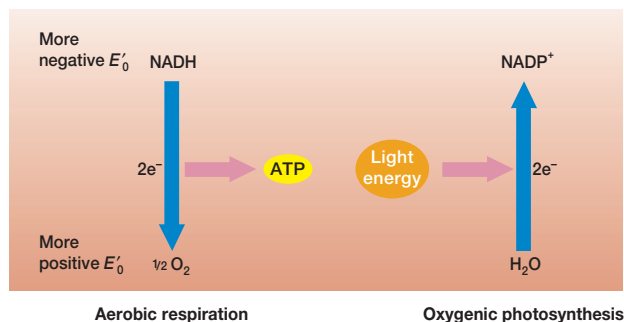
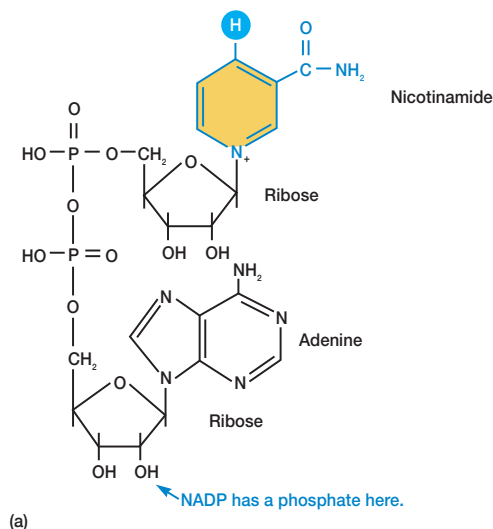
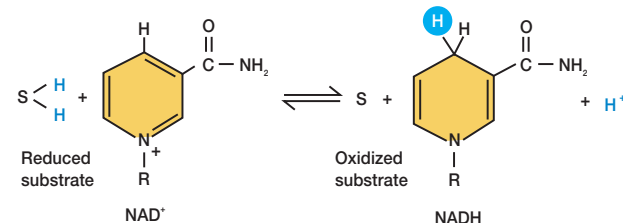


Figure 8.8 Energy Flow in Metabolism. Examples of the relationship between electron flow and energy in metabolism. Oxygen and NADP^+ serve as electron acceptors for NADH and water, respectively.

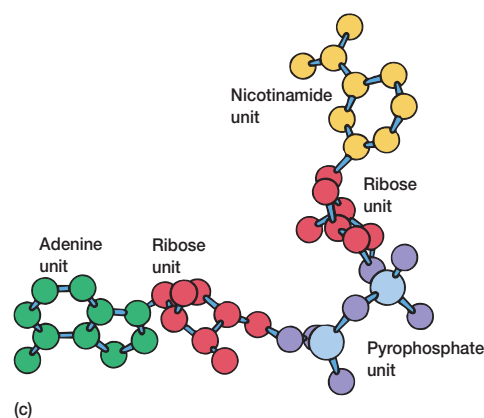


(a)

Figure 8.9 The Structure and Function of NAD. (a) The structure of NAD and NADP. NADP differs from NAD in having an extra phosphate on one of its ribose sugar units. (b) NAD can accept electrons and a hydrogen from a reduced substrate (SH_2). These are carried on the nicotinamide ring. (c) Model of NAD^+ when bound to the enzyme lactate dehydrogenase.



(b)

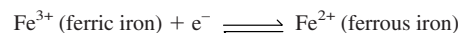


(c)

NADP^+ , that have more negative reduction potentials. These electrons can then flow back to more positive acceptors and provide energy for ATP production during photosynthesis. Photoautotrophs use ATP and NADPH to synthesize complex molecules from CO_2 (see section 10.2). Chemoheterotrophs also make use of energy released during the movement of electrons by oxidizing complex nutrients during respiration to produce NADH. NADH subsequently donates its electrons to O_2 , and the energy released during electron transfer is trapped in the form of ATP. The energy from sunlight is made available to all living organisms because of this relationship between electron flow and energy. [Photosynthesis \(pp. 195–201\); Respiration and electron transport \(pp. 184–89\)](#)

Electron transport is important in aerobic respiration, anaerobic respiration, chemolithotrophy, and photosynthesis. Electron movement in cells requires the participation of carriers such as NAD^+ and NADP^+ , both of which can transport electrons between different locations. The nicotinamide ring of NAD^+ and NADP^+ (figure 8.9) accepts two electrons and one proton from a donor, while a second proton is released. There are several other electron carriers of importance in microbial metabolism (table 8.1), and they carry electrons in a variety of ways. **Flavin adenine dinucleotide (FAD)** and **flavin mononucleotide (FMN)** bear two electrons and two protons on the complex ring system shown in figure 8.10. Proteins bearing FAD and FMN are often called flavoproteins. **Coenzyme Q (CoQ)** or **ubiquinone** is a quinone that transports electrons and protons in many respiratory electron transport chains (figure 8.11). **Cytochromes** and several other

carriers use iron atoms to transport electrons by reversible oxidation and reduction reactions.



In the cytochromes these iron atoms are part of a heme group (figure 8.12) or other similar iron-porphyrin rings. Several different cytochromes, each of which consists of a protein and an iron-porphyrin ring, are a prominent part of respiratory electron transport chains. Some iron containing electron-carrying proteins lack a heme group and are called **nonheme iron proteins**. **Ferredoxin** is a nonheme iron protein active in photosynthetic electron transport and several other electron transport processes. Even though its iron atoms are not bound to a heme group, they still undergo reversible oxidation and reduction reactions. Although all the previously discussed molecules function in electron transport chains, some bear two electrons (NAD, FAD, and CoQ) while others carry only one electron at a time (cytochromes and nonheme iron proteins). This difference in the number of electrons carried is of great importance in the operation of electron transport chains (see pp. 184–87).

1. Why is ATP called a high-energy molecule? What is its role in the cell and how does it fulfill this role?
2. Write a generalized equation for a redox reaction. Define reductant, oxidant, and standard reduction potential.
3. How is the direction of electron flow between redox couples related to the standard reduction potential and the release of free energy? Name and briefly describe the major electron carriers found in cells.

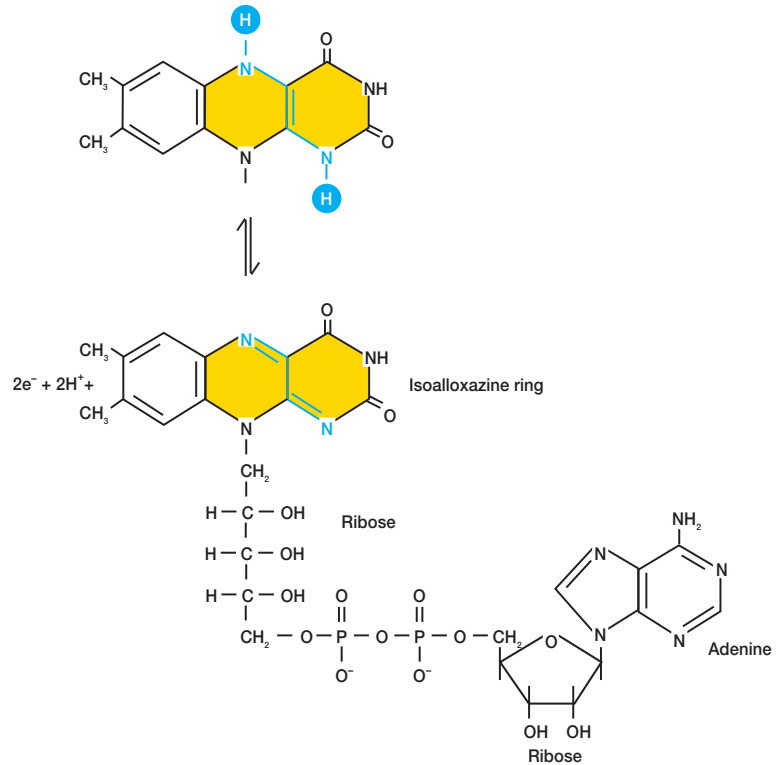


Figure 8.10 The Structure and Function of FAD. The vitamin riboflavin is composed of the isoalloxazine ring and its attached ribose sugar. FMN is riboflavin phosphate. The portion of the ring directly involved in oxidation-reduction reactions is in color.

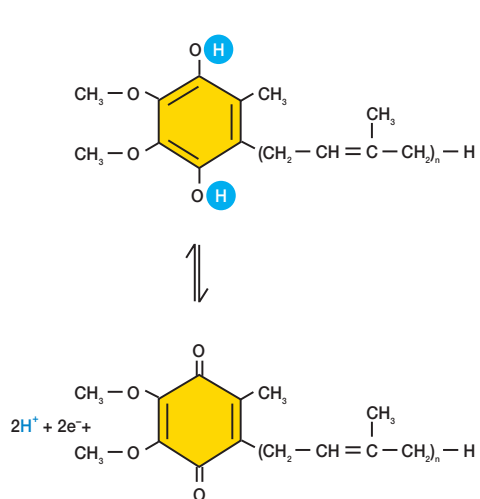


Figure 8.11 The Structure and Function of Coenzyme Q or Ubiquinone. The length of the side chain varies among organisms from $n = 6$ to $n = 10$.

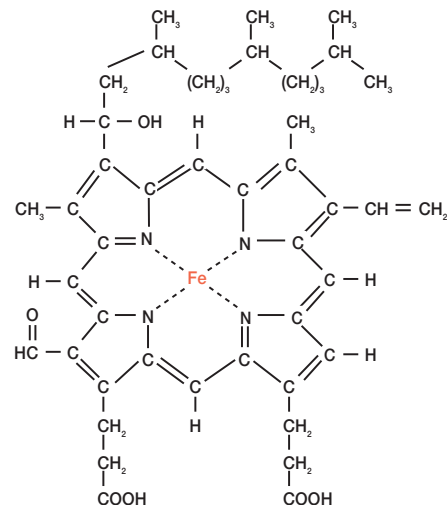


Figure 8.12 The Structure of Heme. Heme is composed of a porphyrin ring and an attached iron atom. It is the nonprotein component of many cytochromes. The iron atom alternatively accepts and releases an electron.

Table 8.2 Enzyme Classification

Type of Enzyme	Reaction Catalyzed by Enzyme	Example of Reaction
Oxidoreductase	Oxidation-reduction reactions	Lactate dehydrogenase: $\text{Pyruvate} + \text{NADH} + \text{H}^+ \rightleftharpoons \text{lactate} + \text{NAD}^+$
Transferase	Reactions involving the transfer of groups between molecules	Aspartate carbamoyltransferase: $\text{Aspartate} + \text{carbamoylphosphate} \rightleftharpoons \text{carbamoylaspartate} + \text{phosphate}$
Hydrolase	Hydrolysis of molecules	Glucose-6-phosphatase: $\text{Glucose 6-phosphate} + \text{H}_2\text{O} \rightarrow \text{glucose} + \text{P}_i$
Lyase	Removal of groups to form double bonds or addition of groups to double bonds	Fumarate hydratase: $\text{L-malate} \rightleftharpoons \text{fumarate} + \text{H}_2\text{O}$
Isomerase	Reactions involving isomerizations	Alanine racemase: $\text{L-alanine} \rightleftharpoons \text{D-alanine}$
Ligase	Joining of two molecules using ATP energy (or that of other nucleoside triphosphates)	Glutamine synthetase: $\text{Glutamate} + \text{NH}_3 + \text{ATP} \rightarrow \text{glutamine} + \text{ADP} + \text{P}_i$

8.6 Enzymes

Recall that an exergonic reaction is one with a negative ΔG° and an equilibrium constant greater than one. An exergonic reaction will proceed to completion in the direction written (that is, toward the right of the equation). Nevertheless, one often can combine the reactants for an exergonic reaction with no obvious result, even though products should be formed. It is precisely in these reactions that enzymes play their part.

Structure and Classification of Enzymes

Enzymes may be defined as protein catalysts that have great specificity for the reaction catalyzed and the molecules acted on. A **catalyst** is a substance that increases the rate of a chemical reaction without being permanently altered itself. Thus enzymes speed up cellular reactions. The reacting molecules are called **substrates**, and the substances formed are the **products**. [Protein structure and properties \(appendix I\)](#)

Many enzymes are indeed pure proteins. However, many enzymes consist of a protein, the **apoenzyme**, and also a non-protein component, a **cofactor**, required for catalytic activity. The complete enzyme consisting of the apoenzyme and its cofactor is called the **holoenzyme**. If the cofactor is firmly attached to the apoenzyme it is a **prosthetic group**. Often the cofactor is loosely attached to the apoenzyme. It can even dissociate from the enzyme protein after products have been formed and carry one of these products to another enzyme (**figure 8.13**). Such a loosely bound cofactor is called a **coenzyme**. For example, NAD^+ is a coenzyme that carries electrons within the cell. Many vitamins that humans require serve as coenzymes or as their precursors. Niacin is incorporated into NAD^+ and riboflavin into FAD. Metal ions may also be bound to apoenzymes and act as cofactors.

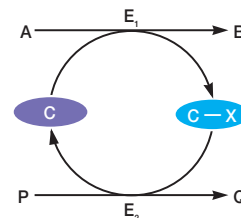
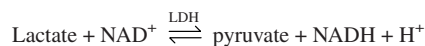


Figure 8.13 Coenzymes as Carriers. The function of a coenzyme in carrying substances around the cell. Coenzyme C participates with enzyme E_1 in the conversion of A to product B. During the reaction, it acquires X from the substrate A. The coenzyme can donate X to substrate P in a second reaction. This will convert it back to its original form, ready to accept another X. The coenzyme is not only participating in both reactions, but is also transporting X to various points in the cell.

Despite the large number and bewildering diversity of enzymes present in cells, they may be placed in one of six general classes (**table 8.2**). Enzymes usually are named in terms of the substrates they act on and the type of reaction catalyzed. For example, lactate dehydrogenase (LDH) removes hydrogens from lactate.



Lactate dehydrogenase can also be given a more complete and detailed name, L-lactate: NAD oxidoreductase. This name describes the substrates and reaction type with even more precision.

The Mechanism of Enzyme Reactions

It is important to keep in mind that enzymes increase the rates of reactions but do not alter their equilibrium constants. If a reaction is endergonic, the presence of an enzyme will not shift its equilibrium

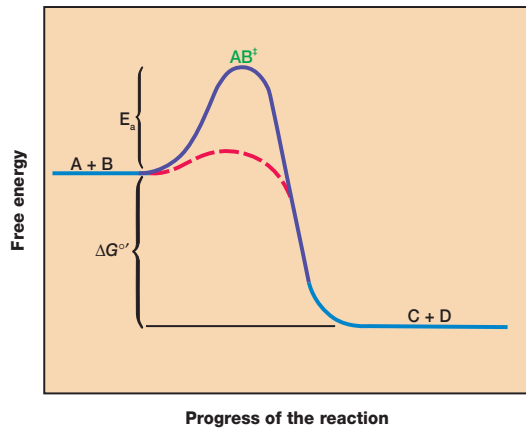
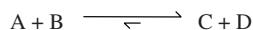


Figure 8.14 Enzymes Lower the Energy of Activation. This figure traces the course of a chemical reaction in which A and B are converted to C and D. The transition-state complex is represented by AB^+ , and the activation energy required to reach it, by E_a . The colored line represents the course of the reaction in the presence of an enzyme. Note that the activation energy is much lower in the enzyme-catalyzed reaction.

so that more products can be formed. Enzymes simply speed up the rate at which a reaction proceeds toward its final equilibrium.

How do enzymes catalyze reactions? Although a complete answer would be long and complex, some understanding of the mechanism can be gained by considering the course of a normal exergonic chemical reaction.



When molecules A and B approach each other to react, they form a **transition-state complex**, which resembles both the substrates and the products (figure 8.14). The **activation energy** is required to bring the reacting molecules together in the correct way to reach the transition state. The transition-state complex can then decompose to yield the products C and D. The difference in free energy level between reactants and products is $\Delta G^o'$. Thus the equilibrium in our example will lie toward the products because $\Delta G^o'$ is negative (i.e., the products are at a lower energy level than the substrates).

Clearly A and B will not be converted to C and D in figure 8.14 if they are not supplied with an amount of energy equivalent to the activation energy. Enzymes accelerate reactions by lowering the activation energy; therefore more substrate molecules will have sufficient energy to come together and form products. Even though the equilibrium constant (or $\Delta G^o'$) is unchanged, equilibrium will be reached more rapidly in the presence of an enzyme because of this decrease in the activation energy.

Researchers have expended much effort in discovering how enzymes lower the activation energy of reactions, and the process is becoming clearer. Enzymes bring substrates together at a special place on their surface called the **active site** or **catalytic site** to form an **enzyme-substrate complex** (figures 8.15, 8.16; see also *AI.19*). The enzyme can interact with a substrate in two general

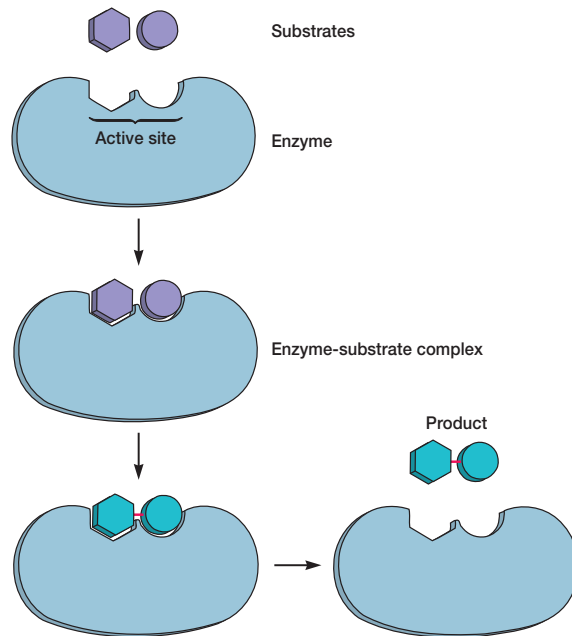


Figure 8.15 Enzyme Function. The formation of the enzyme-substrate complex and its conversion to products is shown.

ways. It may be rigid and shaped to precisely fit the substrate so that the correct substrate binds specifically and is positioned properly for reaction. This mechanism is referred to as the **lock-and-key model**. An enzyme also may change shape when it binds the substrate so that the active site surrounds and precisely fits the substrate. This has been called the **induced fit model** and is used by hexokinase and many other enzymes (figure 8.16). The formation of an enzyme-substrate complex can lower the activation energy in many ways. For example, by bringing the substrates together at the active site, the enzyme is, in effect, concentrating them and speeding up the reaction. An enzyme does not simply concentrate its substrates, however. It also binds them so that they are correctly oriented with respect to each other in order to form a transition-state complex. Such an orientation lowers the amount of energy that the substrates require to reach the transition state. These and other catalytic site activities speed up a reaction hundreds of thousands of times, even though microorganisms grow between -20°C and approximately 113°C . These temperatures are not high enough to favor most organic reactions in the absence of enzyme catalysis, yet cells cannot survive at the high temperatures used by an organic chemist in routine organic syntheses. Enzymes make life possible by accelerating specific reactions at low temperatures.

The Effect of Environment on Enzyme Activity

Enzyme activity varies greatly with changes in environmental factors, one of the most important being the substrate concentration. As will be emphasized later, substrate concentrations are

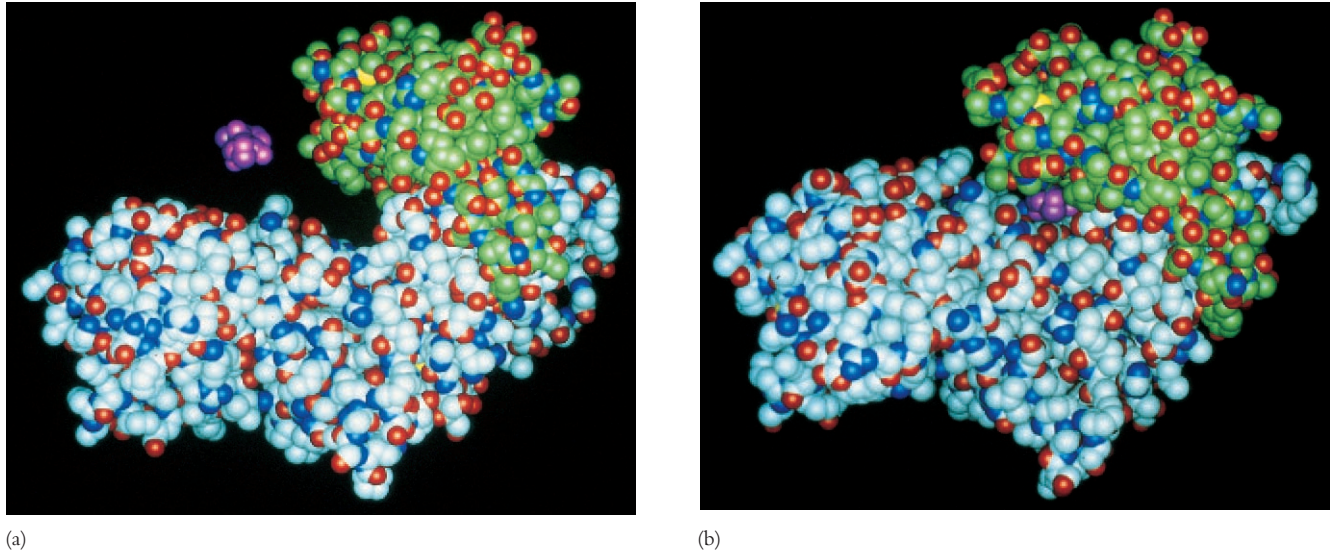
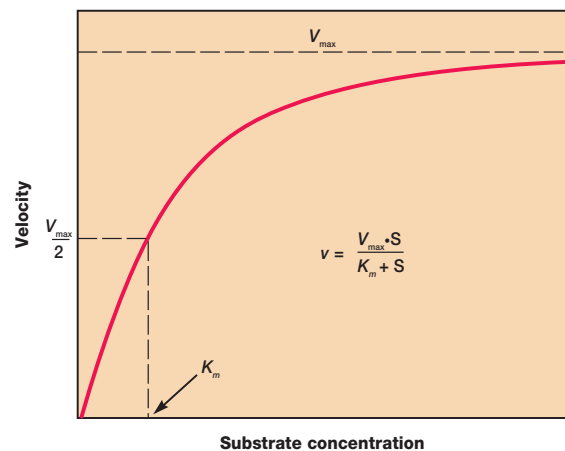


Figure 8.16 An Example of Enzyme-Substrate Complex Formation. (a) A space-filling model of yeast hexokinase and its substrate glucose (purple). The active site is in the cleft formed by the enzyme's small lobe (green) and large lobe (gray). (b) When glucose binds to form the enzyme-substrate complex, hexokinase changes shape and surrounds the substrate.

usually low within cells. At very low substrate concentrations, an enzyme makes product slowly because it seldom contacts a substrate molecule. If more substrate molecules are present, an enzyme binds substrate more often, and the reaction velocity (usually expressed in terms of the rate of product formation) is greater than at a lower substrate concentration. Thus the rate of an enzyme-catalyzed reaction increases with substrate concentration (figure 8.17). Eventually further increases in substrate concentration do not result in a greater reaction velocity because the available enzyme molecules are binding substrate and converting it to product as rapidly as possible. That is, the enzyme is saturated with substrate and operating at maximal velocity ($V_{\max}$). The resulting substrate concentration curve is a hyperbola (figure 8.17). It is useful to know the substrate concentration an enzyme needs to function adequately. Usually the **Michaelis constant** (K_m), the substrate concentration required for the enzyme to achieve half maximal velocity, is used as a measure of the apparent affinity of an enzyme for its substrate. The lower the K_m value, the lower the substrate concentration at which an enzyme catalyzes its reaction.

Enzymes also change activity with alterations in pH and temperature (figure 8.18). Each enzyme functions most rapidly at a specific pH optimum. When the pH deviates too greatly from an enzyme's optimum, activity slows and the enzyme may be damaged. Enzymes likewise have temperature optima for maximum activity. If the temperature rises too much above the optimum, an enzyme's structure will be disrupted and its activity lost. This phenomenon, known as **denaturation**, may be caused by



K_m = the substrate concentration required by the enzyme to operate at half its maximum velocity

$V_{\max}$ = the rate of product formation when the enzyme is saturated with substrate and operating as fast as possible

Figure 8.17 Michaelis-Menten Kinetics. The dependence of enzyme activity upon substrate concentration. This substrate curve fits the Michaelis-Menten equation given in the figure, which relates reaction velocity (v) to the substrate concentration (S) using the maximum velocity and the Michaelis constant (K_m).

Figure 8.18 pH, Temperature, and Enzyme Activity. The variation of enzyme activity with changes in pH and temperature. The ranges in pH and temperature are only representative. Enzymes differ from one another with respect to the location of their optima and the shape of their pH and temperature curves.

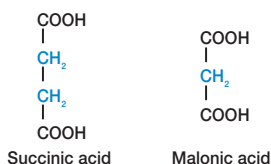
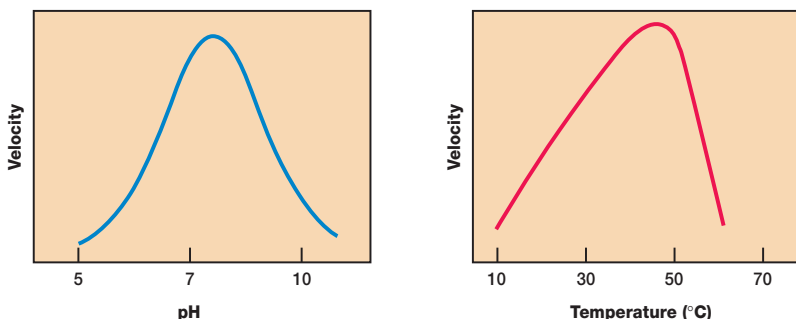


Figure 8.19 Competitive Inhibition of Succinate Dehydrogenase. A comparison of succinic acid and the competitive inhibitor, malonic acid. The colored atoms indicate the parts of the two molecules that differ.

extremes of pH and temperature or by other factors. The pH and temperature optima of a microorganism's enzymes often reflect the pH and temperature of its habitat. Not surprisingly bacteria growing best at high temperatures often have enzymes with high temperature optima and great heat stability. [Temperature and growth](#) (pp. 125–27); [Heat stable enzymes in biotechnology](#) (p. 626)

Enzyme Inhibition

Microorganisms can be poisoned by a variety of chemicals, and many of the most potent poisons are enzyme inhibitors. A **competitive inhibitor** directly competes with the substrate at an enzyme's catalytic site and prevents the enzyme from forming product. A classic example of this behavior is seen with the enzyme succinate dehydrogenase, which catalyzes the oxidation of succinate to fumarate in the tricarboxylic acid cycle (*see section 9.4*). Malonic acid is an effective competitive inhibitor of succinate dehydrogenase because it so closely resembles succinate, the normal substrate (**figure 8.19**). After malonate binds to the enzyme, it cannot be oxidized and the formation of fumarate is blocked. Competitive inhibitors usually resemble normal substrates, but they cannot be converted to products.

Competitive inhibitors are important in the treatment of many microbial diseases. Sulfa drugs like sulfanilamide resemble *p*-aminobenzoate, a molecule used in the formation of the coenzyme folic acid. The drugs compete with *p*-aminobenzoate for the catalytic site of an enzyme involved in folic acid synthesis. This blocks the production of folic acid and inhibits bacterial growth (*see section 35.6*). Humans are not harmed because they cannot synthesize folic acid and must obtain it in their diet. [Destruction of microorganisms by physical and chemical agents](#) (chapter 7)

Inhibitors also can affect enzyme activity by binding to the enzyme at some location other than at the active site. This alters the enzyme's shape, rendering it inactive or less active. These inhibitors are often called **noncompetitive** because they do not directly compete with the substrate. Heavy metal poisons like mercury frequently are noncompetitive inhibitors of enzymes.

1. What is an enzyme and how does it speed up reactions? How are enzymes named? Define apoenzyme, holoenzyme, cofactor, coenzyme, prosthetic group, active or catalytic site, and activation energy.
2. How does enzyme activity change with substrate concentration, pH, and temperature? Define the terms Michaelis constant, maximum velocity, and denaturation.
3. What are competitive and noncompetitive inhibitors and how do they inhibit enzymes?

8.7 The Nature and Significance of Metabolic Regulation

The task of the regulatory machinery is exceptionally complex and difficult. Pathways must be regulated and coordinated so effectively that all cell components are present in precisely the correct amounts. Furthermore, a microbial cell must be able to respond effectively to environmental changes by using those nutrients present at the moment and by switching on new catabolic pathways when different nutrients become available. Because all chemical components of a cell usually are not present in the surroundings, microorganisms also must synthesize unavailable components and alter biosynthetic activity in response to changes in nutrient availability. The chemical composition of a cell's surroundings is constantly changing, and these regulatory processes are dynamic and continuously responding to altered conditions.

Regulation is essential for the cell to conserve microbial energy and material and to maintain metabolic balance. If a particular energy source is unavailable, the enzymes required for its use are not needed and their further synthesis is a waste of carbon, nitrogen, and energy. Similarly it would be extremely wasteful for a microorganism to synthesize the enzymes required to manufacture a certain end product if that end product were already pres-

ent in adequate amounts. Thus both catabolism and anabolism are regulated in such a way as to maximize efficiency of operation.

[Catabolism and anabolism \(p. 173\)](#)

The drive to maintain balance and conserve energy and material is evident in the regulatory responses of a bacterium like *E. coli*. If the bacterium is grown in a very simple medium containing only glucose as a carbon and energy source, it will synthesize the required cell components in balanced amounts. Addition of a biosynthetic end product (the amino acid tryptophan, for example) to the medium will result in the immediate inhibition of the pathway synthesizing that end product; synthesis of the pathway's enzymes also will slow or cease. If *E. coli* is transferred to medium containing only the sugar lactose, it will synthesize the enzymes required for catabolism of this nutrient. In contrast, when *E. coli* grows in a medium possessing both glucose and lactose, glucose (the sugar supporting most rapid growth) is catabolized first. The culture will use lactose only after the glucose supply has been exhausted.

The flow of carbon through a pathway may be regulated in three major ways.

1. The localization of metabolites and enzymes in different parts of a cell, a phenomenon called **metabolic channeling**, influences pathway activity.
2. Critical enzymes often are directly stimulated or inhibited to alter pathway activity rapidly.
3. The number of enzyme molecules also may be controlled. The more catalyst molecules present, the greater the pathway's activity. In bacteria regulation is usually exerted at the level of transcription. Control of mRNA synthesis is slower than direct regulation of enzyme activity but does result in the saving of much energy and raw material because enzymes are not synthesized when not required.

Each of these mechanisms is described in detail. This chapter introduces the first two: metabolic channeling and direct control of enzyme activity. The discussion of gene expression regulation follows a description of DNA, RNA, and protein synthesis in chapters 11 and 12. [Regulation of gene expression \(pp. 275–83\)](#)

8.8 Metabolic Channeling

One of the most common channeling mechanisms is that of **compartmentation**, the differential distribution of enzymes and metabolites among separate cell structures or organelles. Compartmentation is particularly important in eucaryotic microorganisms with their many membrane-bound organelles. For example, fatty acid oxidation is located within the mitochondrion, whereas fatty acid synthesis occurs in the cytoplasmic matrix. The periplasm in procaryotes can also be considered an example of compartmentation (*see p. 55*). Compartmentation makes possible the simultaneous, but separate, operation and regulation of similar pathways. Furthermore, pathway activities can be coordinated through regulation of the transport of metabolites and coenzymes between cell compartments. Suppose two pathways in different cell compartments require NAD for activity. The distribu-

tion of NAD between the two compartments will then determine the relative activity of these competing pathways, and the pathway with access to the most NAD will be favored.

Channeling also occurs within compartments such as the cytoplasmic matrix. The matrix is a structured dense material with many subcompartments. In eucaryotes it also is subdivided by the endoplasmic reticulum and cytoskeleton (*see chapter 4*). Metabolites and coenzymes do not diffuse rapidly in such an environment, and metabolite gradients will build up near localized enzymes or enzyme systems. This occurs because enzymes at a specific site convert their substrates to products, resulting in decreases in the concentration of one or more metabolites and increases in others. For example, product concentrations will be high near an enzyme and decrease with increasing distance from it.

Channeling can generate marked variations in metabolite concentrations and therefore directly affect enzyme activity. Substrate levels are generally around 10^{-3} moles/liter (M) to 10^{-6} M or even lower. Thus they may be in the same range as enzyme concentrations and equal to or less than the Michaelis constants (K_m) of many enzymes. Under these conditions the concentration of an enzyme's substrate may control its activity because the substrate concentration is in the rising portion of the hyperbolic substrate saturation curve (**figure 8.20**). As the substrate level increases, it is converted to product more rapidly; a decline in substrate concentration automatically leads to lower enzyme activity. If two enzymes in different pathways use the same metabolite, they may directly compete for it. The pathway winning this competition—the one with the enzyme having the lowest K_m value for the metabolite—will operate closer to full capacity. Thus channeling within a cell compartment can regulate and coordinate metabolism through variations in metabolite and coenzyme levels. [Enzyme kinetics and the substrate saturation curve \(pp. 162–63\)](#)

-
1. Give three ways in which the flow of carbon through a pathway may be regulated.
 2. Define the terms metabolic channeling and compartmentation. How are they involved in the regulation of metabolism?
-

8.9 Control of Enzyme Activity

Adjustment of the activity of regulatory enzymes controls the functioning of many metabolic pathways. This section describes these enzymes and discusses their role in regulating pathway activity.

Allosteric Regulation

Usually regulatory enzymes are **allosteric enzymes**. The activity of an allosteric enzyme is altered by a small molecule known as an **effector** or **modulator**. The effector binds reversibly by non-covalent forces to a **regulatory site** separate from the catalytic site and causes a change in the shape or conformation of the enzyme (**figure 8.21**). The activity of the catalytic site is altered as

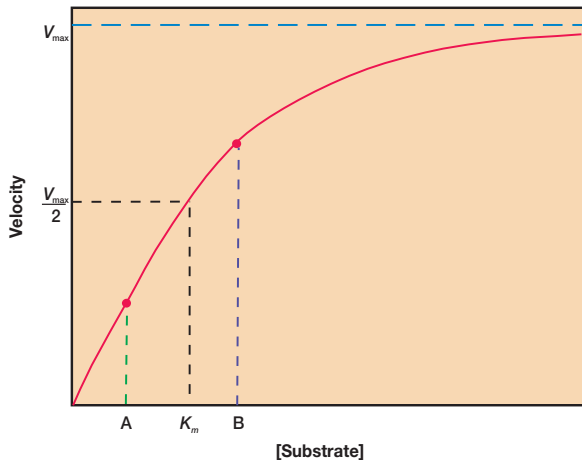


Figure 8.20 Control of Enzyme Activity by Substrate Concentration. An enzyme-substrate saturation curve with the Michaelis constant (K_m) and the velocity equivalent to half the maximum velocity (V_{max}) indicated. The initial velocity of the reaction (v) is plotted against the substrate concentration [Substrate]. The maximum velocity is the greatest velocity attainable with a fixed amount of enzyme under defined conditions. When the substrate concentration is equal to or less than the K_m , the enzyme's activity will vary almost linearly with the substrate concentration. Suppose the substrate increases in concentration from level A to B. Because these concentrations are in the range of the K_m , a significant increase in enzyme activity results. A drop in concentration from B to A will lower the rate of product formation.

a result. A positive effector increases enzyme activity, whereas a negative effector decreases activity or inhibits the enzyme. These changes in activity often result from alterations in the apparent affinity of the enzyme for its substrate, but changes in maximum velocity also can occur.

The kinetic characteristics of nonregulatory enzymes show that the Michaelis constant (K_m) is the substrate concentration required for an enzyme to operate at half its maximal velocity. This constant applies only to hyperbolic substrate saturation curves, not to the sigmoidal curves often seen with allosteric enzymes (figure 8.23). The substrate concentration required for half maximal velocity with allosteric enzymes having sigmoidal substrate curves is called the $[S]_{0.5}$ or $K_{0.5}$ value.

One of the best-studied allosteric regulatory enzymes is the aspartate carbamoyltransferase (ACTase) from *E. coli*. The enzyme catalyzes the condensation of carbamoyl phosphate with aspartate to form carbamoylaspartate (figure 8.22). ACTase catalyzes the rate-determining reaction of the pyrimidine biosynthetic pathway in *E. coli*. The substrate saturation curve is sigmoidal when the concentration of either substrate is varied

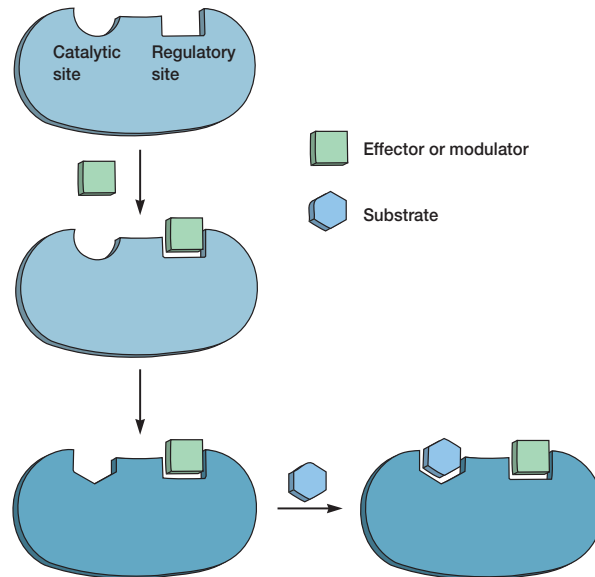


Figure 8.21 Allosteric Regulation. The structure and function of an allosteric enzyme. In this example the effector or modulator first binds to a separate regulatory site and causes a change in enzyme conformation that results in an alteration in the shape of the active site. The active site can now more effectively bind the substrate. This effector is a positive effector because it stimulates substrate binding and catalytic activity.

(figure 8.23). The enzyme has more than one active site, and the binding of a substrate molecule to an active site increases the binding of substrate at the other sites. In addition, cytidine triphosphate (CTP), an end product of pyrimidine biosynthesis, inhibits the enzyme and the purine ATP activates it. Both effectors alter the $K_{0.5}$ value of the enzyme but not its maximum velocity. CTP inhibits by increasing $K_{0.5}$ or by shifting the substrate saturation curve to higher values. This allows the enzyme to operate more slowly at a particular substrate concentration when CTP is present. ATP activates by moving the curve to lower substrate concentration values so that the enzyme is maximally active over a wider substrate concentration range. Thus when the pathway is so active that the CTP concentration rises too high, ACTase activity decreases and the rate of end product formation slows. In contrast, when the purine end product ATP increases in concentration relative to CTP, it stimulates CTP synthesis through its effects on ACTase. [Pyrimidine and purine biosynthesis \(pp. 216–18\)](#)

E. coli aspartate carbamoyltransferase provides a clear example of separate regulatory and catalytic sites in allosteric en-

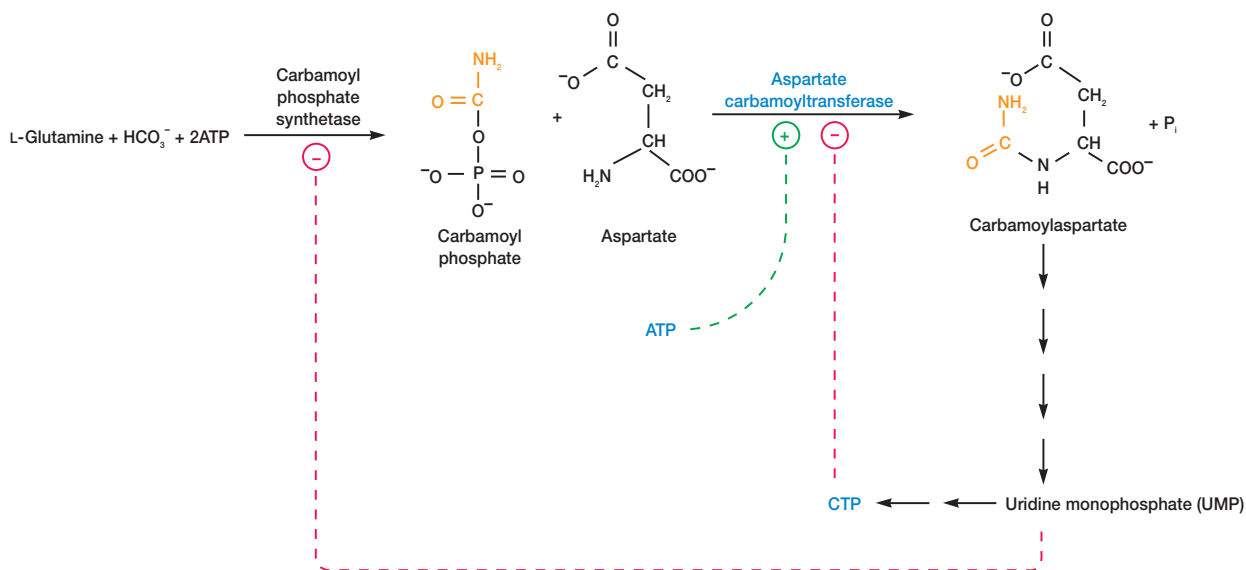


Figure 8.22 ACTase Regulation. The aspartate carbamoyltransferase reaction and its role in the regulation of pyrimidine biosynthesis. The end product CTP inhibits its activity (–) while ATP activates the enzyme (+). Carbamoyl phosphate synthetase is also inhibited by pathway end products such as UMP.

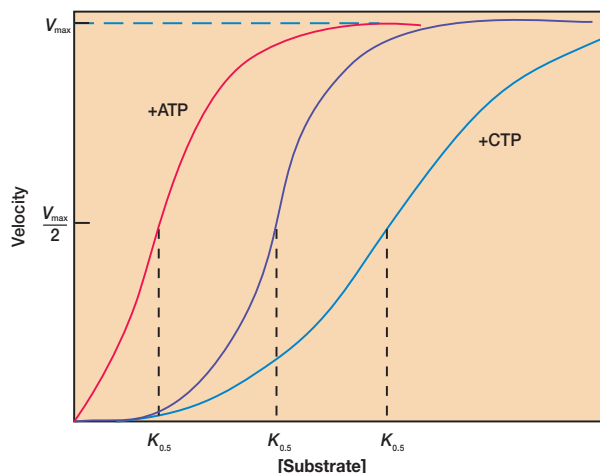


Figure 8.23 The Kinetics of *E. coli* Aspartate Carbamoyltransferase. CTP, a negative effector, increases the $K_{0.5}$ value while ATP, a positive effector, lowers the $K_{0.5}$. The $V_{\max}$ remains constant.

zymes. The enzyme is a large protein composed of two catalytic subunits and three regulatory subunits (figure 8.24a). The catalytic subunits contain only catalytic sites and are unaffected by CTP and ATP. Regulatory subunits do not catalyze the reaction but do possess regulatory sites to which CTP and ATP bind. When these effectors bind to the complete enzyme, they cause conformational changes in the regulatory subunits and subsequently in the catalytic subunits and their catalytic sites. The enzyme can change reversibly between a less active T form and a more active R form (figure 8.24b,c). Thus the regulatory site influences a catalytic site about 6.0 nm distant.

Covalent Modification of Enzymes

Regulatory enzymes also can be switched on and off by **reversible covalent modification**. Usually this occurs through the addition and removal of a particular group, one form of the enzyme being more active than the other. For example, glycogen phosphorylase of the bread mold *Neurospora crassa* exists in phosphorylated and dephosphorylated forms called phosphorylase *a* and phosphorylase *b*, respectively (figure 8.25). Phosphorylase *b* is inactive

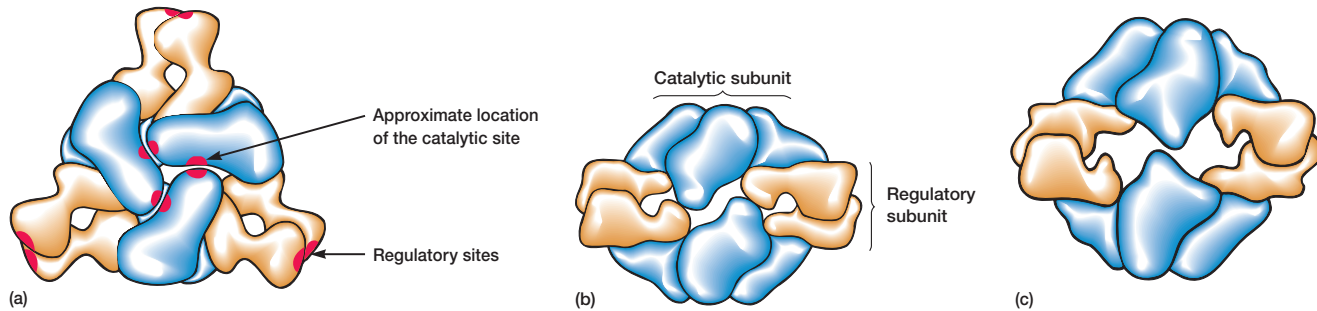


Figure 8.24 The Structure and Regulation of *E. coli* Aspartate Carbamoyltransferase. (a) A schematic diagram of the enzyme showing the six catalytic polypeptide chains (blue), the six regulatory chains (orange), and the catalytic and regulatory sites. The enzyme is viewed from the top. Each catalytic subunit contains three catalytic chains, and each regulatory subunit has two chains. (b) The less active T state of ACTase viewed from the side. (c) The more active R state of ACTase. The regulatory subunits have rotated and pushed the catalytic subunits apart.

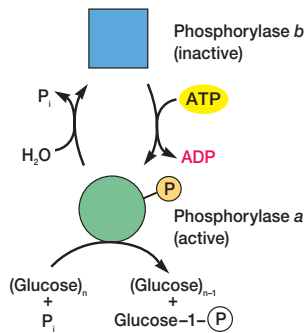
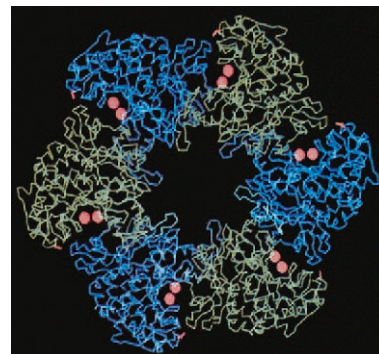
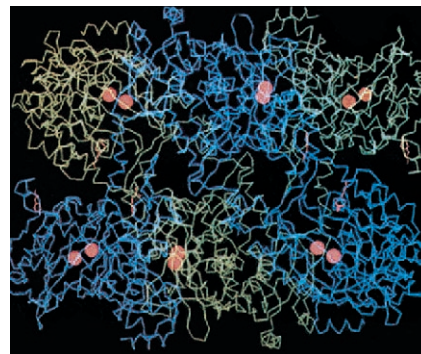


Figure 8.25 Reversible Covalent Modification of Glycogen Phosphorylase. The active form, phosphorylase *a*, is produced by phosphorylation and is inactivated when the phosphate is removed hydrolytically to produce inactive phosphorylase *b*.



(a)



(b)

Figure 8.26 The Structure of *E. coli* Glutamine Synthetase. The enzyme contains 12 subunits in the shape of a hexagonal prism. For clarity the subunits are colored alternating green and blue. Each of the six catalytic sites has a pair of Mn^{2+} ions (red). The tyrosine residues to which adenylyl groups can be attached are colored red. (a) Top view of molecule. (b) Side view showing the six nearest subunits.

because its required activator AMP is usually not present at sufficiently high levels. Phosphorylase *a*, the phosphorylated form, is active even without AMP. Glycogen phosphorylase is stimulated by phosphorylation of phosphorylase *b* to produce phosphorylase *a*. The attachment of phosphate changes the enzyme's conformation to an active form. Phosphorylation and dephosphorylation are catalyzed by separate enzymes, which also are regulated. [Phosphorylase and glycogen degradation \(p. 192\)](#)

Enzymes can be regulated through the attachment of groups other than phosphate. One of the most intensively studied regulatory enzymes is *E. coli* glutamine synthetase, a large, complex enzyme existing in two forms (**figure 8.26**). When an adenylic acid residue is attached to each of its 12 subunits forming an adenyly-